

# JESS Thermodynamic Database v8.9

C13

24-Jan-23

| Reaction No. 2889                                         |    |       |                                |                |   |       |        |
|-----------------------------------------------------------|----|-------|--------------------------------|----------------|---|-------|--------|
| 2BenzoylAm4MePentan-1 + Zn+2 = Zn+2_2BenzoylAm4MePentan-1 |    |       |                                |                |   |       |        |
| No                                                        | t  | I-Str | Medium                         | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.1   | Water:Dioxan<br>1:1Vol<br>KNO3 | lgK:2.74 (3SF) | 5 | MGL   | 2261   |
| 2                                                         | 35 | 0.1   | Water:Dioxan<br>1:1Vol<br>KNO3 | dH:2. (1SF)    | 4 | ENS   | 2261   |

| Reaction No. 2912                                         |    |       |                                |                |   |       |        |
|-----------------------------------------------------------|----|-------|--------------------------------|----------------|---|-------|--------|
| 2BenzoylAm4MePentan-1 + Pb+2 = Pb+2_2BenzoylAm4MePentan-1 |    |       |                                |                |   |       |        |
| No                                                        | t  | I-Str | Medium                         | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.1   | Water:Dioxan<br>1:1Vol<br>KNO3 | lgK:3.32 (3SF) | 5 | MGL   | 2261   |
| 2                                                         | 35 | 0.1   | Water:Dioxan<br>1:1Vol<br>KNO3 | dH:-4.5 (2SF)  | 4 | ENS   | 2261   |

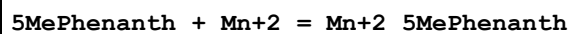
| Reaction No. 2913                                         |    |       |                                |                |   |       |        |
|-----------------------------------------------------------|----|-------|--------------------------------|----------------|---|-------|--------|
| 2BenzoylAm4MePentan-1 + Cd+2 = Cd+2_2BenzoylAm4MePentan-1 |    |       |                                |                |   |       |        |
| No                                                        | t  | I-Str | Medium                         | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.1   | Water:Dioxan<br>1:1Vol<br>KNO3 | lgK:2.72 (3SF) | 5 | MGL   | 2261   |
| 2                                                         | 35 | 0.1   | Water:Dioxan<br>1:1Vol<br>KNO3 | dH:3.4 (2SF)   | 5 | ENS   | 2261   |

| Reaction No. 3067                                       |    |       |                        |                |   |       |        |
|---------------------------------------------------------|----|-------|------------------------|----------------|---|-------|--------|
| H+1 + 2BenzoylAm4MePentan-1 = H+1_2BenzoylAm4MePentan-1 |    |       |                        |                |   |       |        |
| No                                                      | t  | I-Str | Medium                 | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 25 |       | Water:Dioxan<br>1:1Vol | lgK:5.50 (3SF) | 5 | MGL   | 2261   |

| Reaction No. 3600                   |  |  |  |  |  |  |  |
|-------------------------------------|--|--|--|--|--|--|--|
| H+1 + 5MePhenanth = H+1_5MePhenanth |  |  |  |  |  |  |  |

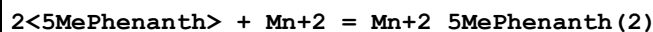
| No | t     | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|-------|-------|---------------|-------------------|---|-------|--------|
| 1  | 15    | 0     | Inf. Dilution | lgK:5.22 (0.02SD) | 4 | MSP   | 999    |
| 2  | 20    | 0     | Inf. Dilution | lgK:5.23 (3SF)    | 6 | MGL   | 999    |
| 3  | 25    | 0     | Inf. Dilution | lgK:5.11 (0.01SD) | 4 | MSP   | 999    |
| 4  | 25    | 0     | Unknown       | lgK:5.17 (3SF)    | 6 | ENS   | 773    |
| 5  | 25    | 0.1   | KCl           | lgK:5.28 (3SF)    | 6 | MGL   | 999    |
| 6  | 25    | 0.1   | KNO3          | lgK:5.26 (3SF)    | 6 | MGL   | 999    |
| 7  | 25    | 0.1   | Unknown       | lgK:5.25 (3SF)    | 6 | MSP   | 999    |
| 8  | 35    | 0     | Inf. Dilution | lgK:4.99 (0.02SD) | 4 | MSP   | 999    |
| 9  | 15:35 | 0     | Inf. Dilution | dH:-4. (0.3SD)    | 5 | MCL   | 999    |

## Reaction No. 3605



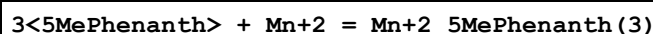
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:4.28 (3SF) | 6 | MDS   | 3077   |

## Reaction No. 3606



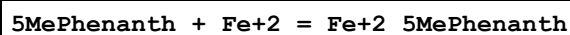
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:7.61 (3SF) | 5 | MDS   | 3077   |

## Reaction No. 3607



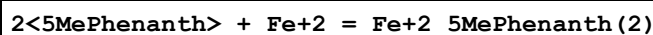
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl     | lgK:11.21 (4SF) | 4 | MDS   | 3077   |
| 2  | 25 | 0.1   | Unknown | lgK:15.3 (3SF)  | 0 | CRV   | 2556   |

## Reaction No. 3608



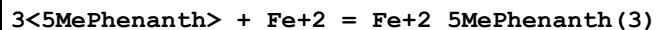
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 22 | 0     | Inf. Dilution | lgK:6.11 (3SF)   | 6 | MSP   | 999    |
| 2  | 25 | 0.1   | KCl           | lgK:6.46 (3SF)   | 6 | MDS   | 3077   |
| 3  | 25 | 0.1   | Unknown       | lgK:6.00 (3SF)   | 6 | ENS   | 773    |
| 4  | 25 | 0.15  | KNO3          | lgK:6.0 (0.05SD) | 3 | MSP   | 906    |

## Reaction No. 3609



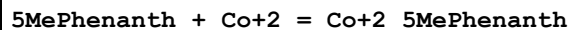
| No | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:11.45 (4SF)   | 6 | ENS   | 773    |
| 2  | 25 | 0.15  | KNO3    | lgK:11.5 (0.06SD) | 3 | MSP   | 906    |

## Reaction No. 3610



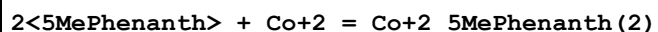
| No | t     | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|-------|-------|---------------|-------------------|---|-------|--------|
| 1  | 15    | 0     | Inf. Dilution | lgK:21.9 (0.04SD) | 2 | MSP   | 999    |
| 2  | 22    | 0     | Inf. Dilution | lgK:21.33 (4SF)   | 2 | MSP   | 999    |
| 3  | 25    | 0.1   | KCl           | lgK:21.94 (4SF)   | 2 | MDS   | 999    |
| 4  | 25    | 0.1   | KCl           | lgK:21.94 (4SF)   | 2 | ENS   | 3077   |
| 5  | 25    | 0.15  | KNO3          | lgK:16.6 (0.05SD) | 0 | MSP   | 906    |
| 6  | 35    | 0     | Inf. Dilution | lgK:20.2 (0.03SD) | 2 | MSP   | 999    |
| 7  | 45    | 0     | Inf. Dilution | lgK:19.45 (4SF)   | 2 | MSP   | 999    |
| 8  | 15:45 | 0     | Inf. Dilution | dH:-34. (0.7SD)   | 2 | MCL   | 999    |

## Reaction No. 3611



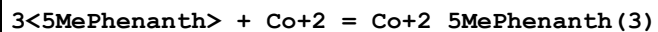
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:7.14 (3SF) | 6 | MDS   | 3077   |

## Reaction No. 3612



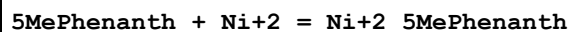
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:14.0 (3SF) | 4 | MDS   | 3077   |

## Reaction No. 3613



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:20.60 (4SF) | 4 | MDS   | 3077   |

## Reaction No. 3614



| No | t  | I-Str | Medium                            | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|-----------------------------------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl                               | lgK:8.30 (3SF)   | 6 | MDS   | 3077   |
| 2  | 25 | 0.1   | Ethanol:Water<br>50:50Wgt<br>KNO3 | lgK:6.1 (0.05SD) | 4 | MAG   | 906    |

| Reaction No. 3615                           |    |       |                                   |                  |   |       |        |
|---------------------------------------------|----|-------|-----------------------------------|------------------|---|-------|--------|
| 2<5MePhenanth> + Ni+2 = Ni+2_5MePhenanth(2) |    |       |                                   |                  |   |       |        |
| No                                          | t  | I-Str | Medium                            | Value (Dev)      | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KCl                               | lgK:16.95(4SF)   | 6 | MDS   | 3077   |
| 2                                           | 25 | 0.1   | Ethanol:Water<br>50:50Wgt<br>KNO3 | lgK:12.1(0.05SD) | 4 | MAG   | 906    |

| Reaction No. 3616                           |    |       |                                   |                |   |       |        |
|---------------------------------------------|----|-------|-----------------------------------|----------------|---|-------|--------|
| 3<5MePhenanth> + Ni+2 = Ni+2_5MePhenanth(3) |    |       |                                   |                |   |       |        |
| No                                          | t  | I-Str | Medium                            | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KCl                               | lgK:24.68(4SF) | 6 | MDS   | 3077   |
| 2                                           | 25 | 0.1   | Ethanol:Water<br>50:50Wgt<br>KNO3 | lgK:17.(0.2SD) | 4 | MAG   | 906    |

| Reaction No. 3617                     |    |       |        |               |   |       |        |
|---------------------------------------|----|-------|--------|---------------|---|-------|--------|
| 5MePhenanth + Cu+2 = Cu+2_5MePhenanth |    |       |        |               |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KCl    | lgK:8.55(3SF) | 6 | MDS   | 3077   |

| Reaction No. 3618                           |    |       |        |                |   |       |        |
|---------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 2<5MePhenanth> + Cu+2 = Cu+2_5MePhenanth(2) |    |       |        |                |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KCl    | lgK:15.02(4SF) | 6 | MDS   | 3077   |

| Reaction No. 3619                           |    |       |        |                |   |       |        |
|---------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 3<5MePhenanth> + Cu+2 = Cu+2_5MePhenanth(3) |    |       |        |                |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KCl    | lgK:20.12(4SF) | 6 | MDS   | 3077   |

| Reaction No. 3620                     |    |       |                                   |               |   |       |        |
|---------------------------------------|----|-------|-----------------------------------|---------------|---|-------|--------|
| 5MePhenanth + Ag+1 = Ag+1_5MePhenanth |    |       |                                   |               |   |       |        |
| No                                    | t  | I-Str | Medium                            | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | Unknown                           | lgK:7.30(3SF) | 6 | ENS   | 773    |
| 2                                     | 25 | 0.1   | Ethanol:Water<br>50:50Wgt<br>KNO3 | lgK:5.34(3SF) | 4 | MAG   | 906    |

| Reaction No. 3621                           |    |       |                                   |                |   |       |        |
|---------------------------------------------|----|-------|-----------------------------------|----------------|---|-------|--------|
| 2<5MePhenanth> + Ag+1 = Ag+1_5MePhenanth(2) |    |       |                                   |                |   |       |        |
| No                                          | t  | I-Str | Medium                            | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | Unknown                           | lgK:13.39(4SF) | 6 | ENS   | 773    |
| 2                                           | 25 | 0.1   | Ethanol:Water<br>50:50Wgt<br>KNO3 | lgK:10.25(4SF) | 4 | MAG   | 906    |

| Reaction No. 3622                     |    |       |                                   |                  |   |       |        |
|---------------------------------------|----|-------|-----------------------------------|------------------|---|-------|--------|
| 5MePhenanth + Zn+2 = Zn+2_5MePhenanth |    |       |                                   |                  |   |       |        |
| No                                    | t  | I-Str | Medium                            | Value (Dev)      | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KCl                               | lgK:6.62(3SF)    | 6 | MDS   | 3077   |
| 2                                     | 25 | 0.1   | Ethanol:Water<br>50:50Wgt<br>KNO3 | lgK:5.82(0.01SD) | 4 | MAG   | 906    |

| Reaction No. 3623                           |    |       |                                   |                   |   |       |        |
|---------------------------------------------|----|-------|-----------------------------------|-------------------|---|-------|--------|
| 2<5MePhenanth> + Zn+2 = Zn+2_5MePhenanth(2) |    |       |                                   |                   |   |       |        |
| No                                          | t  | I-Str | Medium                            | Value (Dev)       | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KCl                               | lgK:12.58(4SF)    | 6 | MDS   | 3077   |
| 2                                           | 25 | 0.1   | Ethanol:Water<br>50:50Wgt<br>KNO3 | lgK:11.53(0.02SD) | 4 | MAG   | 906    |

| Reaction No. 3624                           |    |       |                                   |                  |   |       |        |
|---------------------------------------------|----|-------|-----------------------------------|------------------|---|-------|--------|
| 3<5MePhenanth> + Zn+2 = Zn+2_5MePhenanth(3) |    |       |                                   |                  |   |       |        |
| No                                          | t  | I-Str | Medium                            | Value (Dev)      | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KCl                               | lgK:18.25(4SF)   | 6 | MDS   | 3077   |
| 2                                           | 25 | 0.1   | Ethanol:Water<br>50:50Wgt<br>KNO3 | lgK:16.5(0.04SD) | 4 | MAG   | 906    |

| Reaction No. 3625                     |    |       |                                   |                 |   |       |        |
|---------------------------------------|----|-------|-----------------------------------|-----------------|---|-------|--------|
| 5MePhenanth + Cd+2 = Cd+2_5MePhenanth |    |       |                                   |                 |   |       |        |
| No                                    | t  | I-Str | Medium                            | Value (Dev)     | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KNO3                              | lgK:6.13(3SF)   | 6 | MDS   | 3077   |
| 2                                     | 25 | 0.1   | Ethanol:Water<br>50:50Wgt<br>KNO3 | lgK:5.3(0.04SD) | 4 | MAG   | 906    |

| Reaction No. 3626 |  |  |  |  |  |  |  |
|-------------------|--|--|--|--|--|--|--|
|-------------------|--|--|--|--|--|--|--|

| 2<5MePhenanth> + Cd+2 = Cd+2_5MePhenanth(2) |    |       |                                   |                  |   |       |        |
|---------------------------------------------|----|-------|-----------------------------------|------------------|---|-------|--------|
| No                                          | t  | I-Str | Medium                            | Value (Dev)      | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KNO3                              | lgK:11.03(4SF)   | 6 | MDS   | 3077   |
| 2                                           | 25 | 0.1   | Ethanol:Water<br>50:50Wgt<br>KNO3 | lgK:10.6(0.05SD) | 4 | MAG   | 906    |

| Reaction No. 6372                                   |    |       |        |               |   |       |        |
|-----------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| H+1 + ProLeuGlyHydroxam-1 = H+1_ProLeuGlyHydroxam-1 |    |       |        |               |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.2   | KCl    | lgK:8.22(3SF) | 6 | MSP   | 972    |

| Reaction No. 7513                                          |    |       |        |               |   |       |        |
|------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| H+1_ProLeuGlyHydroxam-1 + H+1 = H+1(2)_ProLeuGlyHydroxam-1 |    |       |        |               |   |       |        |
| No                                                         | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                          | 25 | 0.2   | KCl    | lgK:9.06(3SF) | 6 | MSP   | 972    |

| Reaction No. 8401                                               |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + ProLeuGlyHydroxam-1 + Co+2 = Co+2_H+1_ProLeuGlyHydroxam-1 |    |       |        |                  |   |       |        |
| No                                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                               | 25 | 0.2   | KCl    | lgK:12.5(0.04SD) | 4 | MSP   | 972    |

| Reaction No. 8402                                     |    |       |        |                  |   |       |        |
|-------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| ProLeuGlyHydroxam-1 + Co+2 = Co+2_ProLeuGlyHydroxam-1 |    |       |        |                  |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.2   | KCl    | lgK:4.96(0.02SD) | 4 | MSP   | 972    |

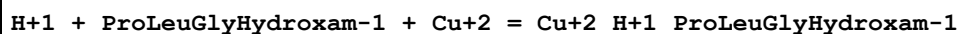
| Reaction No. 8403                                           |    |       |        |                 |   |       |        |
|-------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<ProLeuGlyHydroxam-1> + Co+2 = Co+2_ProLeuGlyHydroxam-1(2) |    |       |        |                 |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.2   | KCl    | lgK:8.2(0.08SD) | 4 | MSP   | 972    |

| Reaction No. 8404                                               |    |       |        |                 |   |       |        |
|-----------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| H+1 + ProLeuGlyHydroxam-1 + Zn+2 = Zn+2_H+1_ProLeuGlyHydroxam-1 |    |       |        |                 |   |       |        |
| No                                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                               | 25 | 0.2   | KCl    | lgK:12.5(0.1SD) | 4 | MSP   | 972    |

| Reaction No. 8405                                     |  |  |  |  |  |  |  |
|-------------------------------------------------------|--|--|--|--|--|--|--|
| ProLeuGlyHydroxam-1 + Zn+2 = Zn+2_ProLeuGlyHydroxam-1 |  |  |  |  |  |  |  |

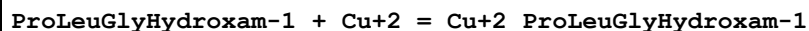
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:4.75 (0.02SD) | 4 | MSP   | 972    |

## Reaction No. 8406



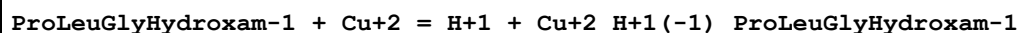
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:14.9 (0.1SD) | 4 | MSP   | 972    |

## Reaction No. 8407



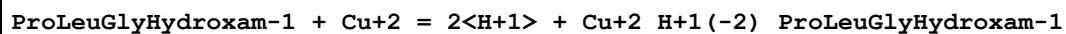
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:9.5 (0.06SD) | 3 | MSP   | 972    |

## Reaction No. 8408



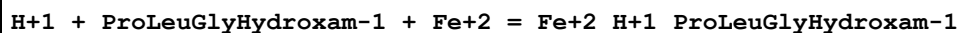
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:3.4 (0.05SD) | 4 | MSP   | 972    |

## Reaction No. 8409



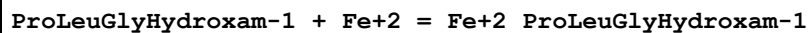
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:-3.9 (0.03SD) | 4 | MSP   | 972    |

## Reaction No. 8410



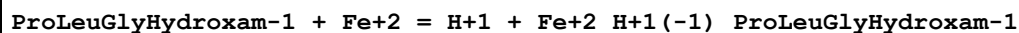
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:17.5 (0.03SD) | 6 | MSP   | 972    |

## Reaction No. 8411



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:13.7 (0.05SD) | 4 | MSP   | 972    |

## Reaction No. 8412



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:8.4 (0.07SD) | 4 | MSP   | 972    |

| Reaction No. 8413                                           |    |       |        |                 |   |       |        |
|-------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<ProLeuGlyHydroxam-1> + Fe+2 = Fe+2_ProLeuGlyHydroxam-1(2) |    |       |        |                 |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.2   | KCl    | lgK:21.0(0.1SD) | 3 | MSP   | 972    |

| Reaction No. 8414                                                         |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<ProLeuGlyHydroxam-1> + Fe+2 = H+1 + Fe+2_H+1(-1)_ProLeuGlyHydroxam-1(2) |    |       |        |                 |   |       |        |
| No                                                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0.2   | KCl    | lgK:12.5(0.1SD) | 4 | MSP   | 972    |

| Reaction No. 8415                                                            |    |       |        |                 |   |       |        |
|------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<ProLeuGlyHydroxam-1> + Fe+2 = 2<H+1> + Fe+2_H+1(-2)_ProLeuGlyHydroxam-1(2) |    |       |        |                 |   |       |        |
| No                                                                           | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                            | 25 | 0.2   | KCl    | lgK:3.7(0.09SD) | 4 | MSP   | 972    |

| Reaction No. 9005                   |    |       |        |                  |   |       |        |
|-------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + GlyTyrGly-2 = H+1_GlyTyrGly-2 |    |       |        |                  |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | NaClO4 | lgK:9.8(0.05SD)  | 5 | MGL   | 2450   |
| 2                                   | 25 | 0.2   | KCl    | lgK:9.78(0.01SD) | 6 | MGL   | 1018   |
| 3                                   | 25 | 0.1   | NaClO4 | dG:-13.3(3SF)    | 5 | MGL   | 2450   |
| 4                                   | 25 | 0.1   | NaClO4 | dH:-6.7(2SF)     | 5 | MCL   | 2450   |
| 5                                   | 25 | 0.1   | NaClO4 | dS:22.3(3SF)     | 5 | EFE   | 2450   |

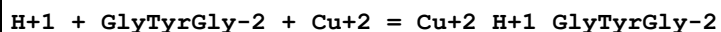
| Reaction No. 9006                          |    |       |        |                  |   |       |        |
|--------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1_GlyTyrGly-2 + H+1 = H+1(2)_GlyTyrGly-2 |    |       |        |                  |   |       |        |
| No                                         | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                          | 25 | 0.1   | NaClO4 | lgK:8.1(0.03SD)  | 5 | MGL   | 2450   |
| 2                                          | 25 | 0.2   | KCl    | lgK:7.92(0.01SD) | 4 | MGL   | 1018   |
| 3                                          | 25 | 0.1   | NaClO4 | dG:-11.0(3SF)    | 5 | MGL   | 2450   |
| 4                                          | 25 | 0.1   | NaClO4 | dH:-8.3(2SF)     | 5 | MCL   | 2450   |
| 5                                          | 25 | 0.1   | NaClO4 | dS:9.0(2SF)      | 5 | EFE   | 2450   |

| Reaction No. 9007                             |    |       |        |                 |   |       |        |
|-----------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| H+1(2)_GlyTyrGly-2 + H+1 = H+1(3)_GlyTyrGly-2 |    |       |        |                 |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | NaClO4 | lgK:3.1(0.04SD) | 5 | MGL   | 2450   |



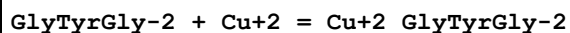
|   |    |     |     |                  |   |     |      |
|---|----|-----|-----|------------------|---|-----|------|
| 2 | 25 | 0.2 | KCl | lgK:3.36(0.01SD) | 6 | MGL | 1018 |
|---|----|-----|-----|------------------|---|-----|------|

## Reaction No. 9008



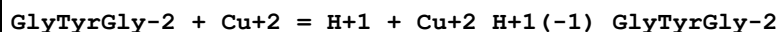
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:14.8(0.03SD) | 4 | MGL   | 1018   |

## Reaction No. 9009



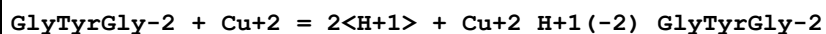
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:9.72(0.01SD) | 6 | MGL   | 1018   |

## Reaction No. 9010



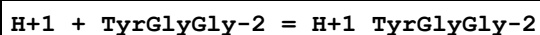
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:3.52(0.01SD) | 6 | MGL   | 1018   |

## Reaction No. 9011



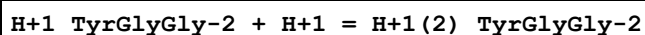
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.2   | KCl    | lgK:-6.41(0.01SD) | 6 | MGL   | 1018   |

## Reaction No. 9012



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:9.8(0.06SD)   | 5 | MGL   | 2450   |
| 2  | 25 | 0.2   | KCl    | lgK:10.09(0.02SD) | 3 | MGL   | 1018   |
| 3  | 25 | 0.1   | NaClO4 | dG:-13.3(3SF)     | 5 | MGL   | 2450   |
| 4  | 25 | 0.1   | NaClO4 | dH:-5.4(2SF)      | 5 | MCL   | 2450   |
| 5  | 25 | 0.1   | NaClO4 | dS:26.5(3SF)      | 5 | EFE   | 2450   |

## Reaction No. 9013



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:7.75(0.02SD) | 5 | MGL   | 2450   |
| 2  | 25 | 0.2   | KCl    | lgK:7.37(0.01SD) | 6 | MGL   | 1018   |
| 3  | 25 | 0.1   | NaClO4 | dG:-10.5(3SF)    | 5 | MGL   | 2450   |

|   |    |     |        |               |   |     |      |
|---|----|-----|--------|---------------|---|-----|------|
| 4 | 25 | 0.1 | NaClO4 | dH:-7.5 (2SF) | 5 | MCL | 2450 |
| 5 | 25 | 0.1 | NaClO4 | dS:10.3 (3SF) | 5 | EFE | 2450 |

| Reaction No. 9014                             |    |       |        |                   |   |       |        |
|-----------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| H+1(2)_TyrGlyGly-2 + H+1 = H+1(3)_TyrGlyGly-2 |    |       |        |                   |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | NaClO4 | lgK:3.2 (0.03SD)  | 5 | MGL   | 2450   |
| 2                                             | 25 | 0.2   | KCl    | lgK:3.19 (0.01SD) | 6 | MGL   | 1018   |

| Reaction No. 9015                               |    |       |        |                  |   |       |        |
|-------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + TyrGlyGly-2 + Cu+2 = Cu+2_H+1_TyrGlyGly-2 |    |       |        |                  |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                               | 25 | 0.2   | KCl    | lgK:14.8 (0.1SD) | 4 | MGL   | 1018   |

| Reaction No. 9016                     |    |       |        |                   |   |       |        |
|---------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| TyrGlyGly-2 + Cu+2 = Cu+2_TyrGlyGly-2 |    |       |        |                   |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                     | 25 | 0.2   | KCl    | lgK:9.77 (0.02SD) | 3 | MGL   | 1018   |

| Reaction No. 9017                                   |    |       |        |                   |   |       |        |
|-----------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| TyrGlyGly-2 + Cu+2 = H+1 + Cu+2_H+1(-1)_TyrGlyGly-2 |    |       |        |                   |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.2   | KCl    | lgK:3.57 (0.02SD) | 6 | MGL   | 1018   |

| Reaction No. 9018                                      |    |       |        |                   |   |       |        |
|--------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| TyrGlyGly-2 + Cu+2 = 2<H+1> + Cu+2_H+1(-2)_TyrGlyGly-2 |    |       |        |                   |   |       |        |
| No                                                     | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                      | 25 | 0.2   | KCl    | lgK:-6.7 (0.03SD) | 4 | MGL   | 1018   |

| Reaction No. 11072              |    |       |        |                 |   |       |        |
|---------------------------------|----|-------|--------|-----------------|---|-------|--------|
| H+1 + 1PrEDTA-4 = H+1_1PrEDTA-4 |    |       |        |                 |   |       |        |
| No                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                               | 20 | 0.1   | KNO3   | lgK:10.97 (4SF) | 6 | MPL   | 2011   |

| Reaction No. 11073                     |    |       |        |                |   |       |        |
|----------------------------------------|----|-------|--------|----------------|---|-------|--------|
| H+1_1PrEDTA-4 + H+1 = H+1(2)_1PrEDTA-4 |    |       |        |                |   |       |        |
| No                                     | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                      | 20 | 0.1   | KNO3   | lgK:6.05 (3SF) | 6 | MPL   | 2011   |

| Reaction No. 11074                                               |    |       |        |               |   |       |        |
|------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $\text{H+1(2)_1PrEDTA-4} + \text{H+1} = \text{H+1(3)_1PrEDTA-4}$ |    |       |        |               |   |       |        |
| No                                                               | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                | 20 | 0.1   | KNO3   | lgK:3.05(3SF) | 6 | MPL   | 2011   |

| Reaction No. 11075                                               |    |       |        |              |   |       |        |
|------------------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| $\text{H+1(3)_1PrEDTA-4} + \text{H+1} = \text{H+1(4)_1PrEDTA-4}$ |    |       |        |              |   |       |        |
| No                                                               | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                | 20 | 0.1   | KNO3   | lgK:2.1(2SF) | 4 | MPL   | 2011   |

| Reaction No. 11076                                            |    |       |        |                |   |       |        |
|---------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{H+1} + 2\text{PrEDTA-4} = \text{H+1}_2\text{PrEDTA-4}$ |    |       |        |                |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                             | 20 | 0.1   | KNO3   | lgK:11.25(4SF) | 6 | MPT   | 2011   |

| Reaction No. 11077                                                          |    |       |        |               |   |       |        |
|-----------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $\text{H+1}_2\text{PrEDTA-4} + \text{H+1} = \text{H+1(2)}_2\text{PrEDTA-4}$ |    |       |        |               |   |       |        |
| No                                                                          | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                           | 20 | 0.1   | KNO3   | lgK:5.64(3SF) | 6 | MPT   | 2011   |

| Reaction No. 11078                                                             |    |       |        |               |   |       |        |
|--------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $\text{H+1(2)}_2\text{PrEDTA-4} + \text{H+1} = \text{H+1(3)}_2\text{PrEDTA-4}$ |    |       |        |               |   |       |        |
| No                                                                             | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                              | 20 | 0.1   | KNO3   | lgK:3.33(3SF) | 6 | MPT   | 2011   |

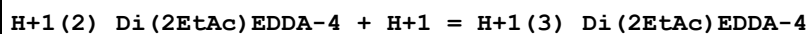
| Reaction No. 11079                                                             |    |       |        |              |   |       |        |
|--------------------------------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| $\text{H+1(3)}_2\text{PrEDTA-4} + \text{H+1} = \text{H+1(4)}_2\text{PrEDTA-4}$ |    |       |        |              |   |       |        |
| No                                                                             | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                              | 20 | 0.1   | KNO3   | lgK:2.4(2SF) | 5 | MPT   | 2011   |

| Reaction No. 11108                                                          |    |       |         |                |   |       |        |
|-----------------------------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| $\text{H+1} + \text{Di(2EtAc)EDDA-4} = \text{H+1}_{\text{Di(2EtAc)EDDA-4}}$ |    |       |         |                |   |       |        |
| No                                                                          | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                           | 20 | 0.1   | Unknown | lgK:10.42(4SF) | 6 | ENS   | 141    |

| Reaction No. 11109                                                                          |  |  |  |  |  |  |  |
|---------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $\text{H+1}_{\text{Di(2EtAc)EDDA-4}} + \text{H+1} = \text{H+1(2)}_{\text{Di(2EtAc)EDDA-4}}$ |  |  |  |  |  |  |  |

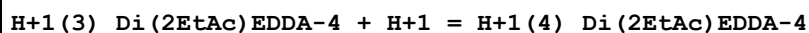
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown | lgK:6.09 (3SF) | 6 | ENS   | 141    |

## Reaction No. 11110



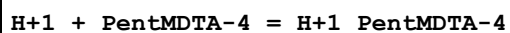
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown | lgK:2.69 (3SF) | 6 | ENS   | 141    |

## Reaction No. 11111



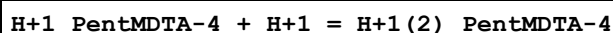
| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown | lgK:1.8 (2SF) | 4 | ENS   | 141    |

## Reaction No. 11128



| No | t  | I-Str | Medium   | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|----------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl      | lgK:10.58 (4SF) | 6 | MHY   | 1957   |
| 2  | 20 | 0.1   | KNO3     | lgK:10.70 (4SF) | 6 | MGL   | 1956   |
| 3  | 20 | 0.1   | Me4NC1   | lgK:10.75 (4SF) | 6 | MGL   | 188    |
| 4  | 20 | 1     | KNO3     | lgK:10.50 (4SF) | 6 | MGL   | 188    |
| 5  | 20 | 1     | Me4NC1   | lgK:10.49 (4SF) | 6 | MHY   | 188    |
| 6  | 20 | 1     | NaBr     | lgK:10.27 (4SF) | 6 | MMX   | 1562   |
| 7  | 25 | 0.1   | K+1 salt | lgK:10.60 (4SF) | 6 | ENS   | 233    |
| 8  | 20 | 0.1   | KNO3     | dH:-7.5 (2SF)   | 5 | MCL   | 1956   |

## Reaction No. 11129



| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl     | lgK:9.50 (3SF) | 6 | MHY   | 1957   |
| 2  | 20 | 0.1   | KNO3    | lgK:9.52 (3SF) | 6 | MGL   | 1956   |
| 3  | 20 | 0.1   | Me4NC1  | lgK:9.58 (3SF) | 6 | MGL   | 188    |
| 4  | 20 | 1     | KNO3    | lgK:9.46 (3SF) | 6 | MGL   | 188    |
| 5  | 20 | 1     | Me4NC1  | lgK:9.42 (3SF) | 6 | MHY   | 188    |
| 6  | 20 | 1     | NaBr    | lgK:9.36 (3SF) | 6 | MMX   | 1562   |
| 7  | 25 | 0.1   | Unknown | lgK:9.43 (3SF) | 6 | ENS   | 233    |
| 8  | 20 | 0.1   | KNO3    | dH:-6.3 (2SF)  | 5 | MCL   | 1956   |

| Reaction No. 11130                          |    |       |         |               |   |       |        |
|---------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| H+1(2)_PentMDTA-4 + H+1 = H+1(3)_PentMDTA-4 |    |       |         |               |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | KCl     | lgK:2.70(3SF) | 6 | MHY   | 1957   |
| 2                                           | 20 | 0.1   | KNO3    | lgK:2.7(2SF)  | 5 | MGL   | 1956   |
| 3                                           | 20 | 1     | KNO3    | lgK:2.6(2SF)  | 5 | MGL   | 188    |
| 4                                           | 20 | 1     | Me4NC1  | lgK:2.6(2SF)  | 5 | MHY   | 188    |
| 5                                           | 20 | 1     | NaBr    | lgK:2.62(3SF) | 6 | MMX   | 1562   |
| 6                                           | 25 | 0.1   | Unknown | lgK:2.70(3SF) | 6 | ENS   | 233    |

| Reaction No. 11131                          |    |       |         |               |   |       |        |
|---------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| H+1(3)_PentMDTA-4 + H+1 = H+1(4)_PentMDTA-4 |    |       |         |               |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | KCl     | lgK:2.3(2SF)  | 4 | MHY   | 1957   |
| 2                                           | 20 | 0.1   | KNO3    | lgK:2.3(2SF)  | 4 | MGL   | 1956   |
| 3                                           | 20 | 1     | KNO3    | lgK:2.4(2SF)  | 4 | MGL   | 188    |
| 4                                           | 20 | 1     | Me4NC1  | lgK:2.2(2SF)  | 4 | MHY   | 188    |
| 5                                           | 20 | 1     | NaBr    | lgK:1.99(3SF) | 6 | MMX   | 1562   |
| 6                                           | 25 | 0.1   | Unknown | lgK:2.2(2SF)  | 4 | ENS   | 233    |

| Reaction No. 11136                          |    |       |        |                |   |       |        |
|---------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| H+1 + 13DiMeTriMDTA-4 = H+1_13DiMeTriMDTA-4 |    |       |        |                |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | KNO3   | lgK:10.84(4SF) | 6 | ENS   | 233    |

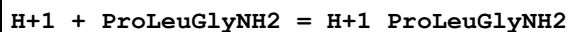
| Reaction No. 11137                                 |    |       |        |               |   |       |        |
|----------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| H+1_13DiMeTriMDTA-4 + H+1 = H+1(2)_13DiMeTriMDTA-4 |    |       |        |               |   |       |        |
| No                                                 | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                  | 20 | 0.1   | KNO3   | lgK:8.54(3SF) | 6 | ENS   | 233    |

| Reaction No. 11138                                    |    |       |        |               |   |       |        |
|-------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| H+1(2)_13DiMeTriMDTA-4 + H+1 = H+1(3)_13DiMeTriMDTA-4 |    |       |        |               |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                     | 20 | 0.1   | KNO3   | lgK:2.42(3SF) | 6 | ENS   | 233    |

| Reaction No. 11139                                    |  |  |  |  |  |  |  |
|-------------------------------------------------------|--|--|--|--|--|--|--|
| H+1(3)_13DiMeTriMDTA-4 + H+1 = H+1(4)_13DiMeTriMDTA-4 |  |  |  |  |  |  |  |

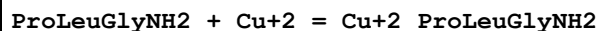
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:2.09(3SF) | 6 | ENS   | 233    |

## Reaction No. 12064



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:8.8(0.03SD) | 4 | MGL   | 1225   |

## Reaction No. 12065



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:5.69(0.02SD) | 6 | MGL   | 1225   |

## Reaction No. 12066



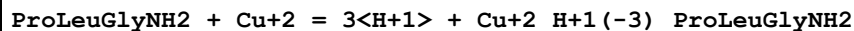
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:0.02(0.01SD) | 6 | MGL   | 1225   |

## Reaction No. 12067



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:-7.23(0.02SD) | 4 | MGL   | 1225   |

## Reaction No. 12068



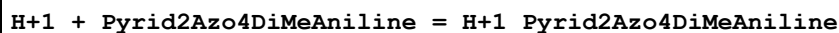
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:-16.30(0.02SD) | 6 | MGL   | 1225   |

## Reaction No. 12069



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:-18.1(0.1SD) | 4 | MGL   | 1225   |

## Reaction No. 14607



| No | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|--------|--------------|---|-------|--------|
| 1  | 25 | 0.15  | NaNO3  | lgK:4.5(2SF) | 3 | MSP   | 5703   |

| Reaction No. 14608                                             |    |       |        |               |   |       |        |
|----------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| H+1_Pyrid2Azo4DiMeAniline + H+1 = H+1(2)_Pyrid2Azo4DiMeAniline |    |       |        |               |   |       |        |
| No                                                             | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                              | 25 | 0.15  | NaNO3  | lgK:2.0 (2SF) | 3 | MSP   | 5703   |

| Reaction No. 14609                                        |    |       |        |                 |   |       |        |
|-----------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Pyrid2Azo4DiMeAniline + Co+2 = Co+2_Pyrid2Azo4DiMeAniline |    |       |        |                 |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                         | 16 | 0.1   | KNO3   | lgK:3.8 (2SF)   | 4 | MKN   | 999    |
| 2                                                         | 25 | 0.15  | NaNO3  | lgK:3.33 (3SF)  | 5 | MSP   | 5703   |
| 3                                                         | 25 | 0.3   | NaNO3  | lgK:3.32 (3SF)  | 5 | MKN   | 999    |
| 4                                                         | 25 | 0.3   | NaNO3  | lgK:3.4 (0.1SD) | 5 | MSP   | 999    |

| Reaction No. 14610                                        |    |       |        |               |   |       |        |
|-----------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Pyrid2Azo4DiMeAniline + Ca+2 = Ca+2_Pyrid2Azo4DiMeAniline |    |       |        |               |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.15  | NaNO3  | lgK:0.0 (1SF) | 4 | MSP   | 5703   |

| Reaction No. 14611                                        |    |       |        |                |   |       |        |
|-----------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Pyrid2Azo4DiMeAniline + Cu+2 = Cu+2_Pyrid2Azo4DiMeAniline |    |       |        |                |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                         | 16 | 0.1   | KNO3   | lgK:5.0 (2SF)  | 4 | MKN   | 999    |
| 2                                                         | 25 | 0.15  | NaNO3  | lgK:5.21 (3SF) | 6 | MSP   | 5703   |

| Reaction No. 14612                                        |    |       |        |                |   |       |        |
|-----------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Pyrid2Azo4DiMeAniline + Hg+2 = Hg+2_Pyrid2Azo4DiMeAniline |    |       |        |                |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.15  | NaNO3  | lgK:5.06 (3SF) | 6 | MSP   | 5703   |

| Reaction No. 14613                                        |    |       |        |               |   |       |        |
|-----------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Pyrid2Azo4DiMeAniline + Mg+2 = Mg+2_Pyrid2Azo4DiMeAniline |    |       |        |               |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.15  | NaNO3  | lgK:0.0 (1SF) | 4 | MSP   | 5703   |

| Reaction No. 14614                                        |   |       |        |             |   |       |        |
|-----------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Pyrid2Azo4DiMeAniline + Mn+2 = Mn+2_Pyrid2Azo4DiMeAniline |   |       |        |             |   |       |        |
| No                                                        | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|   |    |      |       |              |   |     |      |
|---|----|------|-------|--------------|---|-----|------|
| 1 | 25 | 0.15 | NaNO3 | lgK:0.7(1SF) | 4 | MSP | 5703 |
|---|----|------|-------|--------------|---|-----|------|

| Reaction No. 14615                                        |    |       |        |                 |   |       |        |
|-----------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Pyrid2Azo4DiMeAniline + Ni+2 = Ni+2_Pyrid2Azo4DiMeAniline |    |       |        |                 |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.15  | NaNO3  | lgK:4.24(3SF)   | 5 | MSP   | 5703   |
| 2                                                         | 25 | 0.3   | NaNO3  | lgK:4.1(0.05SD) | 5 | MSP   | 4859   |
| 3                                                         | 25 |       | DMSO   | lgK:2.699(4SF)  | 5 | MKN   | 4862   |
| 4                                                         | 25 |       | DMSO   | dH:-5.3(2SF)    | 4 | MCL   | 4862   |

| Reaction No. 14616                                        |    |       |        |                 |   |       |        |
|-----------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Pyrid2Azo4DiMeAniline + Zn+2 = Zn+2_Pyrid2Azo4DiMeAniline |    |       |        |                 |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                         | 16 | 0.1   | KNO3   | lgK:2.62(3SF)   | 5 | MKN   | 999    |
| 2                                                         | 25 | 0.15  | NaNO3  | lgK:2.36(3SF)   | 5 | MSP   | 5703   |
| 3                                                         | 25 | 0.3   | NaNO3  | lgK:2.4(0.07SD) | 5 | MKN   | 999    |
| 4                                                         | 25 | 0.3   | NaNO3  | lgK:2.3(0.06SD) | 4 | MSP   | 999    |

| Reaction No. 14617                                        |    |       |        |              |   |       |        |
|-----------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Pyrid2Azo4DiMeAniline + Cd+2 = Cd+2_Pyrid2Azo4DiMeAniline |    |       |        |              |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                         | 16 | 0.1   | KNO3   | lgK:2.7(2SF) | 4 | MKN   | 999    |

| Reaction No. 15418                                          |    |       |        |                  |   |       |        |
|-------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + PimelBisIPrDiHydroxam-2 = H+1_PimelBisIPrDiHydroxam-2 |    |       |        |                  |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                           | 25 | 0.1   | KNO3   | lgK:9.83(0.02SD) | 3 | MGL   | 1329   |

| Reaction No. 15419                                                 |    |       |        |                  |   |       |        |
|--------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1_PimelBisIPrDiHydroxam-2 + H+1 = H+1(2)_PimelBisIPrDiHydroxam-2 |    |       |        |                  |   |       |        |
| No                                                                 | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                  | 25 | 0.1   | KNO3   | lgK:9.16(0.02SD) | 4 | MGL   | 1329   |

| Reaction No. 15420                                            |    |       |        |                  |   |       |        |
|---------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| PimelBisIPrDiHydroxam-2 + Fe+3 = Fe+3_PimelBisIPrDiHydroxam-2 |    |       |        |                  |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.1   | KNO3   | lgK:22.8(0.06SD) | 4 | MGL   | 1329   |



| Reaction No. 15421                                                                                        |    |       |        |                  |   |       |        |
|-----------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $3\text{<PimelBisIPrDiHydroxam-2>} + 2\text{<Fe+3>} = \text{Fe+3(2)}_{\text{PimelBisIPrDiHydroxam-2(3)}}$ |    |       |        |                  |   |       |        |
| No                                                                                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                         | 25 | 0.1   | KNO3   | lgK:62.4 (0.2SD) | 3 | MGL   | 1329   |

| Reaction No. 16811                                                                        |    |       |        |                  |   |       |        |
|-------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $[\text{10}]\text{N3:Acet*3-3} + \text{H+1} = \text{H+1}_{[\text{10}]\text{N3:Acet*3-3}}$ |    |       |        |                  |   |       |        |
| No                                                                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                         | 25 | 0.1   | KCl    | lgK:14.8 (0.2SD) | 4 | MMR   | 1325   |

| Reaction No. 16812                                                                                        |    |       |        |                   |   |       |        |
|-----------------------------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $\text{H+1}_{[\text{10}]\text{N3:Acet*3-3}} + \text{H+1} = \text{H+1(2)}_{[\text{10}]\text{N3:Acet*3-3}}$ |    |       |        |                   |   |       |        |
| No                                                                                                        | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                         | 25 | 0.1   | KCl    | lgK:6.12 (0.02SD) | 4 | MGL   | 1325   |

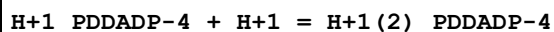
| Reaction No. 16813                                                                                           |    |       |        |                   |   |       |        |
|--------------------------------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $\text{H+1(2)}_{[\text{10}]\text{N3:Acet*3-3}} + \text{H+1} = \text{H+1(3)}_{[\text{10}]\text{N3:Acet*3-3}}$ |    |       |        |                   |   |       |        |
| No                                                                                                           | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                            | 25 | 0.1   | KCl    | lgK:3.69 (0.02SD) | 6 | MGL   | 1325   |

| Reaction No. 16814                                                                          |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Gd+3} + [\text{10}]\text{N3:Acet*3-3} = \text{Gd+3}_{[\text{10}]\text{N3:Acet*3-3}}$ |    |       |        |                 |   |       |        |
| No                                                                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                           | 25 | 0.1   | KCl    | lgK:15. (0.3SD) | 4 | MMR   | 1325   |

| Reaction No. 16837                                            |    |       |        |                 |   |       |        |
|---------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{H+1} + \text{PDDADP-4} = \text{H+1}_{\text{PDDADP-4}}$ |    |       |        |                 |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                             | 15 | 0.1   | NaNO3  | lgK:10.16 (4SF) | 6 | MGL   | 1308   |
| 2                                                             | 20 | 0.1   | KNO3   | lgK:10.21 (4SF) | 6 | MGL   | 1308   |
| 3                                                             | 20 | 0.1   | NaNO3  | lgK:10.12 (4SF) | 6 | MGL   | 1308   |
| 4                                                             | 25 | 0.1   | NaNO3  | lgK:10.06 (4SF) | 6 | MGL   | 1308   |
| 5                                                             | 25 | 0.5   | NaNO3  | lgK:9.95 (3SF)  | 6 | MGL   | 1308   |
| 6                                                             | 25 | 1     | NaNO3  | lgK:9.92 (3SF)  | 6 | MGL   | 1308   |
| 7                                                             | 30 | 0.1   | KNO3   | lgK:10.11 (4SF) | 6 | MGL   | 1308   |
| 8                                                             | 30 | 0.1   | NaNO3  | lgK:10.00 (4SF) | 6 | MGL   | 1308   |
| 9                                                             | 35 | 0.1   | NaNO3  | lgK:9.93 (3SF)  | 6 | MGL   | 1308   |

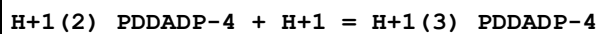
|    |    |     |       |                |   |     |      |
|----|----|-----|-------|----------------|---|-----|------|
| 10 | 25 | 0.1 | NaNO3 | dH: -4.5 (2SF) | 5 | MCL | 1308 |
|----|----|-----|-------|----------------|---|-----|------|

## Reaction No. 16838



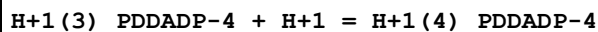
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 15 | 0.1   | NaNO3  | lgK: 8.34 (3SF) | 6 | MGL   | 1308   |
| 2  | 20 | 0.1   | KNO3   | lgK: 8.30 (3SF) | 6 | MGL   | 1308   |
| 3  | 20 | 0.1   | NaNO3  | lgK: 8.27 (3SF) | 6 | MGL   | 1308   |
| 4  | 25 | 0.1   | NaNO3  | lgK: 8.24 (3SF) | 6 | MGL   | 1308   |
| 5  | 25 | 0.5   | NaNO3  | lgK: 8.07 (3SF) | 6 | MGL   | 1308   |
| 6  | 25 | 1     | NaNO3  | lgK: 7.84 (3SF) | 6 | MGL   | 1308   |
| 7  | 30 | 0.1   | KNO3   | lgK: 8.11 (3SF) | 6 | MGL   | 1308   |
| 8  | 30 | 0.1   | NaNO3  | lgK: 8.17 (3SF) | 6 | MGL   | 1308   |
| 9  | 35 | 0.1   | NaNO3  | lgK: 8.13 (3SF) | 6 | MGL   | 1308   |
| 10 | 25 | 0.1   | NaNO3  | dH: -4.1 (2SF)  | 5 | MCL   | 1308   |

## Reaction No. 16839



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 15 | 0.1   | NaNO3  | lgK: 3.69 (3SF) | 6 | MGL   | 1308   |
| 2  | 20 | 0.1   | KNO3   | lgK: 3.76 (3SF) | 6 | MGL   | 1308   |
| 3  | 20 | 0.1   | NaNO3  | lgK: 3.72 (3SF) | 6 | MGL   | 1308   |
| 4  | 25 | 0.1   | NaNO3  | lgK: 3.73 (3SF) | 6 | MGL   | 1308   |
| 5  | 25 | 0.5   | NaNO3  | lgK: 3.55 (3SF) | 6 | MGL   | 1308   |
| 6  | 25 | 1     | NaNO3  | lgK: 3.41 (3SF) | 6 | MGL   | 1308   |
| 7  | 30 | 0.1   | KNO3   | lgK: 3.75 (3SF) | 6 | MGL   | 1308   |
| 8  | 30 | 0.1   | NaNO3  | lgK: 3.77 (3SF) | 6 | MGL   | 1308   |
| 9  | 35 | 0.1   | NaNO3  | lgK: 3.79 (3SF) | 6 | MGL   | 1308   |
| 10 | 25 | 0.1   | NaNO3  | dH: 1.7 (2SF)   | 5 | MCL   | 1308   |

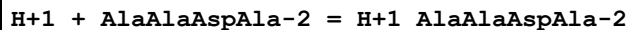
## Reaction No. 16840



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 15 | 0.1   | NaNO3  | lgK: 2.94 (3SF) | 6 | MGL   | 1308   |
| 2  | 20 | 0.1   | KNO3   | lgK: 2.79 (3SF) | 6 | MGL   | 1308   |
| 3  | 20 | 0.1   | NaNO3  | lgK: 2.97 (3SF) | 6 | MGL   | 1308   |
| 4  | 25 | 0.1   | NaNO3  | lgK: 2.98 (3SF) | 6 | MGL   | 1308   |

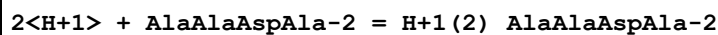
|    |    |     |       |                |   |     |      |
|----|----|-----|-------|----------------|---|-----|------|
| 5  | 25 | 0.5 | NaNO3 | lgK:2.79 (3SF) | 6 | MGL | 1308 |
| 6  | 25 | 1   | NaNO3 | lgK:2.87 (3SF) | 6 | MGL | 1308 |
| 7  | 30 | 0.1 | KNO3  | lgK:2.85 (3SF) | 6 | MGL | 1308 |
| 8  | 30 | 0.1 | NaNO3 | lgK:3.01 (3SF) | 6 | MGL | 1308 |
| 9  | 35 | 0.1 | NaNO3 | lgK:3.00 (3SF) | 6 | MGL | 1308 |
| 10 | 25 | 0.1 | NaNO3 | dH:0.1 (1SF)   | 5 | MCL | 1308 |

## Reaction No. 16841



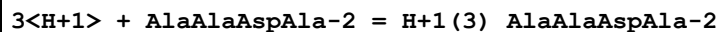
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:8.29 (0.01SD) | 6 | MGL   | 1294   |

## Reaction No. 16842



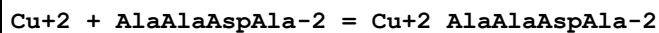
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:12.51 (0.01SD) | 6 | MGL   | 1294   |

## Reaction No. 16843



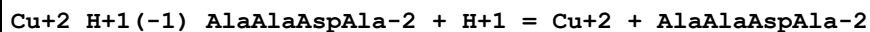
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:15.80 (0.01SD) | 6 | MGL   | 1294   |

## Reaction No. 16844



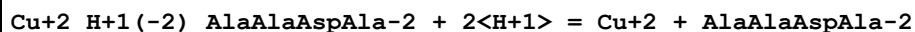
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:5.4 (0.08SD) | 4 | MGL   | 1294   |

## Reaction No. 16845



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:0.57 (0.02SD) | 6 | MGL   | 1294   |

## Reaction No. 16846



| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:-4.73 (0.01SD) | 6 | MGL   | 1294   |

## Reaction No. 20384

| imBenzHis-1 + H+1 = H+1_imBenzHis-1 |    |       |        |                    |   |       |        |
|-------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| No                                  | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | KNO3   | lgK:9.206(0.001SD) | 6 | MGL   | 1620   |

| Reaction No. 20385                         |    |       |        |                    |   |       |        |
|--------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| H+1_imBenzHis-1 + H+1 = H+1(2)_imBenzHis-1 |    |       |        |                    |   |       |        |
| No                                         | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                          | 25 | 0.1   | KNO3   | lgK:5.525(0.002SD) | 6 | MGL   | 1620   |

| Reaction No. 20386                            |    |       |        |                  |   |       |        |
|-----------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1(2)_imBenzHis-1 + H+1 = H+1(3)_imBenzHis-1 |    |       |        |                  |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | KNO3   | lgK:1.94(0.02SD) | 4 | MGL   | 1620   |

| Reaction No. 20387                    |    |       |        |                    |   |       |        |
|---------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| imBenzHis-1 + Ni+2 = Ni+2_imBenzHis-1 |    |       |        |                    |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KNO3   | lgK:8.823(0.001SD) | 6 | MGL   | 1620   |

| Reaction No. 20388                          |    |       |        |                    |   |       |        |
|---------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| 2<imBenzHis-1> + Ni+2 = Ni+2_imBenzHis-1(2) |    |       |        |                    |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KNO3   | lgK:16.11(0.004SD) | 6 | MGL   | 1620   |

| Reaction No. 20389                              |    |       |        |                  |   |       |        |
|-------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + imBenzHis-1 + Ni+2 = Ni+2_H+1_imBenzHis-1 |    |       |        |                  |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | KNO3   | lgK:11.5(0.07SD) | 4 | MGL   | 1620   |

| Reaction No. 20390                                    |    |       |        |                  |   |       |        |
|-------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + 2<imBenzHis-1> + Ni+2 = Ni+2_H+1_imBenzHis-1(2) |    |       |        |                  |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | KNO3   | lgK:20.2(0.09SD) | 3 | MGL   | 1620   |

| Reaction No. 20391                    |    |       |        |                     |   |       |        |
|---------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| imBenzHis-1 + Cu+2 = Cu+2_imBenzHis-1 |    |       |        |                     |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KNO3   | lgK:10.194(0.001SD) | 6 | MGL   | 1620   |

| Reaction No. 20392                                                                |    |       |        |                     |   |       |        |
|-----------------------------------------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| $2\text{<imBenzHis-1>} + \text{Cu}^{+2} = \text{Cu}^{+2}_{\text{imBenzHis-1}}(2)$ |    |       |        |                     |   |       |        |
| No                                                                                | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                 | 25 | 0.1   | KNO3   | lgK:18.519(0.001SD) | 6 | MGL   | 1620   |

| Reaction No. 20393                                                                                          |    |       |        |                    |   |       |        |
|-------------------------------------------------------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| $\text{H}^{+1} + \text{imBenzHis-1} + \text{Cu}^{+2} = \text{Cu}^{+2}_{\text{H}^{+1}_{\text{imBenzHis-1}}}$ |    |       |        |                    |   |       |        |
| No                                                                                                          | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                                           | 25 | 0.1   | KNO3   | lgK:13.74(0.003SD) | 6 | MGL   | 1620   |

| Reaction No. 20394                                                                                                |    |       |        |                     |   |       |        |
|-------------------------------------------------------------------------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| $\text{H}^{+1} + 2\text{<imBenzHis-1>} + \text{Cu}^{+2} = \text{Cu}^{+2}_{\text{H}^{+1}_{\text{imBenzHis-1}}}(2)$ |    |       |        |                     |   |       |        |
| No                                                                                                                | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                                                                 | 25 | 0.1   | KNO3   | lgK:23.623(0.001SD) | 5 | MGL   | 1620   |
| 2                                                                                                                 | 25 | 0.1   | KNO3   | lgK:23.64(0.01SD)   | 6 | MGL   | 1643   |

| Reaction No. 20395                                                                                                      |    |       |        |                   |   |       |        |
|-------------------------------------------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $2\text{<H}^{+1}> + 2\text{<imBenzHis-1>} + \text{Cu}^{+2} = \text{Cu}^{+2}_{\text{H}^{+1}}(2)_{\text{imBenzHis-1}}(2)$ |    |       |        |                   |   |       |        |
| No                                                                                                                      | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                       | 25 | 0.1   | KNO3   | lgK:27.05(0.01SD) | 4 | MGL   | 1620   |

| Reaction No. 20396                                                          |    |       |        |                    |   |       |        |
|-----------------------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| $\text{imBenzHis-1} + \text{Zn}^{+2} = \text{Zn}^{+2}_{\text{imBenzHis-1}}$ |    |       |        |                    |   |       |        |
| No                                                                          | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                           | 25 | 0.1   | KNO3   | lgK:6.580(0.001SD) | 6 | MGL   | 1620   |

| Reaction No. 20397                                                                                                |    |       |        |                   |   |       |        |
|-------------------------------------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $\text{H}^{+1} + 2\text{<imBenzHis-1>} + \text{Zn}^{+2} = \text{Zn}^{+2}_{\text{H}^{+1}_{\text{imBenzHis-1}}}(2)$ |    |       |        |                   |   |       |        |
| No                                                                                                                | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                                 | 25 | 0.1   | KNO3   | lgK:12.17(0.01SD) | 6 | MGL   | 1620   |

| Reaction No. 20412                                                                                          |    |       |        |                    |   |       |        |
|-------------------------------------------------------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| $\text{imBenzHis-1} + \text{Ni}^{+2} + \text{DHis-1} = \text{Ni}^{+2}_{\text{DHis-1}_{\text{imBenzHis-1}}}$ |    |       |        |                    |   |       |        |
| No                                                                                                          | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                                                           | 25 | 0.1   | KNO3   | lgK:16.52(0.003SD) | 6 | MGL   | 1620   |

| Reaction No. 20413 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| imBenzHis-1 + Ni+2 + His-1 = Ni+2_His-1_imBenzHis-1 |    |       |        |                    |   |       |        |
|-----------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| No                                                  | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | KNO3   | lgK:16.06(0.008SD) | 4 | MGL   | 1620   |

| Reaction No. 20414                                              |    |       |        |                     |   |       |        |
|-----------------------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| H+1 + imBenzHis-1 + Cu+2 + DHis-1 = Cu+2_H+1_DHis-1_imBenzHis-1 |    |       |        |                     |   |       |        |
| No                                                              | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                               | 25 | 0.1   | KNO3   | lgK:23.993(0.001SD) | 4 | MGL   | 1620   |

| Reaction No. 20415                                            |    |       |        |                     |   |       |        |
|---------------------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| H+1 + imBenzHis-1 + Cu+2 + His-1 = Cu+2_H+1_His-1_imBenzHis-1 |    |       |        |                     |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.1   | KNO3   | lgK:24.111(0.002SD) | 6 | MGL   | 1620   |

| Reaction No. 20416                                    |    |       |        |                     |   |       |        |
|-------------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| imBenzHis-1 + Cu+2 + DHis-1 = Cu+2_DHis-1_imBenzHis-1 |    |       |        |                     |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | KNO3   | lgK:18.587(0.001SD) | 6 | MGL   | 1620   |

| Reaction No. 20417                                  |    |       |        |                     |   |       |        |
|-----------------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| imBenzHis-1 + Cu+2 + His-1 = Cu+2_His-1_imBenzHis-1 |    |       |        |                     |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | KNO3   | lgK:18.586(0.002SD) | 6 | MGL   | 1620   |

| Reaction No. 20418                                    |    |       |        |                    |   |       |        |
|-------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| imBenzHis-1 + Zn+2 + DHis-1 = Zn+2_DHis-1_imBenzHis-1 |    |       |        |                    |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                     | 25 | 0.1   | KNO3   | lgK:12.58(0.009SD) | 6 | MGL   | 1620   |

| Reaction No. 20419                                  |    |       |        |                   |   |       |        |
|-----------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| imBenzHis-1 + Zn+2 + His-1 = Zn+2_His-1_imBenzHis-1 |    |       |        |                   |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | KNO3   | lgK:12.27(0.01SD) | 6 | MGL   | 1620   |

| Reaction No. 20582                            |    |       |        |                    |   |       |        |
|-----------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| H+1 + [14]2233N3O:Me*3 = H+1_[14]2233N3O:Me*3 |    |       |        |                    |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | KNO3   | lgK:10.10(0.003SD) | 6 | MGL   | 1743   |

| Reaction No. 20583                                                                            |    |       |        |                   |   |       |        |
|-----------------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $\text{H+1}_{[14]2233\text{N30:Me*3}} + \text{H+1} = \text{H+1(2)}_{[14]2233\text{N30:Me*3}}$ |    |       |        |                   |   |       |        |
| No                                                                                            | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                             | 25 | 0.1   | KNO3   | lgK:8.12(0.004SD) | 6 | MGL   | 1743   |

| Reaction No. 20584                                                                               |    |       |        |                   |   |       |        |
|--------------------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $\text{H+1(2)}_{[14]2233\text{N30:Me*3}} + \text{H+1} = \text{H+1(3)}_{[14]2233\text{N30:Me*3}}$ |    |       |        |                   |   |       |        |
| No                                                                                               | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                | 25 | 0.1   | KNO3   | lgK:5.37(0.005SD) | 6 | MGL   | 1743   |

| Reaction No. 20585                                                                     |    |       |        |                |   |       |        |
|----------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $[\text{14}]2233\text{N30:Me*3} + \text{Co+2} = \text{Co+2}_{[14]2233\text{N30:Me*3}}$ |    |       |        |                |   |       |        |
| No                                                                                     | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                      | 25 | 0.1   | KNO3   | lgK:5.8(0.1SD) | 4 | MGL   | 1743   |

| Reaction No. 20586                                                                                                                                      |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{Co+2} + [\text{14}]2233\text{N30:Me*3} + 2\langle\text{H2O}\rangle = \text{Co+2}_{[14]2233\text{N30:Me*3\_OH-1(2)}} + 2\langle\text{H+1}\rangle$ |    |       |        |                  |   |       |        |
| No                                                                                                                                                      | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                                                       | 25 | 0.1   | KNO3   | lgK:-9.7(0.07SD) | 3 | MGL   | 1743   |

| Reaction No. 20587                                                                     |    |       |        |                 |   |       |        |
|----------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $[\text{14}]2233\text{N30:Me*3} + \text{Ni+2} = \text{Ni+2}_{[14]2233\text{N30:Me*3}}$ |    |       |        |                 |   |       |        |
| No                                                                                     | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                      | 25 | 0.1   | KNO3   | lgK:6.4(0.07SD) | 4 | MGL   | 1743   |

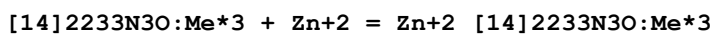
| Reaction No. 20588                                                                                                                                      |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{Ni+2} + [\text{14}]2233\text{N30:Me*3} + 2\langle\text{H2O}\rangle = \text{Ni+2}_{[14]2233\text{N30:Me*3\_OH-1(2)}} + 2\langle\text{H+1}\rangle$ |    |       |        |                  |   |       |        |
| No                                                                                                                                                      | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                                                       | 25 | 0.1   | KNO3   | lgK:-8.5(0.06SD) | 4 | MGL   | 1743   |

| Reaction No. 20589                                                                     |    |       |        |                  |   |       |        |
|----------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $[\text{14}]2233\text{N30:Me*3} + \text{Cu+2} = \text{Cu+2}_{[14]2233\text{N30:Me*3}}$ |    |       |        |                  |   |       |        |
| No                                                                                     | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                      | 25 | 0.1   | KNO3   | lgK:13.6(0.03SD) | 4 | MGL   | 1743   |

| Reaction No. 20590                                                                                                                                      |  |  |  |  |  |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $\text{Cu+2} + [\text{14}]2233\text{N30:Me*3} + 2\langle\text{H2O}\rangle = \text{Cu+2}_{[14]2233\text{N30:Me*3\_OH-1(2)}} + 2\langle\text{H+1}\rangle$ |  |  |  |  |  |  |  |

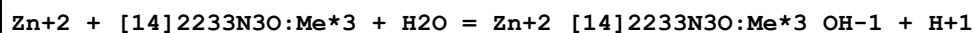
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:-1.6(0.06SD) | 3 | MGL   | 1743   |

## Reaction No. 20591



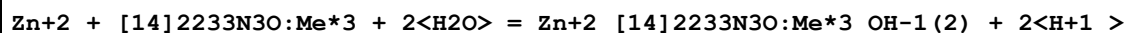
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:6.36(0.02SD) | 6 | MGL   | 1743   |

## Reaction No. 20592



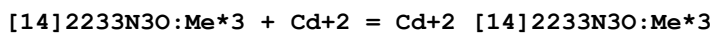
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:-1.58(0.02SD) | 6 | MGL   | 1743   |

## Reaction No. 20593



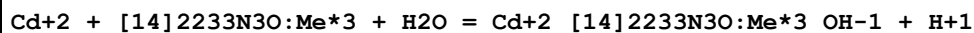
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:-12.5(0.09SD) | 4 | MGL   | 1743   |

## Reaction No. 20594



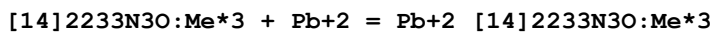
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:5.(0.3SD) | 3 | MGL   | 1743   |

## Reaction No. 20595



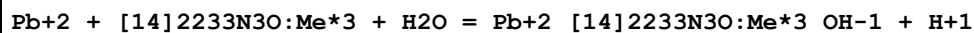
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:-5.0(0.06SD) | 4 | MGL   | 1743   |

## Reaction No. 20596



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:6.61(0.02SD) | 6 | MGL   | 1743   |

## Reaction No. 20597



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:-2.2(0.07SD) | 4 | MGL   | 1743   |



| Reaction No. 20628                                                                                        |    |       |        |                 |   |       |        |
|-----------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Co}+2_{[14]2233\text{N3O:Me*3}} + 2\text{<OH-1>} = \text{Co}+2_{[14]2233\text{N3O:Me*3\_OH-1(2)}}$ |    |       |        |                 |   |       |        |
| No                                                                                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                         | 25 | 0.1   | KNO3   | lgK:12.17 (4SF) | 6 | MGL   | 1743   |

| Reaction No. 20629                                                                                        |    |       |        |                 |   |       |        |
|-----------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Ni}+2_{[14]2233\text{N3O:Me*3}} + 2\text{<OH-1>} = \text{Ni}+2_{[14]2233\text{N3O:Me*3\_OH-1(2)}}$ |    |       |        |                 |   |       |        |
| No                                                                                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                         | 25 | 0.1   | KNO3   | lgK:12.71 (4SF) | 6 | MGL   | 1743   |

| Reaction No. 20630                                                                                        |    |       |        |                 |   |       |        |
|-----------------------------------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{Cu}+2_{[14]2233\text{N3O:Me*3}} + 2\text{<OH-1>} = \text{Cu}+2_{[14]2233\text{N3O:Me*3\_OH-1(2)}}$ |    |       |        |                 |   |       |        |
| No                                                                                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                         | 25 | 0.1   | KNO3   | lgK:12.42 (4SF) | 6 | MGL   | 1743   |

| Reaction No. 20631                                                                                  |    |       |        |                |   |       |        |
|-----------------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Zn}+2_{[14]2233\text{N3O:Me*3}} + \text{OH-1} = \text{Zn}+2_{[14]2233\text{N3O:Me*3\_OH-1}}$ |    |       |        |                |   |       |        |
| No                                                                                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                   | 25 | 0.1   | KNO3   | lgK:5.88 (3SF) | 6 | MGL   | 1743   |

| Reaction No. 20632                                                                                           |    |       |        |                |   |       |        |
|--------------------------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Zn}+2_{[14]2233\text{N3O:Me*3\_OH-1}} + \text{OH-1} = \text{Zn}+2_{[14]2233\text{N3O:Me*3\_OH-1(2)}}$ |    |       |        |                |   |       |        |
| No                                                                                                           | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                            | 25 | 0.1   | KNO3   | lgK:2.91 (3SF) | 6 | MGL   | 1743   |

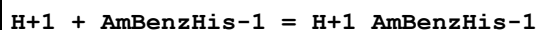
| Reaction No. 20633                                                                                        |    |       |        |                |   |       |        |
|-----------------------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Zn}+2_{[14]2233\text{N3O:Me*3}} + 2\text{<OH-1>} = \text{Zn}+2_{[14]2233\text{N3O:Me*3\_OH-1(2)}}$ |    |       |        |                |   |       |        |
| No                                                                                                        | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                         | 25 | 0.1   | KNO3   | lgK:8.79 (3SF) | 6 | MGL   | 1743   |

| Reaction No. 20634                                                                                  |    |       |        |                |   |       |        |
|-----------------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Cd}+2_{[14]2233\text{N3O:Me*3}} + \text{OH-1} = \text{Cd}+2_{[14]2233\text{N3O:Me*3\_OH-1}}$ |    |       |        |                |   |       |        |
| No                                                                                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                   | 25 | 0.1   | KNO3   | lgK:4.17 (3SF) | 6 | MGL   | 1743   |

| Reaction No. 20635                                                                                  |   |       |        |             |   |       |        |
|-----------------------------------------------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| $\text{Pb}+2_{[14]2233\text{N3O:Me*3}} + \text{OH-1} = \text{Pb}+2_{[14]2233\text{N3O:Me*3\_OH-1}}$ |   |       |        |             |   |       |        |
| No                                                                                                  | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

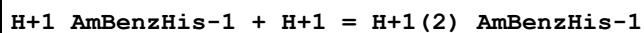
|   |    |     |      |                |   |     |      |
|---|----|-----|------|----------------|---|-----|------|
| 1 | 25 | 0.1 | KNO3 | lgK:4.99 (3SF) | 6 | MGL | 1743 |
|---|----|-----|------|----------------|---|-----|------|

## Reaction No. 21126



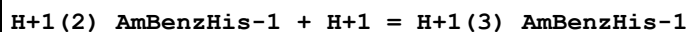
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:8.32 (3SF) | 6 | MGL   | 2752   |

## Reaction No. 21127



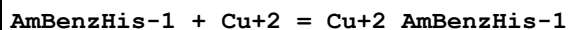
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:5.94 (3SF) | 6 | MGL   | 2752   |

## Reaction No. 21128



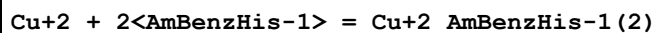
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:1.30 (3SF) | 4 | MGL   | 2752   |

## Reaction No. 21129



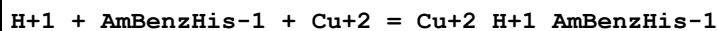
| No | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|--------|---------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:8.866 (0.002SD) | 6 | MGL   | 2752   |

## Reaction No. 21130



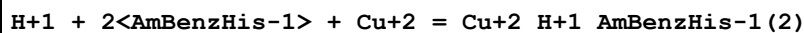
| No | t  | I-Str | Medium  | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|---------|----------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3    | lgK:15.874 (0.002SD) | 6 | MGL   | 2752   |
| 2  | 25 | 0.1   | Unknown | lgK:15.79 (4SF)      | 4 | CRV   | 552    |

## Reaction No. 21131



| No | t  | I-Str | Medium | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|--------|----------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:12.786 (0.002SD) | 6 | MGL   | 2752   |

## Reaction No. 21132



| No | t  | I-Str | Medium | Value (Dev)          | W | Tech. | Ref. # |
|----|----|-------|--------|----------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:21.203 (0.002SD) | 6 | MGL   | 2752   |

| Reaction No. 21133                    |    |       |        |                     |   |       |        |
|---------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| AmBenzHis-1 + Co+2 = Co+2_AmBenzHis-1 |    |       |        |                     |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KCl    | lgK:6.242 (0.001SD) | 6 | MGL   | 2652   |

| Reaction No. 21134                           |    |       |         |                      |   |       |        |
|----------------------------------------------|----|-------|---------|----------------------|---|-------|--------|
| 2<AmBenzHis-1> + Co+2 = Co+2_AmBenzHis-1 (2) |    |       |         |                      |   |       |        |
| No                                           | t  | I-Str | Medium  | Value (Dev)          | W | Tech. | Ref. # |
| 1                                            | 25 | 0.1   | KCl     | lgK:11.885 (0.001SD) | 6 | MGL   | 2652   |
| 2                                            | 25 | 0.1   | Unknown | lgK:11.99 (4SF)      | 4 | CRV   | 552    |

| Reaction No. 21135                    |    |       |        |                     |   |       |        |
|---------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| AmBenzHis-1 + Ni+2 = Ni+2_AmBenzHis-1 |    |       |        |                     |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KCl    | lgK:7.870 (0.001SD) | 6 | MGL   | 2652   |

| Reaction No. 21136                           |    |       |         |                      |   |       |        |
|----------------------------------------------|----|-------|---------|----------------------|---|-------|--------|
| 2<AmBenzHis-1> + Ni+2 = Ni+2_AmBenzHis-1 (2) |    |       |         |                      |   |       |        |
| No                                           | t  | I-Str | Medium  | Value (Dev)          | W | Tech. | Ref. # |
| 1                                            | 25 | 0.1   | KCl     | lgK:14.465 (0.001SD) | 6 | MGL   | 2652   |
| 2                                            | 25 | 0.1   | Unknown | lgK:14.81 (4SF)      | 4 | CRV   | 552    |

| Reaction No. 21137                    |    |       |        |                     |   |       |        |
|---------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| AmBenzHis-1 + Zn+2 = Zn+2_AmBenzHis-1 |    |       |        |                     |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KCl    | lgK:5.790 (0.001SD) | 6 | MGL   | 2652   |

| Reaction No. 21138                           |    |       |         |                      |   |       |        |
|----------------------------------------------|----|-------|---------|----------------------|---|-------|--------|
| 2<AmBenzHis-1> + Zn+2 = Zn+2_AmBenzHis-1 (2) |    |       |         |                      |   |       |        |
| No                                           | t  | I-Str | Medium  | Value (Dev)          | W | Tech. | Ref. # |
| 1                                            | 25 | 0.1   | KCl     | lgK:11.060 (0.001SD) | 6 | MGL   | 2652   |
| 2                                            | 25 | 0.1   | Unknown | lgK:11.2 (3SF)       | 4 | CRV   | 552    |

| Reaction No. 21139                           |    |       |        |                    |   |       |        |
|----------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| 2<imBenzHis-1> + Zn+2 = Zn+2_imBenzHis-1 (2) |    |       |        |                    |   |       |        |
| No                                           | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                            | 25 | 0.1   | KNO3   | lgK:12.17 (0.01SD) | 6 | EIU   | 904    |

| Reaction No. 21304              |    |       |        |                  |   |       |        |
|---------------------------------|----|-------|--------|------------------|---|-------|--------|
| PDDADP-4 + Ni+2 = Ni+2_PDDADP-4 |    |       |        |                  |   |       |        |
| No                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                               | 25 | 0.5   | NaNO3  | lgK:14.2(0.03SD) | 4 | MGL   | 1792   |

| Reaction No. 21305                        |    |       |        |                  |   |       |        |
|-------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + PDDADP-4 + Ni+2 = Ni+2_H+1_PDDADP-4 |    |       |        |                  |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                         | 25 | 0.5   | NaNO3  | lgK:17.6(0.03SD) | 4 | MGL   | 1792   |

| Reaction No. 21307                            |    |       |        |                 |   |       |        |
|-----------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| PDDADP-4 + Ni+2 = H+1 + Ni+2_H+1(-1)_PDDADP-4 |    |       |        |                 |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                             | 25 | 0.5   | NaNO3  | lgK:3.3(0.08SD) | 4 | MGL   | 1792   |

| Reaction No. 21308              |    |       |        |                  |   |       |        |
|---------------------------------|----|-------|--------|------------------|---|-------|--------|
| PDDADP-4 + Cu+2 = Cu+2_PDDADP-4 |    |       |        |                  |   |       |        |
| No                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                               | 25 | 0.5   | NaNO3  | lgK:15.0(0.06SD) | 4 | MGL   | 1792   |

| Reaction No. 21309                        |    |       |        |                  |   |       |        |
|-------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| H+1 + PDDADP-4 + Cu+2 = Cu+2_H+1_PDDADP-4 |    |       |        |                  |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                         | 25 | 0.5   | NaNO3  | lgK:19.0(0.06SD) | 4 | MGL   | 1792   |

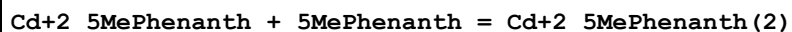
| Reaction No. 21310                              |    |       |        |                  |   |       |        |
|-------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 2<H+1> + PDDADP-4 + Cu+2 = Cu+2_H+1(2)_PDDADP-4 |    |       |        |                  |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                               | 25 | 0.5   | NaNO3  | lgK:22.3(0.04SD) | 4 | MGL   | 1792   |

| Reaction No. 21311                            |    |       |        |                 |   |       |        |
|-----------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| PDDADP-4 + Cu+2 = H+1 + Cu+2_H+1(-1)_PDDADP-4 |    |       |        |                 |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                             | 25 | 0.5   | NaNO3  | lgK:3.4(0.08SD) | 4 | MGL   | 1792   |

| Reaction No. 22214                  |   |       |        |             |   |       |        |
|-------------------------------------|---|-------|--------|-------------|---|-------|--------|
| PentMDTA-4 + Pd+2 = Pd+2_PentMDTA-4 |   |       |        |             |   |       |        |
| No                                  | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

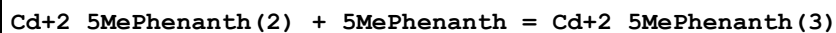
|   |    |   |      |                |   |     |      |
|---|----|---|------|----------------|---|-----|------|
| 1 | 20 | 1 | NaBr | lgK:26.4 (3SF) | 4 | MMX | 1562 |
|---|----|---|------|----------------|---|-----|------|

## Reaction No. 24099



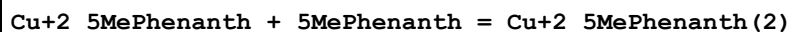
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:4.90 (3SF) | 6 | MDS   | 3077   |

## Reaction No. 24100



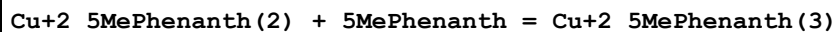
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:5.00 (3SF) | 6 | MDS   | 3077   |

## Reaction No. 24101



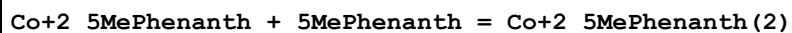
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:6.47 (3SF) | 6 | MDS   | 3077   |

## Reaction No. 24102



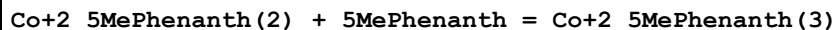
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:5.10 (3SF) | 6 | MDS   | 3077   |

## Reaction No. 24103



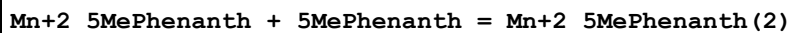
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:6.86 (3SF) | 6 | MDS   | 3077   |

## Reaction No. 24104



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:6.60 (3SF) | 6 | MDS   | 3077   |

## Reaction No. 24105



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:3.33 (3SF) | 6 | MDS   | 3077   |

## Reaction No. 24106

| Mn+2_5MePhenanth(2) + 5MePhenanth = Mn+2_5MePhenanth(3) |    |       |        |               |   |       |        |
|---------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| No                                                      | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | KCl    | lgK:3.60(3SF) | 6 | MDS   | 3077   |

| Reaction No. 24107                                   |    |       |        |               |   |       |        |
|------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Ni+2_5MePhenanth + 5MePhenanth = Ni+2_5MePhenanth(2) |    |       |        |               |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.1   | KCl    | lgK:8.65(3SF) | 6 | MDS   | 3077   |

| Reaction No. 24108                                      |    |       |        |               |   |       |        |
|---------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Ni+2_5MePhenanth(2) + 5MePhenanth = Ni+2_5MePhenanth(3) |    |       |        |               |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | KCl    | lgK:7.73(3SF) | 6 | MDS   | 3077   |

| Reaction No. 24109                                   |    |       |        |               |   |       |        |
|------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Zn+2_5MePhenanth + 5MePhenanth = Zn+2_5MePhenanth(2) |    |       |        |               |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.1   | KCl    | lgK:5.96(3SF) | 6 | MDS   | 3077   |

| Reaction No. 24110                                      |    |       |        |               |   |       |        |
|---------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Zn+2_5MePhenanth(2) + 5MePhenanth = Zn+2_5MePhenanth(3) |    |       |        |               |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | KCl    | lgK:5.67(3SF) | 6 | MDS   | 3077   |

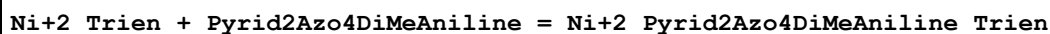
| Reaction No. 24111                                                    |    |       |        |                |   |       |        |
|-----------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Co+2_IDA-2 + Pyrid2Azo4DiMeAniline = Co+2_Pyrid2Azo4DiMeAniline_IDA-2 |    |       |        |                |   |       |        |
| No                                                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                     | 25 | 0.3   | NaNO3  | lgK:3.6(0.1SD) | 3 | MKN   | 999    |
| 2                                                                     | 25 | 0.3   | NaNO3  | lgK:3.(0.3SD)  | 0 | MSP   | 999    |

| Reaction No. 24112                                                    |    |       |        |                 |   |       |        |
|-----------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Co+2_NTA-3 + Pyrid2Azo4DiMeAniline = Co+2_Pyrid2Azo4DiMeAniline_NTA-3 |    |       |        |                 |   |       |        |
| No                                                                    | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                     | 25 | 0.3   | NaNO3  | lgK:2.9(0.1SD)  | 4 | MKN   | 999    |
| 2                                                                     | 25 | 0.3   | NaNO3  | lgK:2.7(0.07SD) | 4 | MSP   | 999    |

| Reaction No. 24113                                                  |  |  |  |  |  |  |  |
|---------------------------------------------------------------------|--|--|--|--|--|--|--|
| Ni+2_Dien + Pyrid2Azo4DiMeAniline = Ni+2_Dien_Pyrid2Azo4DiMeAniline |  |  |  |  |  |  |  |

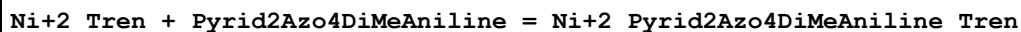
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.3   | NaNO3  | lgK:3.11(0.01SD) | 4 | MKN   | 692    |

## Reaction No. 24114



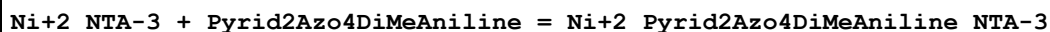
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.3   | NaNO3  | lgK:2.4(0.04SD) | 5 | MKN   | 999    |
| 2  | 25 | 0.3   | NaNO3  | lgK:2.5(0.1SD)  | 5 | MSP   | 999    |

## Reaction No. 24115



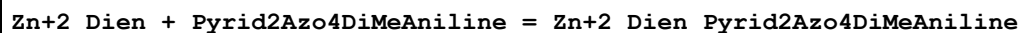
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.3   | NaNO3  | lgK:2.3(0.03SD) | 4 | MKN   | 999    |
| 2  | 25 | 0.3   | NaNO3  | lgK:2.3(0.04SD) | 4 | MSP   | 999    |

## Reaction No. 24116



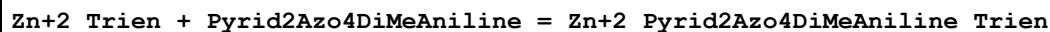
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.3   | NaNO3  | lgK:2.38(0.02SD) | 5 | MKN   | 999    |
| 2  | 25 | 0.3   | NaNO3  | lgK:2.42(0.02SD) | 5 | MSP   | 999    |

## Reaction No. 24117



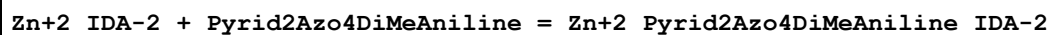
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.3   | NaNO3  | lgK:2.21(3SF)   | 5 | MKN   | 999    |
| 2  | 25 | 0.3   | NaNO3  | lgK:2.1(0.03SD) | 5 | MSP   | 999    |

## Reaction No. 24118



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.3   | NaNO3  | lgK:1.6(0.1SD)  | 4 | MKN   | 999    |
| 2  | 25 | 0.3   | NaNO3  | lgK:1.9(0.04SD) | 4 | MSP   | 999    |

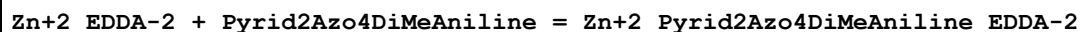
## Reaction No. 24119



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.3   | NaNO3  | lgK:2.5(0.1SD) | 4 | MKN   | 999    |

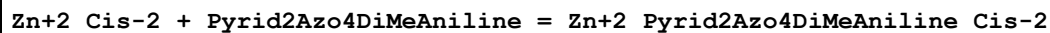
|   |    |     |       |                  |   |     |     |
|---|----|-----|-------|------------------|---|-----|-----|
| 2 | 25 | 0.3 | NaNO3 | lgK:2.4 (0.04SD) | 4 | MSP | 999 |
|---|----|-----|-------|------------------|---|-----|-----|

## Reaction No. 24120



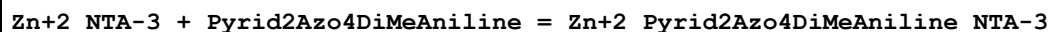
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.3   | NaNO3  | lgK:1.5 (0.07SD) | 4 | MKN   | 999    |
| 2  | 25 | 0.3   | NaNO3  | lgK:1.6 (0.05SD) | 4 | MSP   | 999    |

## Reaction No. 24121



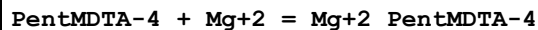
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.3   | NaNO3  | lgK:2. (0.3SD)   | 0 | MSP   | 999    |
| 2  | 25 | 0.3   | NaNO3  | lgK:2.8 (0.03SD) | 4 | MSP   | 999    |

## Reaction No. 24122



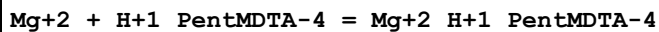
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.3   | NaNO3  | lgK:1.2 (0.05SD) | 3 | MSP   | 999    |

## Reaction No. 24330



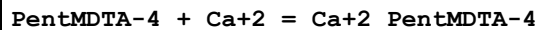
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:5.2 (2SF) | 4 | MGL   | 1956   |

## Reaction No. 24331



| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl    | lgK:3.6 (2SF) | 4 | MGL   | 1957   |
| 2  | 20 | 0.1   | KNO3   | lgK:3.6 (2SF) | 4 | MGL   | 1956   |

## Reaction No. 24332



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl           | lgK:4.60 (3SF) | 2 | MGL   | 1957   |
| 2  | 20 | 0.1   | KNO3          | lgK:5.2 (2SF)  | 2 | MGL   | 1956   |
| 3  | 25 | 0     | Inf. Dilution | lgK:5.2 (2SF)  | 1 | ESC   | 6422   |

## Reaction No. 24333



| Ca+2 + H+1_PentMDTA-4 = Ca+2_H+1_PentMDTA-4 |    |       |        |               |   |       |        |
|---------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| No                                          | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | KCl    | lgK:3.5 (2SF) | 4 | MGL   | 1957   |
| 2                                           | 20 | 0.1   | KNO3   | lgK:3.6 (2SF) | 4 | MGL   | 1956   |

| Reaction No. 24334                  |    |       |        |               |   |       |        |
|-------------------------------------|----|-------|--------|---------------|---|-------|--------|
| PentMDTA-4 + Mn+2 = Mn+2_PentMDTA-4 |    |       |        |               |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 20 | 0.1   | KNO3   | lgK:8.7 (2SF) | 4 | MGL   | 1956   |
| 2                                   | 20 | 0.1   | KNO3   | dH:0.9 (1SF)  | 5 | MCL   | 1956   |

| Reaction No. 24335                          |    |       |        |               |   |       |        |
|---------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Mn+2 + H+1_PentMDTA-4 = Mn+2_H+1_PentMDTA-4 |    |       |        |               |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | KNO3   | lgK:5.6 (2SF) | 4 | MGL   | 1956   |

| Reaction No. 24336                  |    |       |        |                |   |       |        |
|-------------------------------------|----|-------|--------|----------------|---|-------|--------|
| PentMDTA-4 + Fe+2 = Fe+2_PentMDTA-4 |    |       |        |                |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 20 | 0.1   | KNO3   | lgK:10.8 (3SF) | 4 | MGL   | 1956   |

| Reaction No. 24337                          |    |       |        |               |   |       |        |
|---------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Fe+2 + H+1_PentMDTA-4 = Fe+2_H+1_PentMDTA-4 |    |       |        |               |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | KNO3   | lgK:6.4 (2SF) | 4 | MGL   | 1956   |

| Reaction No. 24338                  |    |       |        |                 |   |       |        |
|-------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| PentMDTA-4 + Co+2 = Co+2_PentMDTA-4 |    |       |        |                 |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 20 | 0.1   | KNO3   | lgK:13.38 (4SF) | 6 | MGL   | 1956   |
| 2                                   | 20 | 0.1   | KNO3   | dH:3.1 (2SF)    | 5 | MCL   | 1956   |

| Reaction No. 24339                          |    |       |        |                |   |       |        |
|---------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Co+2 + H+1_PentMDTA-4 = Co+2_H+1_PentMDTA-4 |    |       |        |                |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | KNO3   | lgK:7.94 (3SF) | 6 | MGL   | 1956   |

| Reaction No. 24340 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| PentMDTA-4 + Ni+2 = Ni+2_PentMDTA-4 |    |       |        |               |   |       |        |
|-------------------------------------|----|-------|--------|---------------|---|-------|--------|
| No                                  | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 20 | 0.1   | KNO3   | lgK:13.9(3SF) | 4 | MGL   | 1956   |
| 2                                   | 20 | 0.1   | KNO3   | dH:-6.7(2SF)  | 5 | MCL   | 1956   |

| Reaction No. 24341                          |    |       |        |              |   |       |        |
|---------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Ni+2 + H+1_PentMDTA-4 = Ni+2_H+1_PentMDTA-4 |    |       |        |              |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | KNO3   | lgK:9.0(2SF) | 4 | MGL   | 1956   |

| Reaction No. 24342                  |    |       |        |                |   |       |        |
|-------------------------------------|----|-------|--------|----------------|---|-------|--------|
| PentMDTA-4 + Cu+2 = Cu+2_PentMDTA-4 |    |       |        |                |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 20 | 0.1   | KNO3   | lgK:16.24(4SF) | 6 | MGL   | 1956   |
| 2                                   | 20 | 0.1   | KNO3   | dH:-10.9(3SF)  | 5 | MCL   | 1956   |

| Reaction No. 24343                          |    |       |        |                |   |       |        |
|---------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Cu+2 + H+1_PentMDTA-4 = Cu+2_H+1_PentMDTA-4 |    |       |        |                |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | KNO3   | lgK:11.35(4SF) | 6 | MGL   | 1956   |

| Reaction No. 24344                  |    |       |        |                |   |       |        |
|-------------------------------------|----|-------|--------|----------------|---|-------|--------|
| PentMDTA-4 + Zn+2 = Zn+2_PentMDTA-4 |    |       |        |                |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 20 | 0.1   | KNO3   | lgK:12.67(4SF) | 6 | MGL   | 1956   |
| 2                                   | 20 | 0.1   | KNO3   | dH:-2.7(2SF)   | 5 | MCL   | 1956   |

| Reaction No. 24345                          |    |       |        |              |   |       |        |
|---------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Zn+2 + H+1_PentMDTA-4 = Zn+2_H+1_PentMDTA-4 |    |       |        |              |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | KNO3   | lgK:7.8(2SF) | 4 | MGL   | 1956   |

| Reaction No. 24346                  |    |       |        |               |   |       |        |
|-------------------------------------|----|-------|--------|---------------|---|-------|--------|
| PentMDTA-4 + Cd+2 = Cd+2_PentMDTA-4 |    |       |        |               |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 20 | 0.1   | KNO3   | lgK:11.6(3SF) | 4 | MGL   | 1956   |
| 2                                   | 20 | 0.1   | KNO3   | dH:-4.46(3SF) | 5 | MCL   | 1956   |

| Reaction No. 24347                                                                            |    |       |        |               |   |       |        |
|-----------------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $\text{Cd}+2 + \text{H}+1\_ \text{PentMDTA}-4 = \text{Cd}+2\_ \text{H}+1\_ \text{PentMDTA}-4$ |    |       |        |               |   |       |        |
| No                                                                                            | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                             | 20 | 0.1   | KNO3   | lgK:6.9 (2SF) | 4 | MGL   | 1956   |

| Reaction No. 24348                                                                            |    |       |        |                |   |       |        |
|-----------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Pb}+2 + \text{H}+1\_ \text{PentMDTA}-4 = \text{Pb}+2\_ \text{H}+1\_ \text{PentMDTA}-4$ |    |       |        |                |   |       |        |
| No                                                                                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                             | 20 | 0.1   | KNO3   | lgK:7.83 (3SF) | 6 | MGL   | 1956   |

| Reaction No. 24349                                                                    |    |       |        |               |   |       |        |
|---------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $\text{Ca}+2\_ \text{PentMDTA}-4 + \text{Ca}+2 = \text{Ca}+2 (2)\_ \text{PentMDTA}-4$ |    |       |        |               |   |       |        |
| No                                                                                    | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                     | 20 | 0.1   | KCl    | lgK:2.6 (2SF) | 4 | MGL   | 1957   |

| Reaction No. 24350                                                                            |    |       |               |               |   |       |        |
|-----------------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $\text{Sr}+2 + \text{H}+1\_ \text{PentMDTA}-4 = \text{Sr}+2\_ \text{H}+1\_ \text{PentMDTA}-4$ |    |       |               |               |   |       |        |
| No                                                                                            | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                             | 20 | 0.1   | KCl           | lgK:2.7 (2SF) | 4 | MGL   | 1957   |
| 2                                                                                             | 25 | 0     | Inf. Dilution | lgK:2.7 (2SF) | 1 | ESC   |        |

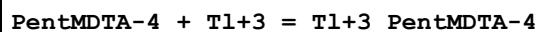
| Reaction No. 24351                                                                            |    |       |               |               |   |       |        |
|-----------------------------------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $\text{Ba}+2 + \text{H}+1\_ \text{PentMDTA}-4 = \text{Ba}+2\_ \text{H}+1\_ \text{PentMDTA}-4$ |    |       |               |               |   |       |        |
| No                                                                                            | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                             | 20 | 0.1   | KCl           | lgK:2.4 (2SF) | 4 | MGL   | 1957   |
| 2                                                                                             | 25 | 0     | Inf. Dilution | lgK:2.6 (2SF) | 1 | ESC   |        |

| Reaction No. 25779                                                  |    |       |               |                |   |       |        |
|---------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{PentMDTA}-4 + \text{Tl}+1 = \text{Tl}+1\_ \text{PentMDTA}-4$ |    |       |               |                |   |       |        |
| No                                                                  | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                   | 20 | 0.1   | KNO3          | lgK:3.73 (3SF) | 6 | MGL   | 2046   |
| 2                                                                   | 25 | 0     | Inf. Dilution | lgK:3.6 (2SF)  | 1 | EES   |        |

| Reaction No. 25780                                                                            |    |       |        |                |   |       |        |
|-----------------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Tl}+1 + \text{H}+1\_ \text{PentMDTA}-4 = \text{H}+1\_ \text{Tl}+1\_ \text{PentMDTA}-4$ |    |       |        |                |   |       |        |
| No                                                                                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                             | 20 | 0.1   | KNO3   | lgK:2.88 (3SF) | 6 | MGL   | 2046   |

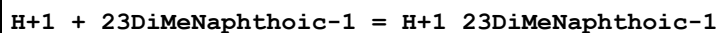
|   |    |   |               |               |   |     |  |
|---|----|---|---------------|---------------|---|-----|--|
| 2 | 25 | 0 | Inf. Dilution | lgK:2.7 (2SF) | 1 | EES |  |
|---|----|---|---------------|---------------|---|-----|--|

## Reaction No. 26368



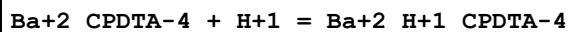
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:31.3 (3SF) | 6 | MGL   | 2046   |

## Reaction No. 27261



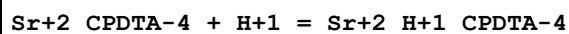
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 15 | 0     | Inf. Dilution | lgK:2.483 (4SF) | 5 | MPS   | 2399   |
| 2  | 20 | 0     | Inf. Dilution | lgK:2.629 (4SF) | 5 | MPS   | 2399   |
| 3  | 25 | 0     | Inf. Dilution | lgK:2.831 (4SF) | 5 | MPS   | 2399   |
| 4  | 30 | 0     | Inf. Dilution | lgK:3.021 (4SF) | 5 | MPS   | 2399   |
| 5  | 35 | 0     | Inf. Dilution | lgK:3.196 (4SF) | 5 | MPS   | 2399   |
| 6  | 40 | 0     | Inf. Dilution | lgK:3.351 (4SF) | 5 | MPS   | 2399   |

## Reaction No. 27289



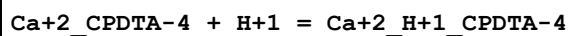
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown       | lgK:6.42 (3SF) | 6 | CRV   | 552    |
| 2  | 25 | 0     | Inf. Dilution | lgK:6.7 (2SF)  | 1 | ESC   |        |

## Reaction No. 27290



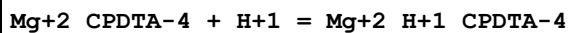
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown       | lgK:5.51 (3SF) | 6 | CRV   | 552    |
| 2  | 25 | 0     | Inf. Dilution | lgK:5.4 (2SF)  | 1 | ESC   |        |

## Reaction No. 27291



| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown | lgK:3.93 (3SF) | 6 | CRV   | 552    |

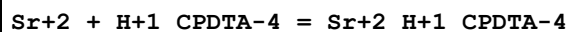
## Reaction No. 27292



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

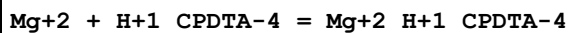
|   |    |     |         |                |   |     |     |
|---|----|-----|---------|----------------|---|-----|-----|
| 1 | 20 | 0.1 | Unknown | lgK:5.36 (3SF) | 6 | CRV | 552 |
|---|----|-----|---------|----------------|---|-----|-----|

## Reaction No. 27293



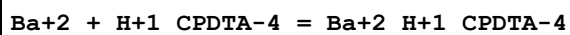
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl           | lgK:4.16 (3SF) | 4 | MGL   | 140    |
| 2  | 25 | 0     | Inf. Dilution | lgK:4.1 (2SF)  | 1 | ESC   |        |

## Reaction No. 27294



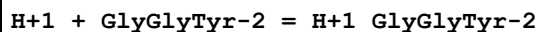
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl    | lgK:4.32 (3SF) | 4 | MGL   | 140    |

## Reaction No. 27295



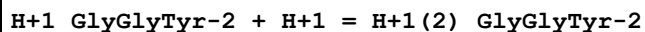
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl           | lgK:3.91 (3SF) | 5 | MGL   | 140    |
| 2  | 25 | 0     | Inf. Dilution | lgK:4.1 (2SF)  | 1 | ESC   |        |

## Reaction No. 28034



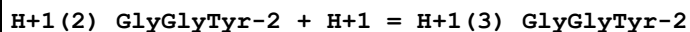
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:9.7 (0.09SD) | 5 | MGL   | 2450   |
| 2  | 25 | 0.1   | NaClO4 | dG:-13.3 (3SF)   | 5 | MGL   | 2450   |
| 3  | 25 | 0.1   | NaClO4 | dH:-4.6 (2SF)    | 5 | MCL   | 2450   |
| 4  | 25 | 0.1   | NaClO4 | dS:29.1 (3SF)    | 5 | EFE   | 2450   |

## Reaction No. 28035



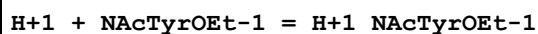
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:8.2 (0.07SD) | 5 | MGL   | 2450   |
| 2  | 25 | 0.1   | NaClO4 | dG:-11.1 (3SF)   | 5 | MGL   | 2450   |
| 3  | 25 | 0.1   | NaClO4 | dH:-7.9 (2SF)    | 5 | MCL   | 2450   |
| 4  | 25 | 0.1   | NaClO4 | dS:10.9 (3SF)    | 5 | EFE   | 2450   |

## Reaction No. 28036



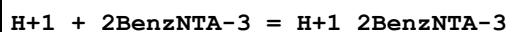
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:3.2 (0.03SD) | 5 | MGL   | 2450   |

## Reaction No. 28464



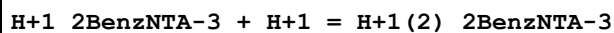
| No | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4  | lgK:10. (0.06SD) | 5 | MGL   | 2450   |
| 2  | 25 | 0.1   | NaClO4  | lgK:9.75 (3SF)   | 5 | MGL   | 2537   |
| 3  | 25 | 0.1   | NaClO4  | dG:-13.6 (3SF)   | 5 | MGL   | 2450   |
| 4  | 20 | 0.01  | Unknown | dH:-5.65 (3SF)   | 5 | MCL   | 774    |
| 5  | 25 | 0.1   | NaClO4  | dH:-4.2 (2SF)    | 5 | MCL   | 2450   |
| 6  | 25 | 0.1   | NaClO4  | dS:31.8 (3SF)    | 5 | EFE   | 2450   |

## Reaction No. 28899



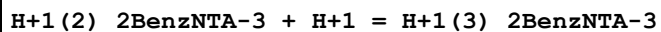
| No | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|----------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:9.59 (3SF) | 6 | CRV   | 2556   |

## Reaction No. 28900



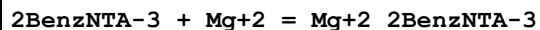
| No | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|----------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:2.51 (3SF) | 6 | CRV   | 2556   |

## Reaction No. 28901



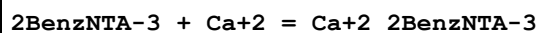
| No | t  | I-Str | Medium   | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|----------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:2.0 (2SF) | 5 | CRV   | 2556   |

## Reaction No. 28902



| No | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|----------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:5.44 (3SF) | 6 | CRV   | 2556   |

## Reaction No. 28903



| No | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|----------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:6.29 (3SF) | 6 | CRV   | 2556   |

| Reaction No. 28904                  |    |       |          |                |   |       |        |
|-------------------------------------|----|-------|----------|----------------|---|-------|--------|
| 2BenzNTA-3 + Sr+2 = Sr+2_2BenzNTA-3 |    |       |          |                |   |       |        |
| No                                  | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | K+1 salt | lgK:4.64 (3SF) | 6 | CRV   | 2556   |

| Reaction No. 28905                  |    |       |          |                |   |       |        |
|-------------------------------------|----|-------|----------|----------------|---|-------|--------|
| 2BenzNTA-3 + Ba+2 = Ba+2_2BenzNTA-3 |    |       |          |                |   |       |        |
| No                                  | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | K+1 salt | lgK:4.40 (3SF) | 6 | CRV   | 2556   |

| Reaction No. 28906                               |    |       |          |                |   |       |        |
|--------------------------------------------------|----|-------|----------|----------------|---|-------|--------|
| Y+3_2BenzNTA-3 + 2BenzNTA-3 = Y+3_2BenzNTA-3 (2) |    |       |          |                |   |       |        |
| No                                               | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                | 25 | 0.1   | K+1 salt | lgK:8.94 (3SF) | 4 | CRV   | 2556   |

| Reaction No. 28907                  |    |       |          |                 |   |       |        |
|-------------------------------------|----|-------|----------|-----------------|---|-------|--------|
| 2BenzNTA-3 + La+3 = La+3_2BenzNTA-3 |    |       |          |                 |   |       |        |
| No                                  | t  | I-Str | Medium   | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | K+1 salt | lgK:10.86 (4SF) | 6 | CRV   | 2556   |

| Reaction No. 28908                                 |    |       |          |                |   |       |        |
|----------------------------------------------------|----|-------|----------|----------------|---|-------|--------|
| La+3_2BenzNTA-3 + 2BenzNTA-3 = La+3_2BenzNTA-3 (2) |    |       |          |                |   |       |        |
| No                                                 | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                  | 25 | 0.1   | K+1 salt | lgK:6.90 (3SF) | 4 | CRV   | 2556   |

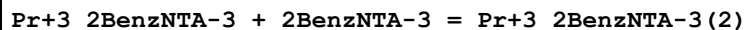
| Reaction No. 28909                  |    |       |          |                |   |       |        |
|-------------------------------------|----|-------|----------|----------------|---|-------|--------|
| 2BenzNTA-3 + Ce+3 = Ce+3_2BenzNTA-3 |    |       |          |                |   |       |        |
| No                                  | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | K+1 salt | lgK:11.2 (3SF) | 6 | CRV   | 2556   |

| Reaction No. 28910                                 |    |       |          |                |   |       |        |
|----------------------------------------------------|----|-------|----------|----------------|---|-------|--------|
| Ce+3_2BenzNTA-3 + 2BenzNTA-3 = Ce+3_2BenzNTA-3 (2) |    |       |          |                |   |       |        |
| No                                                 | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                  | 25 | 0.1   | K+1 salt | lgK:7.67 (3SF) | 4 | CRV   | 2556   |

| Reaction No. 28911                  |   |       |        |             |   |       |        |
|-------------------------------------|---|-------|--------|-------------|---|-------|--------|
| 2BenzNTA-3 + Pr+3 = Pr+3_2BenzNTA-3 |   |       |        |             |   |       |        |
| No                                  | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

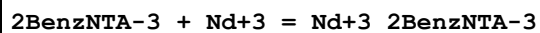
|   |    |     |          |                |   |     |      |
|---|----|-----|----------|----------------|---|-----|------|
| 1 | 25 | 0.1 | K+1 salt | lgK:11.3 (3SF) | 6 | CRV | 2556 |
|---|----|-----|----------|----------------|---|-----|------|

## Reaction No. 28912



| No | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|----------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:7.97 (3SF) | 4 | CRV   | 2556   |

## Reaction No. 28913



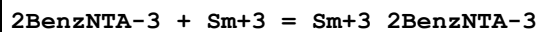
| No | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|----------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:11.5 (3SF) | 6 | CRV   | 2556   |

## Reaction No. 28914



| No | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|----------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:8.28 (3SF) | 4 | CRV   | 2556   |

## Reaction No. 28915



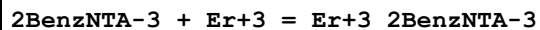
| No | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|----------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:11.6 (3SF) | 6 | CRV   | 2556   |

## Reaction No. 28916



| No | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|----------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:8.86 (3SF) | 4 | CRV   | 2556   |

## Reaction No. 28917



| No | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|----------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:11.8 (3SF) | 6 | CRV   | 2556   |

## Reaction No. 28918



| No | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|----------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:9.16 (3SF) | 4 | CRV   | 2556   |

## Reaction No. 28919



| 2BenzNTA-3 + Mn+2 = Mn+2_2BenzNTA-3 |    |       |          |                |   |       |        |
|-------------------------------------|----|-------|----------|----------------|---|-------|--------|
| No                                  | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | K+1 salt | lgK:7.20 (3SF) | 6 | CRV   | 2556   |

| Reaction No. 28920                  |    |       |          |                 |   |       |        |
|-------------------------------------|----|-------|----------|-----------------|---|-------|--------|
| 2BenzNTA-3 + Co+2 = Co+2_2BenzNTA-3 |    |       |          |                 |   |       |        |
| No                                  | t  | I-Str | Medium   | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | K+1 salt | lgK:10.27 (4SF) | 6 | CRV   | 2556   |

| Reaction No. 28921                  |    |       |          |                 |   |       |        |
|-------------------------------------|----|-------|----------|-----------------|---|-------|--------|
| 2BenzNTA-3 + Ni+2 = Ni+2_2BenzNTA-3 |    |       |          |                 |   |       |        |
| No                                  | t  | I-Str | Medium   | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | K+1 salt | lgK:10.82 (4SF) | 6 | CRV   | 2556   |

| Reaction No. 28922                  |    |       |          |                 |   |       |        |
|-------------------------------------|----|-------|----------|-----------------|---|-------|--------|
| 2BenzNTA-3 + Cu+2 = Cu+2_2BenzNTA-3 |    |       |          |                 |   |       |        |
| No                                  | t  | I-Str | Medium   | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | K+1 salt | lgK:11.15 (4SF) | 6 | CRV   | 2556   |

| Reaction No. 28923                  |    |       |          |                 |   |       |        |
|-------------------------------------|----|-------|----------|-----------------|---|-------|--------|
| 2BenzNTA-3 + Zn+2 = Zn+2_2BenzNTA-3 |    |       |          |                 |   |       |        |
| No                                  | t  | I-Str | Medium   | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | K+1 salt | lgK:10.24 (4SF) | 6 | CRV   | 2556   |

| Reaction No. 29117                          |    |       |         |               |   |       |        |
|---------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Mg+2_PentMDTA-4 + H+1 = Mg+2_H+1_PentMDTA-4 |    |       |         |               |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | Unknown | lgK:9.0 (2SF) | 5 | ENS   | 2373   |

| Reaction No. 29118                          |    |       |         |               |   |       |        |
|---------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Ca+2_PentMDTA-4 + H+1 = Ca+2_H+1_PentMDTA-4 |    |       |         |               |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | Unknown | lgK:9.2 (2SF) | 5 | ENS   | 2373   |

| Reaction No. 29119                          |    |       |         |               |   |       |        |
|---------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Mn+2_PentMDTA-4 + H+1 = Mn+2_H+1_PentMDTA-4 |    |       |         |               |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | Unknown | lgK:7.6 (2SF) | 5 | ENS   | 2373   |

| Reaction No. 29120                          |    |       |         |               |   |       |        |
|---------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Fe+2_PentMDTA-4 + H+1 = Fe+2_H+1_PentMDTA-4 |    |       |         |               |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | Unknown | lgK:5.26(3SF) | 5 | ENS   | 2373   |

| Reaction No. 29121                          |    |       |         |              |   |       |        |
|---------------------------------------------|----|-------|---------|--------------|---|-------|--------|
| Co+2_PentMDTA-4 + H+1 = Co+2_H+1_PentMDTA-4 |    |       |         |              |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | Unknown | lgK:5.8(2SF) | 5 | ENS   | 2373   |

| Reaction No. 29122                          |    |       |         |               |   |       |        |
|---------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Ni+2_PentMDTA-4 + H+1 = Ni+2_H+1_PentMDTA-4 |    |       |         |               |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | Unknown | lgK:5.81(3SF) | 5 | ENS   | 2373   |

| Reaction No. 29123                          |    |       |         |               |   |       |        |
|---------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Cu+2_PentMDTA-4 + H+1 = Cu+2_H+1_PentMDTA-4 |    |       |         |               |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | Unknown | lgK:5.85(3SF) | 5 | ENS   | 2373   |

| Reaction No. 29124                          |    |       |         |              |   |       |        |
|---------------------------------------------|----|-------|---------|--------------|---|-------|--------|
| Zn+2_PentMDTA-4 + H+1 = Zn+2_H+1_PentMDTA-4 |    |       |         |              |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | Unknown | lgK:6.0(2SF) | 5 | ENS   | 2373   |

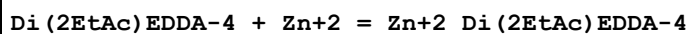
| Reaction No. 29125                          |    |       |         |               |   |       |        |
|---------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Cd+2_PentMDTA-4 + H+1 = Cd+2_H+1_PentMDTA-4 |    |       |         |               |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | Unknown | lgK:8.46(3SF) | 5 | ENS   | 2373   |

| Reaction No. 29130                            |    |       |         |               |   |       |        |
|-----------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Di(2EtAc)EDDA-4 + Mg+2 = Mg+2_Di(2EtAc)EDDA-4 |    |       |         |               |   |       |        |
| No                                            | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                             | 20 | 0.1   | Unknown | lgK:7.83(3SF) | 5 | ENS   | 2373   |

| Reaction No. 29131                            |  |  |  |  |  |  |  |
|-----------------------------------------------|--|--|--|--|--|--|--|
| Di(2EtAc)EDDA-4 + Ca+2 = Ca+2_Di(2EtAc)EDDA-4 |  |  |  |  |  |  |  |

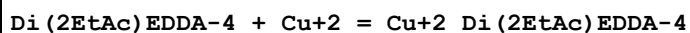
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown | lgK:9.02 (3SF) | 5 | ENS   | 2373   |

## Reaction No. 29132



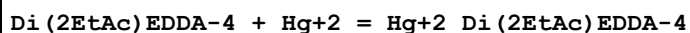
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown | lgK:16.04 (4SF) | 5 | ENS   | 2373   |

## Reaction No. 29133



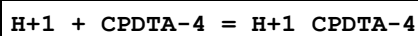
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown | lgK:18.76 (4SF) | 5 | ENS   | 2373   |

## Reaction No. 29143



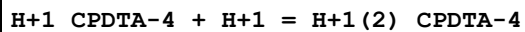
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:20.53 (4SF) | 5 | MPL   | 1993   |

## Reaction No. 29309



| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl     | lgK:10.26 (4SF) | 5 | MGL   | 140    |
| 2  | 20 | 0.1   | NaNO3   | lgK:10.09 (4SF) | 6 | MGL   | 906    |
| 3  | 20 | 0.1   | Unknown | lgK:10.80 (4SF) | 4 | MGL   | 140    |
| 4  | 20 | 0.1   | Unknown | lgK:10.31 (4SF) | 5 | MGL   | 931    |
| 5  | 20 | 1     | KCl     | lgK:10.20 (4SF) | 5 | ENS   | 774    |
| 6  | 20 | 1     | Unknown | lgK:10.13 (4SF) | 6 | CRV   | 552    |
| 7  | 25 | 0.1   | Unknown | lgK:10.01 (4SF) | 6 | CRV   | 552    |
| 8  | 20 | 1     | KCl     | dH:-6.44 (3SF)  | 5 | MCL   | 2018   |

## Reaction No. 29310



| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl     | lgK:7.44 (3SF) | 5 | MGL   | 140    |
| 2  | 20 | 0.1   | NaNO3   | lgK:7.48 (3SF) | 6 | MGL   | 906    |
| 3  | 20 | 0.1   | Unknown | lgK:7.56 (3SF) | 4 | MGL   | 140    |
| 4  | 20 | 0.1   | Unknown | lgK:7.42 (3SF) | 5 | MGL   | 931    |

| Reaction No. 29311                    |    |       |         |               |   |       |        |
|---------------------------------------|----|-------|---------|---------------|---|-------|--------|
| H+1(2)_CPDTA-4 + H+1 = H+1(3)_CPDTA-4 |    |       |         |               |   |       |        |
| No                                    | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 20 | 0.1   | KCl     | lgK:2.63(3SF) | 5 | MGL   | 140    |
| 2                                     | 20 | 0.1   | NaNO3   | lgK:2.44(3SF) | 6 | MGL   | 906    |
| 3                                     | 20 | 0.1   | Unknown | lgK:3.3(2SF)  | 4 | MGL   | 140    |
| 4                                     | 20 | 0.1   | Unknown | lgK:2.92(3SF) | 5 | MGL   | 931    |

| Reaction No. 29312                    |    |       |         |               |   |       |        |
|---------------------------------------|----|-------|---------|---------------|---|-------|--------|
| H+1(3)_CPDTA-4 + H+1 = H+1(4)_CPDTA-4 |    |       |         |               |   |       |        |
| No                                    | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 20 | 0.1   | KCl     | lgK:1.75(3SF) | 5 | MGL   | 140    |
| 2                                     | 20 | 0.1   | NaNO3   | lgK:1.87(3SF) | 4 | MGL   | 906    |
| 3                                     | 20 | 0.1   | Unknown | lgK:2.4(2SF)  | 4 | MGL   | 140    |
| 4                                     | 20 | 0.1   | Unknown | lgK:2.41(3SF) | 5 | MGL   | 931    |

| Reaction No. 29313            |    |       |               |                |   |       |        |
|-------------------------------|----|-------|---------------|----------------|---|-------|--------|
| CPDTA-4 + Ca+2 = Ca+2_CPDTA-4 |    |       |               |                |   |       |        |
| No                            | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                             | 20 | 0.1   | KCl           | lgK:12.2(3SF)  | 0 | MGL   | 140    |
| 2                             | 20 | 0.1   | KCl           | lgK:11.32(4SF) | 4 | MGL   | 140    |
| 3                             | 20 | 0.1   | Unknown       | lgK:11.08(4SF) | 4 | MGL   | 931    |
| 4                             | 20 | 0.1   | Unknown       | lgK:9.45(3SF)  | 0 | ENS   | 2366   |
| 5                             | 20 | 1     | Unknown       | lgK:11.30(4SF) | 6 | CRV   | 552    |
| 6                             | 25 | 0     | Inf. Dilution | lgK:12.6(3SF)  | 1 | EDH   |        |
| 7                             | 25 | 0.1   | Unknown       | lgK:11.20(4SF) | 6 | CRV   | 552    |
| 8                             | 20 | 1     | KCl           | dH:-4.87(3SF)  | 5 | MCL   | 2018   |

| Reaction No. 29319            |    |       |         |               |   |       |        |
|-------------------------------|----|-------|---------|---------------|---|-------|--------|
| CPDTA-4 + Mg+2 = Mg+2_CPDTA-4 |    |       |         |               |   |       |        |
| No                            | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                             | 20 | 0.1   | KCl     | lgK:9.07(3SF) | 4 | MGL   | 140    |
| 2                             | 20 | 0.1   | Unknown | lgK:9.05(3SF) | 4 | MGL   | 931    |
| 3                             | 20 | 1     | Unknown | lgK:9.29(3SF) | 6 | CRV   | 552    |
| 4                             | 25 | 0.1   | Unknown | lgK:9.30(3SF) | 6 | CRV   | 552    |
| 5                             | 20 | 1     | KCl     | dH:4.37(3SF)  | 5 | MCL   | 2018   |

| Reaction No. 29320            |    |       |           |                  |   |       |        |
|-------------------------------|----|-------|-----------|------------------|---|-------|--------|
| CPDTA-4 + Sr+2 = Sr+2_CPDTA-4 |    |       |           |                  |   |       |        |
| No                            | t  | I-Str | Medium    | Value (Dev)      | W | Tech. | Ref. # |
| 1                             | 20 | 0.1   | KCl       | lgK:9.06(3SF)    | 4 | MGL   | 140    |
| 2                             | 20 | 0.1   | NH4+ salt | lgK:10.1(0.03SD) | 4 | MIX   | 931    |
| 3                             | 20 | 0.1   | Unknown   | lgK:9.45(3SF)    | 4 | MGL   | 931    |
| 4                             | 20 | 0.165 | NH4+ salt | lgK:9.71(0.01SD) | 4 | MIX   | 931    |
| 5                             | 20 | 1     | Unknown   | lgK:9.47(3SF)    | 6 | CRV   | 552    |
| 6                             | 25 | 0.1   | Unknown   | lgK:9.42(3SF)    | 6 | CRV   | 552    |
| 7                             | 20 | 1     | KCl       | dH:-0.32(2SF)    | 5 | MCL   | 2018   |

| Reaction No. 29321            |    |       |        |                 |   |       |        |
|-------------------------------|----|-------|--------|-----------------|---|-------|--------|
| CPDTA-4 + Hg+2 = Hg+2_CPDTA-4 |    |       |        |                 |   |       |        |
| No                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                             | 20 | 0.1   | NaNO3  | lgK:23.2(0.1SD) | 4 | MPT   | 5655   |

| Reaction No. 29814            |    |       |         |               |   |       |        |
|-------------------------------|----|-------|---------|---------------|---|-------|--------|
| H+1 + HisPhe-1 = H+1_HisPhe-1 |    |       |         |               |   |       |        |
| No                            | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                             | 25 | 0.1   | Unknown | lgK:7.56(3SF) | 5 | CRV   | 2556   |

| Reaction No. 29815                   |    |       |         |               |   |       |        |
|--------------------------------------|----|-------|---------|---------------|---|-------|--------|
| H+1_HisPhe-1 + H+1 = H+1(2)_HisPhe-1 |    |       |         |               |   |       |        |
| No                                   | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                    | 25 | 0.1   | Unknown | lgK:6.00(3SF) | 5 | CRV   | 2556   |

| Reaction No. 29816                      |    |       |         |               |   |       |        |
|-----------------------------------------|----|-------|---------|---------------|---|-------|--------|
| H+1(2)_HisPhe-1 + H+1 = H+1(3)_HisPhe-1 |    |       |         |               |   |       |        |
| No                                      | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                       | 25 | 0.1   | Unknown | lgK:2.57(3SF) | 5 | CRV   | 2556   |

| Reaction No. 29820            |    |       |         |               |   |       |        |
|-------------------------------|----|-------|---------|---------------|---|-------|--------|
| H+1 + HisTyr-2 = H+1_HisTyr-2 |    |       |         |               |   |       |        |
| No                            | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                             | 25 | 0.1   | Unknown | lgK:9.83(3SF) | 5 | CRV   | 2556   |

| Reaction No. 29821 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| H+1_HisTyr-2 + H+1 = H+1(2)_HisTyr-2 |    |       |         |               |   |       |        |
|--------------------------------------|----|-------|---------|---------------|---|-------|--------|
| No                                   | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                    | 25 | 0.1   | Unknown | lgK:7.44(3SF) | 5 | CRV   | 2556   |

| Reaction No. 29822                      |    |       |         |               |   |       |        |
|-----------------------------------------|----|-------|---------|---------------|---|-------|--------|
| H+1(2)_HisTyr-2 + H+1 = H+1(3)_HisTyr-2 |    |       |         |               |   |       |        |
| No                                      | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                       | 25 | 0.1   | Unknown | lgK:5.91(3SF) | 5 | CRV   | 2556   |

| Reaction No. 29823                      |    |       |         |               |   |       |        |
|-----------------------------------------|----|-------|---------|---------------|---|-------|--------|
| H+1(3)_HisTyr-2 + H+1 = H+1(4)_HisTyr-2 |    |       |         |               |   |       |        |
| No                                      | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                       | 25 | 0.1   | Unknown | lgK:2.55(3SF) | 5 | CRV   | 2556   |

| Reaction No. 29949                          |    |       |         |              |   |       |        |
|---------------------------------------------|----|-------|---------|--------------|---|-------|--------|
| H+1 + 4AcAmAntipyrine = H+1_4AcAmAntipyrine |    |       |         |              |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                                           | 25 | 0.5   | Unknown | lgK:0.7(1SF) | 3 | CRV   | 2556   |

| Reaction No. 29950                |    |       |               |                |   |       |        |
|-----------------------------------|----|-------|---------------|----------------|---|-------|--------|
| H+1 + Pyramidone = H+1_Pyramidone |    |       |               |                |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 25 | 0     | Inf. Dilution | lgK:4.183(4SF) | 4 | ENS   | 774    |
| 2                                 | 25 | 0     | Inf. Dilution | lgK:4.94(3SF)  | 5 | CRV   | 2556   |
| 3                                 | 25 | 0.5   | Unknown       | lgK:5.20(3SF)  | 5 | CRV   | 2556   |
| 4                                 | 25 | 0     | Inf. Dilution | dH:-6.329(4SF) | 4 | MCL   | 774    |
| 5                                 | 25 | 0.5   | Unknown       | dH:-4.1(2SF)   | 4 | CRV   | 2556   |

| Reaction No. 29951                  |    |       |         |               |   |       |        |
|-------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Pyramidone + Mn+2 = Mn+2_Pyramidone |    |       |         |               |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 25 | 0.5   | Unknown | lgK:0.74(2SF) | 4 | CRV   | 2556   |

| Reaction No. 29952                  |    |       |         |               |   |       |        |
|-------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Pyramidone + Co+2 = Co+2_Pyramidone |    |       |         |               |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 25 | 0.5   | Unknown | lgK:0.78(2SF) | 4 | CRV   | 2556   |

| Reaction No. 29953                        |    |       |         |               |   |       |        |
|-------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| 2<Pyramidone> + Co+2 = Co+2_Pyramidone(2) |    |       |         |               |   |       |        |
| No                                        | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                         | 25 | 0.5   | Unknown | lgK:0.8 (1SF) | 3 | CRV   | 2556   |

| Reaction No. 29954                  |    |       |         |                |   |       |        |
|-------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Pyramidone + Ni+2 = Ni+2_Pyramidone |    |       |         |                |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 0.5   | Unknown | lgK:1.07 (3SF) | 4 | CRV   | 2556   |

| Reaction No. 29955                        |    |       |         |               |   |       |        |
|-------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| 2<Pyramidone> + Ni+2 = Ni+2_Pyramidone(2) |    |       |         |               |   |       |        |
| No                                        | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                         | 25 | 0.5   | Unknown | lgK:1.4 (2SF) | 4 | CRV   | 2556   |

| Reaction No. 29956                  |    |       |         |                |   |       |        |
|-------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Pyramidone + Cu+2 = Cu+2_Pyramidone |    |       |         |                |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 0.5   | Unknown | lgK:1.91 (3SF) | 4 | CRV   | 2556   |

| Reaction No. 29957                        |    |       |         |               |   |       |        |
|-------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| 2<Pyramidone> + Cu+2 = Cu+2_Pyramidone(2) |    |       |         |               |   |       |        |
| No                                        | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                         | 25 | 0.5   | Unknown | lgK:3.1 (2SF) | 4 | CRV   | 2556   |

| Reaction No. 29958                        |    |       |         |               |   |       |        |
|-------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| 3<Pyramidone> + Cu+2 = Cu+2_Pyramidone(3) |    |       |         |               |   |       |        |
| No                                        | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                         | 25 | 0.5   | Unknown | lgK:3.5 (2SF) | 4 | CRV   | 2556   |

| Reaction No. 29959                  |    |       |         |                |   |       |        |
|-------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Pyramidone + Zn+2 = Zn+2_Pyramidone |    |       |         |                |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 0.5   | Unknown | lgK:1.00 (3SF) | 4 | CRV   | 2556   |

| Reaction No. 29960                        |   |       |        |             |   |       |        |
|-------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| 2<Pyramidone> + Zn+2 = Zn+2_Pyramidone(2) |   |       |        |             |   |       |        |
| No                                        | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|   |    |     |         |               |   |     |      |
|---|----|-----|---------|---------------|---|-----|------|
| 1 | 25 | 0.5 | Unknown | lgK:1.3 (2SF) | 4 | CRV | 2556 |
|---|----|-----|---------|---------------|---|-----|------|

## Reaction No. 29961

Pyramidone + Cd+2 = Cd+2\_Pyramidone

| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.5   | Unknown | lgK:1.47 (3SF) | 4 | CRV   | 2556   |

## Reaction No. 29962

2&lt;Pyramidone&gt; + Cd+2 = Cd+2\_Pyramidone(2)

| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.5   | Unknown | lgK:2.24 (3SF) | 4 | CRV   | 2556   |

## Reaction No. 30780

CPDTA-4 + Pr+3 = Pr+3\_CPDTA-4

| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:17.44 (4SF) | 6 | CRV   | 552    |
| 2  | 20 | 0.1   | Unknown | dH:-2.5 (2SF)   | 6 | CRV   | 552    |

## Reaction No. 30781

CPDTA-4 + Ce+3 = Ce+3\_CPDTA-4

| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:17.26 (4SF) | 6 | CRV   | 552    |
| 2  | 20 | 0.1   | Unknown | dH:-1.6 (2SF)   | 6 | CRV   | 552    |

## Reaction No. 30782

CPDTA-4 + La+3 = La+3\_CPDTA-4

| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:16.99 (4SF) | 6 | CRV   | 552    |
| 2  | 20 | 0.1   | Unknown | dH:-1.70 (3SF)  | 6 | CRV   | 552    |

## Reaction No. 30784

CPDTA-4 + Tb+3 = Tb+3\_CPDTA-4

| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:18.64 (4SF) | 6 | CRV   | 552    |
| 2  | 20 | 0.1   | Unknown | dH:-0.3 (1SF)   | 6 | CRV   | 552    |

## Reaction No. 30785

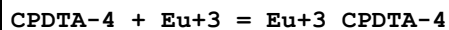
CPDTA-4 + Gd+3 = Gd+3\_CPDTA-4

|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|



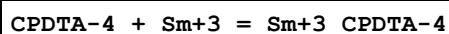
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:18.22 (4SF) | 6 | CRV   | 552    |
| 2  | 20 | 0.1   | Unknown | dH:-1.2 (2SF)   | 6 | CRV   | 552    |

## Reaction No. 30786



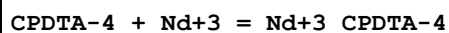
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:18.18 (4SF) | 6 | CRV   | 552    |
| 2  | 20 | 0.1   | Unknown | dH:-1.9 (2SF)   | 6 | CRV   | 552    |

## Reaction No. 30787



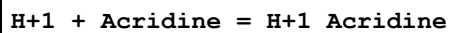
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:18.08 (4SF) | 6 | CRV   | 552    |
| 2  | 20 | 0.1   | Unknown | dH:-2.7 (2SF)   | 6 | CRV   | 552    |

## Reaction No. 30788



| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:17.69 (4SF) | 6 | CRV   | 552    |
| 2  | 20 | 0.1   | Unknown | dH:-2.7 (2SF)   | 6 | CRV   | 552    |

## Reaction No. 32418

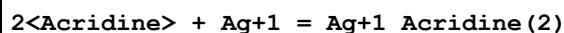


| No | t   | I-Str | Medium                 | Value (Dev)    | W | Tech. | Ref. # |
|----|-----|-------|------------------------|----------------|---|-------|--------|
| 1  | 25  | 0     | P=100<br>Inf. Dilution | lgK:5.53 (3SF) | 4 | MSP   | 22893  |
| 2  | 50  | 0     | P=100<br>Inf. Dilution | lgK:5.17 (3SF) | 4 | MSP   | 22893  |
| 3  | 100 | 0     | P=100<br>Inf. Dilution | lgK:4.59 (3SF) | 4 | MSP   | 22893  |
| 4  | 150 | 0     | P=100<br>Inf. Dilution | lgK:4.25 (3SF) | 4 | MSP   | 22893  |
| 5  | 25  | 0     | P=500<br>Inf. Dilution | lgK:5.54 (3SF) | 4 | MSP   | 22893  |
| 6  | 50  | 0     | P=500<br>Inf. Dilution | lgK:5.17 (3SF) | 4 | MSP   | 22893  |
| 7  | 100 | 0     | P=500<br>Inf. Dilution | lgK:4.59 (3SF) | 4 | MSP   | 22893  |

|    |     |       |                         |                   |   |     |       |
|----|-----|-------|-------------------------|-------------------|---|-----|-------|
| 8  | 150 | 0     | P=500<br>Inf. Dilution  | lgK:4.25 (3SF)    | 4 | MSP | 22893 |
| 9  | 25  | 0     | P=1000<br>Inf. Dilution | lgK:5.55 (3SF)    | 4 | MSP | 22893 |
| 10 | 50  | 0     | P=1000<br>Inf. Dilution | lgK:5.18 (3SF)    | 4 | MSP | 22893 |
| 11 | 100 | 0     | P=1000<br>Inf. Dilution | lgK:4.58 (3SF)    | 4 | MSP | 22893 |
| 12 | 150 | 0     | P=1000<br>Inf. Dilution | lgK:4.24 (3SF)    | 4 | MSP | 22893 |
| 13 | 25  | 0     | P=1500<br>Inf. Dilution | lgK:5.55 (3SF)    | 4 | MSP | 22893 |
| 14 | 50  | 0     | P=1500<br>Inf. Dilution | lgK:5.17 (3SF)    | 4 | MSP | 22893 |
| 15 | 100 | 0     | P=1500<br>Inf. Dilution | lgK:4.58 (3SF)    | 4 | MSP | 22893 |
| 16 | 150 | 0     | P=1500<br>Inf. Dilution | lgK:4.23 (3SF)    | 4 | MSP | 22893 |
| 17 | 25  | 0     | P=2000<br>Inf. Dilution | lgK:5.55 (3SF)    | 4 | MSP | 22893 |
| 18 | 50  | 0     | P=2000<br>Inf. Dilution | lgK:5.17 (3SF)    | 4 | MSP | 22893 |
| 19 | 100 | 0     | P=2000<br>Inf. Dilution | lgK:4.57 (3SF)    | 4 | MSP | 22893 |
| 20 | 150 | 0     | P=2000<br>Inf. Dilution | lgK:4.20 (3SF)    | 4 | MSP | 22893 |
| 21 | 15  |       |                         | lgK:5.65 (3SF)    | 5 | ENS | 774   |
| 22 | 20  | 0.01  | Unknown                 | lgK:5.60 (3SF)    | 5 | MSP | 22896 |
| 23 | 25  | 0     | Inf. Dilution           | lgK:5.60 (3SF)    | 3 | MFL | 2424  |
| 24 | 25  | 0     | Inf. Dilution           | lgK:5.42 (3SF)    | 4 | MSP | 16741 |
| 25 | 25  | 0     | Inf. Dilution           | lgK:5.52 (0.01SD) | 5 | MSP | 22893 |
| 26 | 25  | 0     | Inf. Dilution           | lgK:5.397 (4SF)   | 4 | EOD | 29878 |
| 27 | 25  | 0     | Inf. Dilution           | lgK:5.523 (4SF)   | 4 | EOD | 29878 |
| 28 | 25  | 0.005 | Unknown                 | lgK:5.54 (3SF)    | 5 | MSP | 22897 |
| 29 | 50  | 0     | Inf. Dilution           | lgK:5.15 (0.01SD) | 5 | MSP | 22893 |
| 30 | 50  | 0     | Inf. Dilution           | lgK:5.019 (4SF)   | 3 | EOD | 29878 |
| 31 | 50  | 0     | Inf. Dilution           | lgK:5.157 (4SF)   | 4 | EOD | 29878 |
| 32 | 75  | 0     | Inf. Dilution           | lgK:4.85 (0.01SD) | 5 | MSP | 22893 |
| 33 | 75  | 0     | Inf. Dilution           | lgK:4.697 (4SF)   | 3 | EOD | 29878 |
| 34 | 75  | 0     | Inf. Dilution           | lgK:4.843 (4SF)   | 4 | EOD | 29878 |
| 35 | 100 | 0     | Inf. Dilution           | lgK:4.56 (3SF)    | 5 | MSP | 22893 |

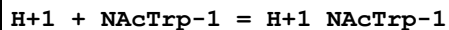
|    |     |   |                                          |                     |   |     |       |
|----|-----|---|------------------------------------------|---------------------|---|-----|-------|
| 36 | 100 | 0 | Inf. Dilution                            | lgK:4.423 (4SF)     | 3 | EOD | 29878 |
| 37 | 100 | 0 | Inf. Dilution                            | lgK:4.571 (4SF)     | 3 | EOD | 29878 |
| 38 | 125 | 0 | Inf. Dilution                            | lgK:4.187 (4SF)     | 3 | EOD | 29878 |
| 39 | 125 | 0 | Inf. Dilution                            | lgK:4.334 (4SF)     | 3 | EOD | 29878 |
| 40 | 150 | 0 | Inf. Dilution                            | lgK:4.17 (3SF)      | 5 | MSP | 22893 |
| 41 | 150 | 0 | Inf. Dilution                            | lgK:3.983 (4SF)     | 3 | EOD | 29878 |
| 42 | 150 | 0 | Inf. Dilution                            | lgK:4.124 (4SF)     | 3 | EOD | 29878 |
| 43 | 175 | 0 | Inf. Dilution                            | lgK:3.807 (4SF)     | 3 | EOD | 29878 |
| 44 | 175 | 0 | Inf. Dilution                            | lgK:3.938 (4SF)     | 3 | EOD | 29878 |
| 45 | 200 | 0 | Inf. Dilution                            | lgK:3.74 (3SF)      | 5 | MSP | 22893 |
| 46 | 200 | 0 | Inf. Dilution                            | lgK:3.655 (4SF)     | 2 | EOD | 29878 |
| 47 | 200 | 0 | Inf. Dilution                            | lgK:3.772 (4SF)     | 3 | EOD | 29878 |
| 48 | 225 | 0 | Inf. Dilution                            | lgK:3.525 (4SF)     | 2 | EOD | 29878 |
| 49 | 225 | 0 | Inf. Dilution                            | lgK:3.622 (4SF)     | 2 | EOD | 29878 |
| 50 | 250 | 0 | Inf. Dilution                            | lgK:3.417 (4SF)     | 2 | EOD | 29878 |
| 51 | 250 | 0 | Inf. Dilution                            | lgK:3.487 (4SF)     | 2 | EOD | 29878 |
| 52 | 275 | 0 | Inf. Dilution                            | lgK:3.332 (4SF)     | 2 | EOD | 29878 |
| 53 | 275 | 0 | Inf. Dilution                            | lgK:3.363 (4SF)     | 2 | EOD | 29878 |
| 54 | 15  |   |                                          | dH:-8.32 (3SF)      | 5 | MTD | 774   |
| 55 | 25  |   | Ethanol:Water<br>50:50Wgt<br>* Not found | lgK:4.07 (3SF)      | 5 | MGL | 4239  |
| 56 | 25  |   | Ethanol:Water<br>50:50Wgt                | dH:-3.480 (0.050SD) | 5 | MCL | 4239  |

## Reaction No. 32419



| No | t  | I-Str | Medium                    | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------------------|----------------|---|-------|--------|
| 1  | 25 |       | Ethanol:Water<br>59:41Vol | lgK:4.41 (3SF) | 5 | MIS   | 140    |

## Reaction No. 32600



| No | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------|-------------------|---|-------|--------|
| 1  | 20 | 0.37  | NaNO3   | lgK:3.37 (0.02SD) | 5 | MGL   | 2616   |
| 2  | 20 | 0.5   | Unknown | lgK:3.23 (3SF)    | 6 | CRV   | 552    |

## Reaction No. 32601

| NacTrp-1 + Cu+2 = Cu+2_NacTrp-1 |    |       |        |                |   |       |        |
|---------------------------------|----|-------|--------|----------------|---|-------|--------|
| No                              | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                               | 20 | 0.37  | NaNO3  | lgK:7.59 (3SF) | 5 | MGL   | 2616   |

| Reaction No. 32602                           |    |       |        |                |   |       |        |
|----------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Cu+2_NacTrp-1 + NacTrp-1 = Cu+2_NacTrp-1 (2) |    |       |        |                |   |       |        |
| No                                           | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                            | 20 | 0.37  | NaNO3  | lgK:4.61 (3SF) | 5 | MGL   | 2616   |

| Reaction No. 32603                                           |    |       |        |                |   |       |        |
|--------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Cu+2_H+1(-1)_NacTrp-1 + NacTrp-1 = Cu+2_H+1(-1)_NacTrp-1 (2) |    |       |        |                |   |       |        |
| No                                                           | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                            | 20 | 0.37  | NaNO3  | lgK:2.93 (3SF) | 5 | MGL   | 2616   |

| Reaction No. 32604                                  |    |       |         |                 |   |       |        |
|-----------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| Cu+2_NacTrp-1 (2) = Cu+2_H+1(-1)_NacTrp-1 (2) + H+1 |    |       |         |                 |   |       |        |
| No                                                  | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                   | 20 | 0.37  | NaNO3   | lgK:-4.93 (3SF) | 5 | MGL   | 2616   |
| 2                                                   | 20 | 0.5   | Unknown | lgK:-4.79 (3SF) | 6 | CRV   | 552    |

| Reaction No. 32683                          |    |       |         |                 |   |       |        |
|---------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| Cu+2_NacTrp-1 = Cu+2_H+1(-1)_NacTrp-1 + H+1 |    |       |         |                 |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 20 | 0.37  | NaNO3   | lgK:-4.28 (3SF) | 5 | MGL   | 2616   |
| 2                                           | 20 | 0.5   | Unknown | lgK:-4.14 (3SF) | 6 | CRV   | 552    |

| Reaction No. 32684                                          |    |       |         |                 |   |       |        |
|-------------------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| Cu+2_H+1(-1)_NacTrp-1 (2) = Cu+2_H+1(-2)_NacTrp-1 (2) + H+1 |    |       |         |                 |   |       |        |
| No                                                          | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                           | 20 | 0.37  | NaNO3   | lgK:-7.56 (3SF) | 5 | MGL   | 2616   |
| 2                                                           | 20 | 0.5   | Unknown | lgK:-7.42 (3SF) | 6 | CRV   | 552    |

| Reaction No. 32918                         |    |       |        |                 |   |       |        |
|--------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| H+1 + [17]N4 = H+1_[17]N4                  |    |       |        |                 |   |       |        |
| No                                         | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                          | 25 | 0.2   | NaClO4 | lgK:10.23 (4SF) | 0 | MPT   | 6907   |
| Note (1): Low wgt for internal consistency |    |       |        |                 |   |       |        |

|   |    |     |      |                     |   |     |      |
|---|----|-----|------|---------------------|---|-----|------|
| 2 | 25 | 0.5 | KNO3 | lgK:11.20 (0.01SD)  | 6 | MGL | 2718 |
| 3 | 25 | 0.5 | KNO3 | dH:-43.5 (0.4SD) kJ | 6 | MCL | 2718 |

| Reaction No. 32919               |    |       |               |                     |   |       |        |
|----------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| H+1_[17]N4 + H+1 = H+1(2)_[17]N4 |    |       |               |                     |   |       |        |
| No                               | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                | 25 | 0     | Inf. Dilution | lgK:9.4 (2SF)       | 1 | EDH   |        |
| 2                                | 25 | 0.2   | NaClO4        | lgK:9.66 (3SF)      | 2 | MPT   | 6907   |
| 3                                | 25 | 0.5   | KNO3          | lgK:10.13 (0.01SD)  | 2 | MGL   | 2718   |
| 4                                | 25 | 0.5   | KNO3          | dH:-46.4 (0.4SD) kJ | 3 | MCL   | 2718   |

| Reaction No. 32920                  |    |       |               |                     |   |       |        |
|-------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| H+1(2)_[17]N4 + H+1 = H+1(3)_[17]N4 |    |       |               |                     |   |       |        |
| No                                  | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                   | 25 | 0     | Inf. Dilution | lgK:6.9 (2SF)       | 1 | EDH   |        |
| 2                                   | 25 | 0.2   | NaClO4        | lgK:7.40 (3SF)      | 2 | MPT   | 6907   |
| 3                                   | 25 | 0.5   | KNO3          | lgK:7.96 (0.01SD)   | 2 | MGL   | 2718   |
| 4                                   | 25 | 0.5   | KNO3          | dH:-45.6 (0.8SD) kJ | 3 | MCL   | 2718   |

| Reaction No. 32921                         |    |       |        |                     |   |       |        |
|--------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| H+1(3)_[17]N4 + H+1 = H+1(4)_[17]N4        |    |       |        |                     |   |       |        |
| No                                         | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                          | 25 | 0.2   | NaClO4 | lgK:5.31 (3SF)      | 0 | MPT   | 6907   |
| Note (1): Low wgt for internal consistency |    |       |        |                     |   |       |        |
| 2                                          | 25 | 0.5   | KNO3   | lgK:6.30 (0.01SD)   | 6 | MGL   | 2718   |
| 3                                          | 25 | 0.5   | KNO3   | dH:-45.6 (0.8SD) kJ | 6 | MCL   | 2718   |

| Reaction No. 32948                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Ca+2 = Ca+2_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:11.77 (4SF) | 6 | MPT   | 2011   |

| Reaction No. 32949                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Mg+2 = Mg+2_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:10.30 (4SF) | 6 | MPT   | 2011   |

| Reaction No. 32950                |    |       |               |               |   |       |        |
|-----------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 2PrEDTA-4 + Sr+2 = Sr+2_2PrEDTA-4 |    |       |               |               |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3          | lgK:9.74(3SF) | 6 | MPT   | 2011   |
| 2                                 | 25 | 0     | Inf. Dilution | lgK:9.6(2SF)  | 1 | ESC   |        |

| Reaction No. 32951                |    |       |               |               |   |       |        |
|-----------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 2PrEDTA-4 + Ba+2 = Ba+2_2PrEDTA-4 |    |       |               |               |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3          | lgK:8.60(3SF) | 6 | MPT   | 2011   |
| 2                                 | 25 | 0     | Inf. Dilution | lgK:8.9(2SF)  | 1 | ESC   |        |

| Reaction No. 32952                |    |       |        |                |   |       |        |
|-----------------------------------|----|-------|--------|----------------|---|-------|--------|
| 2PrEDTA-4 + Cd+2 = Cd+2_2PrEDTA-4 |    |       |        |                |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:18.06(4SF) | 6 | MPT   | 2011   |

| Reaction No. 32958                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Pb+2 = Pb+2_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:19.2(0.1SD) | 6 | MPT   | 2012   |

| Reaction No. 32959                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Zn+2 = Zn+2_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:18.5(0.1SD) | 6 | MPT   | 2012   |

| Reaction No. 32960                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Co+2 = Co+2_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:18.3(0.1SD) | 6 | MPT   | 2012   |

| Reaction No. 32961                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Mn+2 = Mn+2_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:15.5(0.1SD) | 6 | MPT   | 2012   |

| Reaction No. 32962 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| 2PrEDTA-4 + Cu+2 = Cu+2_2PrEDTA-4 |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:21.1(0.2SD) | 6 | MPT   | 2012   |

| Reaction No. 32963                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + La+3 = La+3_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:16.4(0.2SD) | 6 | MPT   | 2012   |

| Reaction No. 32964                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Ce+3 = Ce+3_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:17.0(0.1SD) | 6 | MPT   | 2012   |

| Reaction No. 32965                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Pr+3 = Pr+3_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:17.3(0.1SD) | 6 | MPT   | 2012   |

| Reaction No. 32966                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Nd+3 = Nd+3_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:17.6(0.1SD) | 6 | MPT   | 2012   |

| Reaction No. 32967                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Sm+3 = Sm+3_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:18.1(0.1SD) | 6 | MPT   | 2012   |

| Reaction No. 32968                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Eu+3 = Eu+3_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:18.3(0.1SD) | 6 | MPT   | 2012   |

| Reaction No. 32969                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Gd+3 = Gd+3_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:18.4(0.1SD) | 6 | MPT   | 2012   |

| Reaction No. 32970                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Tb+3 = Tb+3_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:19.0(0.1SD) | 6 | MPT   | 2012   |

| Reaction No. 32971                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Dy+3 = Dy+3_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:19.4(0.1SD) | 6 | MPT   | 2012   |

| Reaction No. 32972                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Ho+3 = Ho+3_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:19.7(0.1SD) | 6 | MPT   | 2012   |

| Reaction No. 32973                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Er+3 = Er+3_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:20.0(0.1SD) | 6 | MPT   | 2012   |

| Reaction No. 32974                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Tm+3 = Tm+3_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:20.4(0.1SD) | 6 | MPT   | 2012   |

| Reaction No. 32975                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Yb+3 = Yb+3_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:20.7(0.2SD) | 6 | MPT   | 2012   |

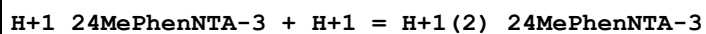
| Reaction No. 32976                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2PrEDTA-4 + Lu+3 = Lu+3_2PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:20.8(0.2SD) | 6 | MPT   | 2012   |

| Reaction No. 33011                      |  |  |  |  |  |  |  |
|-----------------------------------------|--|--|--|--|--|--|--|
| H+1 + 24MePhenNTA-3 = H+1_24MePhenNTA-3 |  |  |  |  |  |  |  |



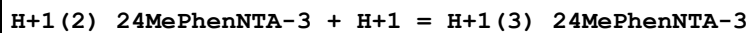
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl    | lgK:9.45 (0.02SD) | 6 | MGL   | 2595   |

## Reaction No. 33012



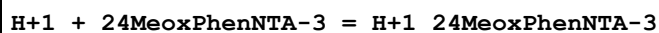
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl    | lgK:2.40 (0.02SD) | 6 | MGL   | 2595   |

## Reaction No. 33013



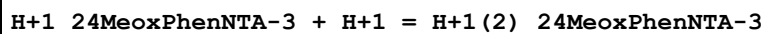
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl    | lgK:1.5 (0.05SD) | 4 | MGL   | 2595   |

## Reaction No. 33014



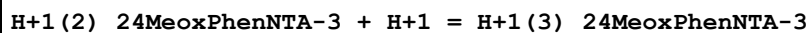
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl    | lgK:9.51 (0.02SD) | 6 | MGL   | 2595   |

## Reaction No. 33015



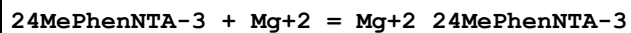
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl    | lgK:2.44 (0.02SD) | 6 | MGL   | 2595   |

## Reaction No. 33016



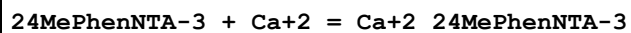
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl    | lgK:1.5 (0.05SD) | 4 | MGL   | 2595   |

## Reaction No. 33027



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl    | lgK:4.74 (0.02SD) | 6 | MGL   | 2595   |

## Reaction No. 33028



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl    | lgK:6.22 (0.01SD) | 6 | MGL   | 2595   |

| Reaction No. 33029                        |    |       |               |                  |   |       |        |
|-------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 24MePhenNTA-3 + Sr+2 = Sr+2_24MePhenNTA-3 |    |       |               |                  |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                         | 20 | 0.1   | KCl           | lgK:4.48(0.02SD) | 6 | MGL   | 2595   |
| 2                                         | 25 | 0     | Inf. Dilution | lgK:4.4(2SF)     | 1 | ESC   |        |

| Reaction No. 33030                        |    |       |               |                 |   |       |        |
|-------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 24MePhenNTA-3 + Ba+2 = Ba+2_24MePhenNTA-3 |    |       |               |                 |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                         | 20 | 0.1   | KCl           | lgK:4.3(0.03SD) | 6 | MGL   | 2595   |
| 2                                         | 25 | 0     | Inf. Dilution | lgK:4.5(2SF)    | 1 | ESC   |        |

| Reaction No. 33031                        |    |       |        |                 |   |       |        |
|-------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 24MePhenNTA-3 + Zn+2 = Zn+2_24MePhenNTA-3 |    |       |        |                 |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                         | 20 | 0.1   | KCl    | lgK:9.9(0.03SD) | 6 | MGL   | 2595   |

| Reaction No. 33032                            |    |       |        |                  |   |       |        |
|-----------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 24MeoxPhenNTA-3 + Mg+2 = Mg+2_24MeoxPhenNTA-3 |    |       |        |                  |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                             | 20 | 0.1   | KCl    | lgK:4.75(0.01SD) | 6 | MGL   | 2595   |

| Reaction No. 33033                            |    |       |        |                  |   |       |        |
|-----------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 24MeoxPhenNTA-3 + Ca+2 = Ca+2_24MeoxPhenNTA-3 |    |       |        |                  |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                             | 20 | 0.1   | KCl    | lgK:6.24(0.01SD) | 6 | MGL   | 2595   |

| Reaction No. 33034                            |    |       |               |                  |   |       |        |
|-----------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 24MeoxPhenNTA-3 + Sr+2 = Sr+2_24MeoxPhenNTA-3 |    |       |               |                  |   |       |        |
| No                                            | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                             | 20 | 0.1   | KCl           | lgK:4.49(0.01SD) | 6 | MGL   | 2595   |
| 2                                             | 25 | 0     | Inf. Dilution | lgK:4.4(2SF)     | 1 | ESC   |        |

| Reaction No. 33035                            |    |       |        |                  |   |       |        |
|-----------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| 24MeoxPhenNTA-3 + Ba+2 = Ba+2_24MeoxPhenNTA-3 |    |       |        |                  |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                             | 20 | 0.1   | KCl    | lgK:4.32(0.01SD) | 6 | MGL   | 2595   |

|   |    |   |               |               |   |     |  |
|---|----|---|---------------|---------------|---|-----|--|
| 2 | 25 | 0 | Inf. Dilution | lgK:4.5 (2SF) | 1 | ESC |  |
|---|----|---|---------------|---------------|---|-----|--|

| Reaction No. 33036                            |    |       |        |                   |   |       |        |
|-----------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| 24MeoxPhenNTA-3 + Zn+2 = Zn+2_24MeoxPhenNTA-3 |    |       |        |                   |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                             | 20 | 0.1   | KCl    | lgK:9.95 (0.02SD) | 6 | MGL   | 2595   |

| Reaction No. 33087          |    |       |        |                   |   |       |        |
|-----------------------------|----|-------|--------|-------------------|---|-------|--------|
| H+1 + CysSG-3 = H+1_CysSG-3 |    |       |        |                   |   |       |        |
| No                          | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                           | 25 | 0.4   | KCl    | lgK:9.30 (0.02SD) | 5 | MGL   | 2421   |

| Reaction No. 33088                  |    |       |        |                   |   |       |        |
|-------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| H+1_CysSG-3 + H+1 = H+1 (2)_CysSG-3 |    |       |        |                   |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                   | 25 | 0.4   | KCl    | lgK:8.55 (0.02SD) | 5 | MGL   | 2421   |

| Reaction No. 33538                                   |    |       |               |               |   |       |        |
|------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Fe+2_5MePhenanth + 5MePhenanth = Fe+2_5MePhenanth(2) |    |       |               |               |   |       |        |
| No                                                   | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                    | 25 | 0     | Inf. Dilution | lgK:5.2 (2SF) | 1 | ELC   |        |
| 2                                                    | 25 | 0.1   | KCl           | lgK:7.0 (2SF) | 0 | MDS   | 3077   |

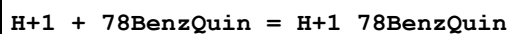
| Reaction No. 33539                                      |    |       |               |                |   |       |        |
|---------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Fe+2_5MePhenanth(2) + 5MePhenanth = Fe+2_5MePhenanth(3) |    |       |               |                |   |       |        |
| No                                                      | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 25 | 0     | Inf. Dilution | lgK:10.3 (3SF) | 1 | ELC   |        |
| 2                                                       | 25 | 0.1   | KCl           | lgK:8.4 (2SF)  | 0 | MDS   | 3077   |

| Reaction No. 33587                        |    |       |        |                 |   |       |        |
|-------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| H+1 + 4AmBenzPhenone = H+1_4AmBenzPhenone |    |       |        |                 |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                         | 25 |       |        | lgK:2.15 (3SF)  | 5 | ENS   | 774    |
| 2                                         | 25 |       |        | dH:-4.541 (4SF) | 5 | MTD   | 774    |

| Reaction No. 33588                |   |       |        |             |   |       |        |
|-----------------------------------|---|-------|--------|-------------|---|-------|--------|
| H+1 + 56BenzQuin = H+1_56BenzQuin |   |       |        |             |   |       |        |
| No                                | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

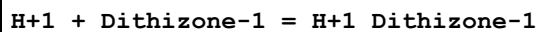
|   |    |  |  |                |   |     |     |
|---|----|--|--|----------------|---|-----|-----|
| 1 | 20 |  |  | lgK:5.00 (3SF) | 5 | ENS | 774 |
| 2 | 20 |  |  | dH:-4.54 (3SF) | 5 | MTD | 774 |

## Reaction No. 33589



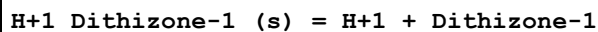
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 20 |       |        | lgK:4.15 (3SF) | 5 | ENS   | 774    |
| 2  | 20 |       |        | dH:-6.09 (3SF) | 5 | MTD   | 774    |

## Reaction No. 34182



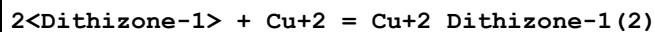
| No | t  | I-Str | Medium                              | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KClO4                               | lgK:4.45 (3SF)   | 3 | MSL   | 140    |
| 2  | 25 | 0.1   | Unknown                             | lgK:4.55 (3SF)   | 5 | MSP   | 140    |
| 3  | 25 | 0.1   | CCl4<br>NaNO3                       | lgK:4.6 (2SF)    | 5 | MDS   | 140    |
| 4  | 25 | 0.1   | CHCl3<br>NaNO3                      | lgK:5.7 (2SF)    | 5 | MDS   | 140    |
| 5  | 25 | 0.1   | Cyclohexane<br>NaNO3                | lgK:4.2 (2SF)    | 5 | MDS   | 140    |
| 6  | 25 | 0.1   | Dioxan:Water<br>50:50Wgt<br>Unknown | lgK:5.77 (3SF)   | 4 | MSP   | 906    |
| 7  | 25 | 0.1   | Dioxan:Water<br>50:50Wgt<br>Unknown | lgK:5.80 (3SF)   | 5 | MSP   | 931    |
| 8  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:5.60 (3SF) w | 4 | MPH   | 906    |

## Reaction No. 34183



| No | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:-11.06 (4SF) | 5 | CRV   | 2556   |

## Reaction No. 34184



| No | t     | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|-------|-------|--------|-------------------|---|-------|--------|
| 1  | 24:26 | 0.1   | NaClO4 | lgK:19.2 (0.07SD) | 0 | MSP   | 906    |
| 2  | 25    | 0.1   | NaCl   | lgK:22.3 (3SF)    | 3 | MDS   | 140    |

| Reaction No. 34185                                                                       |    |       |         |                 |   |       |        |
|------------------------------------------------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| $\text{Cu}+2\_ \text{Dithizone-1(2)}\_ (\text{s}) = \text{Cu}+2 + 2<\text{Dithizone-1}>$ |    |       |         |                 |   |       |        |
| No                                                                                       | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                        | 25 | 0.1   | Unknown | lgK:-30.7 (3SF) | 5 | CRV   | 2556   |

| Reaction No. 34186                                                    |       |       |                                     |                  |   |       |        |
|-----------------------------------------------------------------------|-------|-------|-------------------------------------|------------------|---|-------|--------|
| $\text{Dithizone-1} + \text{Zn}+2 = \text{Zn}+2\_ \text{Dithizone-1}$ |       |       |                                     |                  |   |       |        |
| No                                                                    | t     | I-Str | Medium                              | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                     | 24:26 | 0.1   | NaClO4                              | lgK:6.9 (0.09SD) | 5 | MSP   | 906    |
| 2                                                                     | 25    | 1     | NaClO4                              | lgK:5.05 (3SF)   | 0 | MDS   | 140    |
| 3                                                                     | 25    | 1     | Unknown                             | lgK:7.75 (3SF)   | 4 | CRV   | 2556   |
| 4                                                                     | 25    | 0.1   | Dioxan:Water<br>50:50Wgt<br>Unknown | lgK:6.18 (3SF)   | 4 | MSP   | 140    |
| 5                                                                     | 25    | 0.1   | Dioxan:Water<br>50:50Wgt<br>Unknown | lgK:6.16 (3SF)   | 4 | MSP   | 906    |

| Reaction No. 34187                                                          |       |       |         |                   |   |       |        |
|-----------------------------------------------------------------------------|-------|-------|---------|-------------------|---|-------|--------|
| $2<\text{Dithizone-1}> + \text{Zn}+2 = \text{Zn}+2\_ \text{Dithizone-1(2)}$ |       |       |         |                   |   |       |        |
| No                                                                          | t     | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                           | 24:26 | 0.1   | NaClO4  | lgK:14.0 (0.09SD) | 4 | MSP   | 906    |
| 2                                                                           | 25    | 1     | Unknown | lgK:15.05 (4SF)   | 5 | CRV   | 2556   |

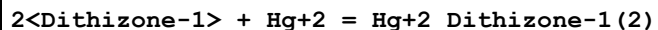
| Reaction No. 34188                                                          |       |       |         |                  |   |       |        |
|-----------------------------------------------------------------------------|-------|-------|---------|------------------|---|-------|--------|
| $2<\text{Dithizone-1}> + \text{Pb}+2 = \text{Pb}+2\_ \text{Dithizone-1(2)}$ |       |       |         |                  |   |       |        |
| No                                                                          | t     | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                           | 23    | 0.1   | NaClO4  | lgK:19.15 (4SF)  | 0 | MDS   | 906    |
| 2                                                                           | 24:26 | 0.1   | NaClO4  | lgK:14.2 (0.1SD) | 3 | MSP   | 906    |
| 3                                                                           | 25    | 0.1   | Unknown | lgK:15.85 (4SF)  | 0 | MDS   | 140    |
| 4                                                                           | 25    | 0.1   | Unknown | lgK:15.2 (3SF)   | 3 | CRV   | 2556   |

| Reaction No. 34189                                                                       |    |       |         |                 |   |       |        |
|------------------------------------------------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| $\text{Pb}+2\_ \text{Dithizone-1(2)}\_ (\text{s}) = \text{Pb}+2 + 2<\text{Dithizone-1}>$ |    |       |         |                 |   |       |        |
| No                                                                                       | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                        | 25 | 0.1   | Unknown | lgK:-23.7 (3SF) | 5 | MDS   | 140    |

| Reaction No. 34190                                                          |  |  |  |  |  |  |  |
|-----------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $2<\text{Dithizone-1}> + \text{Co}+2 = \text{Co}+2\_ \text{Dithizone-1(2)}$ |  |  |  |  |  |  |  |

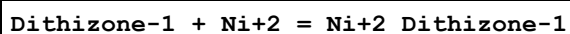
| No | t     | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|-------|-------|--------|------------------|---|-------|--------|
| 1  | 24:26 | 0.1   | NaClO4 | lgK:14.0(0.08SD) | 4 | MSP   | 906    |
| 2  | 25    |       |        | lgK:13.(2SF)     | 4 | MDS   | 140    |

## Reaction No. 34191



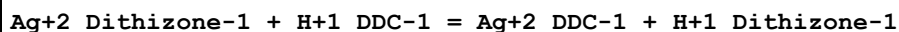
| No | t     | I-Str   | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|-------|---------|--------|----------------|---|-------|--------|
| 1  | 24:26 | 0.1     | NaClO4 | lgK:40.(0.8SD) | 1 | MSP   | 906    |
| 2  | 25    | 0.6:2.7 | HCl    | lgK:40.34(4SF) | 1 | MDS   | 140    |

## Reaction No. 34192



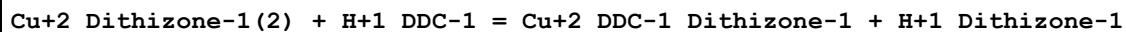
| No | t     | I-Str | Medium                              | Value (Dev)    | W | Tech. | Ref. # |
|----|-------|-------|-------------------------------------|----------------|---|-------|--------|
| 1  | 24:26 | 0.1   | NaClO4                              | lgK:7.4(0.1SD) | 5 | MSP   | 906    |
| 2  | 25    | 0.1   | Dioxan:Water<br>50:50Wgt<br>Unknown | lgK:5.69(3SF)  | 4 | MSP   | 906    |
| 3  | 25    | 0.1   | Dioxan:Water<br>50:50Wgt<br>Unknown | lgK:5.83(3SF)  | 4 | MSP   | 931    |

## Reaction No. 34196



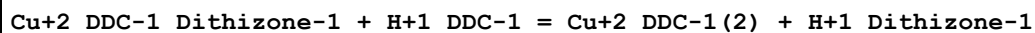
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 |       | CCl4   | lgK:2.6(0.03SD) | 5 | MSP   | 931    |

## Reaction No. 34197



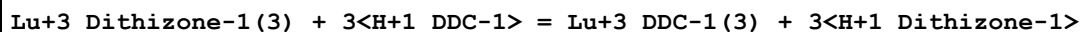
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 20 |       | CCl4   | lgK:2.5(0.1SD) | 5 | MSP   | 931    |

## Reaction No. 34198



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 20 |       | CCl4   | lgK:2.1(0.1SD) | 5 | MSP   | 931    |

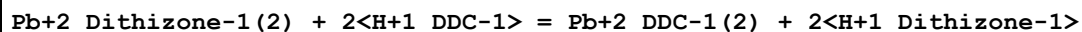
## Reaction No. 34199



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

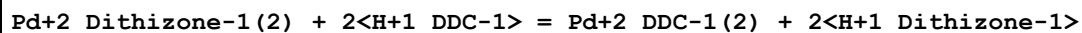
|   |    |  |      |               |   |     |     |
|---|----|--|------|---------------|---|-----|-----|
| 1 | 20 |  | CCl4 | lgK:4.75(3SF) | 5 | MSP | 931 |
|---|----|--|------|---------------|---|-----|-----|

## Reaction No. 34200



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 |       | CCl4   | lgK:5.0(0.05SD) | 5 | MSP   | 931    |

## Reaction No. 34201



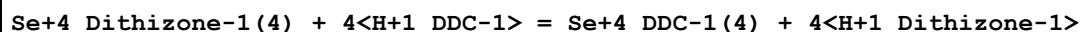
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 20 |       | CCl4   | lgK:1.6(0.2SD) | 5 | MSP   | 931    |

## Reaction No. 34202



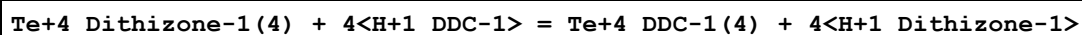
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 |       | CCl4   | lgK:2.5(0.07SD) | 5 | MSP   | 931    |

## Reaction No. 34203



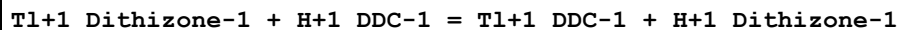
| No | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|--------|--------------|---|-------|--------|
| 1  | 20 |       | CCl4   | lgK:6.5(2SF) | 4 | MSP   | 931    |

## Reaction No. 34204



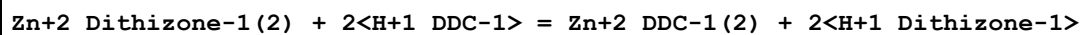
| No | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|--------|--------------|---|-------|--------|
| 1  | 20 |       | CCl4   | lgK:5.5(2SF) | 4 | MSP   | 931    |

## Reaction No. 34205



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 |       | CCl4   | lgK:3.5(0.05SD) | 5 | MSP   | 931    |

## Reaction No. 34206



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 |       | CCl4   | lgK:0.2(0.03SD) | 4 | MSP   | 931    |

## Reaction No. 34348

| As+3_H+1(-1)_DDC-1_Dithizone-1 + 2<H+1_DDC-1> = As+3_DDC-1(3) + H+1_Dithizone-1 |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| No                                                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                               | 20 |       | CCl4   | lgK:7.9(0.06SD) | 5 | MSP   | 931    |

| Reaction No. 34349                                                      |    |       |        |                 |   |       |        |
|-------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Bi+3_DDC-1(3) + 3<H+1_Dithizone-1> = Bi+3_Dithizone-1(3) + 3<H+1_DDC-1> |    |       |        |                 |   |       |        |
| No                                                                      | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                       | 20 |       | CCl4   | lgK:5.7(0.08SD) | 3 | MSP   | 931    |

| Reaction No. 34350                                                      |    |       |        |                 |   |       |        |
|-------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Cd+2_Dithizone-1(2) + 2<H+1_DDC-1> = Cd+2_DDC-1(2) + 2<H+1_Dithizone-1> |    |       |        |                 |   |       |        |
| No                                                                      | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                       | 20 |       | CCl4   | lgK:2.5(0.05SD) | 5 | MSP   | 931    |

| Reaction No. 35252                  |    |       |         |                 |   |       |        |
|-------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| H+1 + 2MePhenanth = H+1_2MePhenanth |    |       |         |                 |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 17 | 0.1   | Unknown | lgK:5.3(2SF)    | 5 | MSP   | 140    |
| 2                                   | 25 | 0.1   | KCl     | lgK:5.42(3SF)   | 5 | MGL   | 931    |
| 3                                   | 25 | 0.1   | KCl     | lgK:5.42(3SF)   | 5 | MGL   | 3076   |
| 4                                   | 25 | 0.3   | K2SO4   | lgK:5.3(0.05SD) | 5 | MGL   | 931    |

| Reaction No. 35253                    |    |       |        |              |   |       |        |
|---------------------------------------|----|-------|--------|--------------|---|-------|--------|
| 2MePhenanth + Fe+2 = Fe+2_2MePhenanth |    |       |        |              |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KCl    | lgK:4.2(2SF) | 5 | MGL   | 140    |

| Reaction No. 35254                                   |    |       |        |              |   |       |        |
|------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Fe+2_2MePhenanth + 2MePhenanth = Fe+2_2MePhenanth(2) |    |       |        |              |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.1   | KCl    | lgK:3.7(2SF) | 5 | MGL   | 140    |

| Reaction No. 35255                                      |    |       |        |              |   |       |        |
|---------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Fe+2_2MePhenanth(2) + 2MePhenanth = Fe+2_2MePhenanth(3) |    |       |        |              |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | KCl    | lgK:2.9(2SF) | 5 | MGL   | 140    |

| Reaction No. 35256 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|



| 2MePhenanth + Co+2 = Co+2_2MePhenanth |    |       |        |               |   |       |        |
|---------------------------------------|----|-------|--------|---------------|---|-------|--------|
| No                                    | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KCl    | lgK:5.1 (2SF) | 5 | MDS   | 3076   |

| Reaction No. 35257                                   |    |       |        |               |   |       |        |
|------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Co+2_2MePhenanth + 2MePhenanth = Co+2_2MePhenanth(2) |    |       |        |               |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.1   | KCl    | lgK:4.9 (2SF) | 5 | MDS   | 3076   |

| Reaction No. 35258                                      |    |       |        |               |   |       |        |
|---------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Co+2_2MePhenanth(2) + 2MePhenanth = Co+2_2MePhenanth(3) |    |       |        |               |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | KCl    | lgK:3.9 (2SF) | 5 | MDS   | 3076   |

| Reaction No. 35259                    |    |       |        |                |   |       |        |
|---------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 2MePhenanth + Cd+2 = Cd+2_2MePhenanth |    |       |        |                |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KCl    | lgK:5.15 (3SF) | 5 | MDS   | 3076   |

| Reaction No. 35260                                   |    |       |        |                |   |       |        |
|------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Cd+2_2MePhenanth + 2MePhenanth = Cd+2_2MePhenanth(2) |    |       |        |                |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.1   | KCl    | lgK:4.50 (3SF) | 5 | MDS   | 3076   |

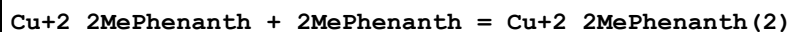
| Reaction No. 35261                                      |    |       |        |                |   |       |        |
|---------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Cd+2_2MePhenanth(2) + 2MePhenanth = Cd+2_2MePhenanth(3) |    |       |        |                |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | KCl    | lgK:3.65 (3SF) | 5 | MDS   | 3076   |

| Reaction No. 35262                          |    |       |        |                 |   |       |        |
|---------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2<2MePhenanth> + Cu+1 = Cu+1_2MePhenanth(2) |    |       |        |                 |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 25 | 0.3   | K2SO4  | lgK:16.95 (4SF) | 5 | MPT   | 931    |

| Reaction No. 35263                    |    |       |        |                |   |       |        |
|---------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 2MePhenanth + Cu+2 = Cu+2_2MePhenanth |    |       |        |                |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KCl    | lgK:7.40 (3SF) | 5 | MDS   | 3076   |

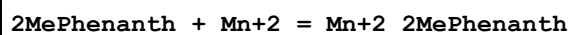
|   |    |     |       |               |   |     |     |
|---|----|-----|-------|---------------|---|-----|-----|
| 2 | 25 | 0.3 | K2SO4 | lgK:7.4 (2SF) | 5 | MPT | 931 |
|---|----|-----|-------|---------------|---|-----|-----|

## Reaction No. 35264



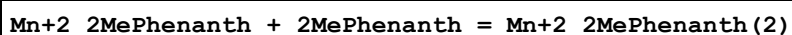
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:6.45 (3SF) | 5 | MDS   | 3076   |
| 2  | 25 | 0.2   | K2SO4  | lgK:6.21 (3SF) | 5 | MSP   | 931    |
| 3  | 25 | 0.3   | K2SO4  | lgK:6.4 (2SF)  | 5 | MPT   | 931    |

## Reaction No. 35265



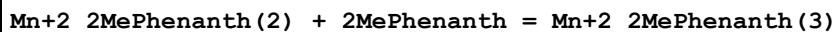
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:3.0 (2SF) | 5 | MDS   | 3076   |

## Reaction No. 35266



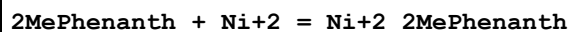
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:2.5 (2SF) | 5 | MDS   | 3076   |

## Reaction No. 35267



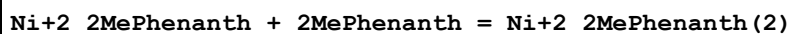
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:2.4 (2SF) | 5 | MDS   | 3076   |

## Reaction No. 35268



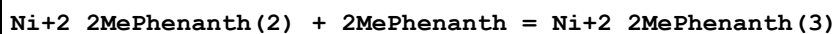
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:5.95 (3SF) | 5 | MDS   | 3076   |

## Reaction No. 35269



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:6.85 (3SF) | 3 | MDS   | 3076   |

## Reaction No. 35270



| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:4.9 (2SF) | 3 | MDS   | 3076   |

| Reaction No. 35271                    |    |       |        |                |   |       |        |
|---------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 2MePhenanth + Zn+2 = Zn+2_2MePhenanth |    |       |        |                |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KCl    | lgK:4.96 (3SF) | 3 | MDS   | 3076   |

| Reaction No. 35272                                   |    |       |        |                |   |       |        |
|------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Zn+2_2MePhenanth + 2MePhenanth = Zn+2_2MePhenanth(2) |    |       |        |                |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.1   | KCl    | lgK:4.40 (3SF) | 3 | MDS   | 3076   |

| Reaction No. 35273                                      |    |       |        |                |   |       |        |
|---------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Zn+2_2MePhenanth(2) + 2MePhenanth = Zn+2_2MePhenanth(3) |    |       |        |                |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | KCl    | lgK:3.35 (3SF) | 3 | MDS   | 3076   |

| Reaction No. 35274                                      |    |       |                                  |                |   |       |        |
|---------------------------------------------------------|----|-------|----------------------------------|----------------|---|-------|--------|
| Cu+2_5MePhenanth(2) + Cu+1 = Cu+1_5MePhenanth(2) + Cu+2 |    |       |                                  |                |   |       |        |
| No                                                      | t  | I-Str | Medium                           | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.3   | Dioxan:Water<br>50:50Wgt<br>KNO3 | lgK:0.81 (2SF) | 5 | MPT   | 931    |

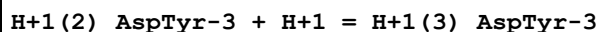
| Reaction No. 35278                  |    |       |               |                |   |       |        |
|-------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| H+1 + 6MePhenanth = H+1_6MePhenanth |    |       |               |                |   |       |        |
| No                                  | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 0     | Inf. Dilution | lgK:5.11 (3SF) | 5 | ENS   | 774    |
| 2                                   | 25 | 0     | Inf. Dilution | dH:-4.40 (3SF) | 5 | MTD   | 774    |

| Reaction No. 35289            |      |       |         |                 |   |       |        |
|-------------------------------|------|-------|---------|-----------------|---|-------|--------|
| H+1 + AspTyr-3 = H+1_AspTyr-3 |      |       |         |                 |   |       |        |
| No                            | t    | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                             | 25   | 0     | Unknown | lgK:10.23 (4SF) | 5 | ENS   | 774    |
| 2                             | 0:25 | 0     | Unknown | dH:-6.20 (3SF)  | 5 | MTD   | 774    |

| Reaction No. 35290                   |    |       |         |                |   |       |        |
|--------------------------------------|----|-------|---------|----------------|---|-------|--------|
| H+1_AspTyr-3 + H+1 = H+1(2)_AspTyr-3 |    |       |         |                |   |       |        |
| No                                   | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                    | 25 | 0     | Unknown | lgK:8.92 (3SF) | 5 | ENS   | 774    |

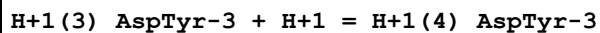
|   |      |   |         |                |   |     |     |
|---|------|---|---------|----------------|---|-----|-----|
| 2 | 0:25 | 0 | Unknown | dH:-10.16(4SF) | 5 | MTD | 774 |
|---|------|---|---------|----------------|---|-----|-----|

## Reaction No. 35291



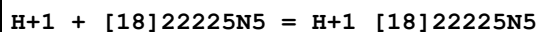
| No | t    | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|------|-------|---------|---------------|---|-------|--------|
| 1  | 25   | 0     | Unknown | lgK:3.57(3SF) | 5 | ENS   | 774    |
| 2  | 0:25 | 0     | Unknown | dH:0.0(1SF)   | 5 | MTD   | 774    |

## Reaction No. 35292



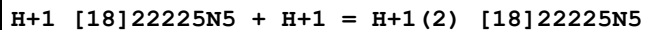
| No | t    | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|------|-------|---------|----------------|---|-------|--------|
| 1  | 25   | 0     | Unknown | lgK:2.13(3SF)  | 5 | ENS   | 774    |
| 2  | 0:25 | 0     | Unknown | dH:-0.750(3SF) | 5 | MTD   | 774    |

## Reaction No. 35678



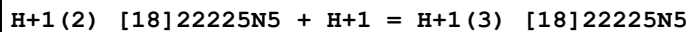
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 35 | 0.2   | NaClO4 | lgK:10.15(4SF) | 5 | MPT   | 3840   |

## Reaction No. 35679



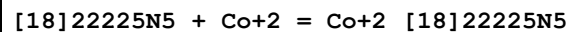
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 35 | 0.2   | NaClO4 | lgK:9.52(3SF) | 5 | MPT   | 3840   |

## Reaction No. 35680



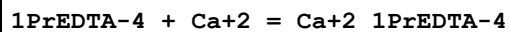
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 35 | 0.2   | NaClO4 | lgK:8.55(3SF) | 5 | MPT   | 3840   |

## Reaction No. 35681



| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 35 | 0.2   | NaClO4 | lgK:11.5(3SF) | 5 | MPT   | 3840   |

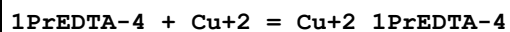
## Reaction No. 36131



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:11.74(4SF) | 6 | MPL   | 2011   |

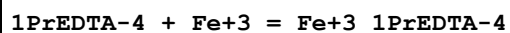
|   |    |     |         |               |   |     |      |
|---|----|-----|---------|---------------|---|-----|------|
| 2 | 25 | 0.1 | Unknown | lgK:11.6(3SF) | 6 | ENS | 1384 |
|---|----|-----|---------|---------------|---|-----|------|

## Reaction No. 36132



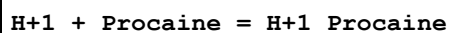
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3    | lgK:20.5(0.1SD) | 6 | MPL   | 1996   |
| 2  | 25 | 0.1   | Unknown | lgK:19.8(3SF)   | 6 | ENS   | 1384   |

## Reaction No. 36133



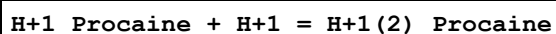
| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:26.0(3SF) | 6 | ENS   | 1384   |

## Reaction No. 37035



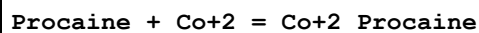
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 1     | KNO3   | lgK:8.8(0.2SD) | 4 | MGL   | 931    |

## Reaction No. 37036



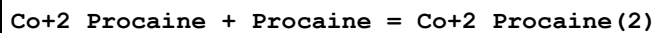
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 1     | KNO3   | lgK:2.3(0.1SD) | 4 | MGL   | 931    |

## Reaction No. 37037



| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 1     | KNO3   | lgK:7.21(3SF) | 4 | MGL   | 931    |

## Reaction No. 37038



| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 1     | KNO3   | lgK:6.18(3SF) | 4 | MGL   | 931    |

## Reaction No. 37039



| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 1     | KNO3   | lgK:5.90(3SF) | 4 | MGL   | 931    |

| Reaction No. 37040                             |    |       |        |                |   |       |        |
|------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Co+2_Procaine(3) + Procaine = Co+2_Procaine(4) |    |       |        |                |   |       |        |
| No                                             | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                              | 25 | 1     | KNO3   | lgK:5.80 (3SF) | 4 | MGL   | 931    |

| Reaction No. 37779                                  |    |       |                               |                 |   |       |        |
|-----------------------------------------------------|----|-------|-------------------------------|-----------------|---|-------|--------|
| 4<Thiocarbanilide> + Hg+2 = Hg+2_Thiocarbanilide(4) |    |       |                               |                 |   |       |        |
| No                                                  | t  | I-Str | Medium                        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.2   | DMF:Water<br>82:18Wgt<br>KNO3 | lgK:23.17 (4SF) | 4 | MPT   | 4129   |

| Reaction No. 37945                          |    |       |                                   |                |   |       |        |
|---------------------------------------------|----|-------|-----------------------------------|----------------|---|-------|--------|
| H+1 + Thiocarbanilide = H+1_Thiocarbanilide |    |       |                                   |                |   |       |        |
| No                                          | t  | I-Str | Medium                            | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 25 | 0.01  | Dioxan:Water<br>50:50Vol<br>HClO4 | lgK:2.91 (3SF) | 4 | MPT   | 4147   |

| Reaction No. 37978                                                                  |    |       |        |                   |   |       |        |
|-------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Ni+2_148Triazaoct + Pyrid2Azo4DiMeAniline = Ni+2_148Triazaoct_Pyrid2Azo4DiMeAniline |    |       |        |                   |   |       |        |
| No                                                                                  | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                   | 25 | 0.3   | NaNO3  | lgK:2.05 (0.01SD) | 4 | MKN   | 692    |
| 2                                                                                   | 25 | 0.3   | NaNO3  | lgK:2.34 (0.02SD) | 4 | MSP   | 692    |

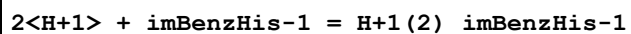
| Reaction No. 37979                                                                  |    |       |        |                   |   |       |        |
|-------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Ni+2_159Triazanon + Pyrid2Azo4DiMeAniline = Ni+2_159Triazanon_Pyrid2Azo4DiMeAniline |    |       |        |                   |   |       |        |
| No                                                                                  | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                   | 25 | 0.3   | NaNO3  | lgK:1.88 (0.01SD) | 4 | MKN   | 692    |
| 2                                                                                   | 25 | 0.3   | NaNO3  | lgK:2.06 (0.01SD) | 4 | MSP   | 692    |

| Reaction No. 38529                                  |    |       |                                   |                  |   |       |        |
|-----------------------------------------------------|----|-------|-----------------------------------|------------------|---|-------|--------|
| 2<Thiocarbanilide> + Hg+2 = Hg+2_Thiocarbanilide(2) |    |       |                                   |                  |   |       |        |
| No                                                  | t  | I-Str | Medium                            | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.1   | Ethanol:Water<br>50:50Vol<br>KNO3 | lgK:21.3 (0.1SD) | 5 | MPL   | 1524   |

| Reaction No. 39672            |  |  |  |  |  |  |  |
|-------------------------------|--|--|--|--|--|--|--|
| CPDTA-4 + Ba+2 = Ba+2_CPDTA-4 |  |  |  |  |  |  |  |

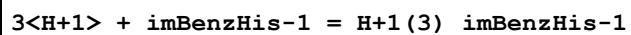
| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl     | lgK:7.75(3SF) | 5 | MGL   | 140    |
| 2  | 20 | 0.1   | Unknown | lgK:7.75(3SF) | 5 | MGL   | 931    |
| 3  | 20 | 1     | Unknown | lgK:8.62(3SF) | 6 | CRV   | 552    |
| 4  | 25 | 0.1   | Unknown | lgK:8.54(3SF) | 6 | CRV   | 552    |
| 5  | 20 | 1     | KCl     | dH:-2.63(3SF) | 5 | MCL   | 2018   |

## Reaction No. 39859



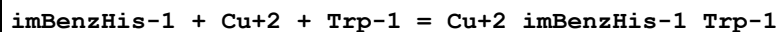
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:14.73(0.003SD) | 5 | MGL   | 1643   |

## Reaction No. 39860



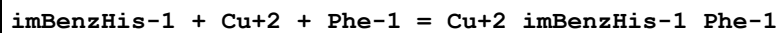
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:16.67(0.02SD) | 5 | MGL   | 1643   |

## Reaction No. 39867



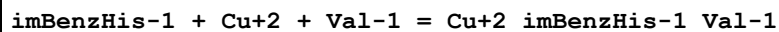
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:18.18(0.009SD) | 6 | MGL   | 1643   |

## Reaction No. 39868



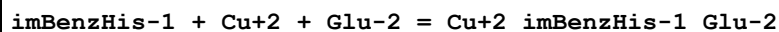
| No | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|--------|---------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:17.716(0.001SD) | 6 | MGL   | 1643   |

## Reaction No. 39869



| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:17.69(0.006SD) | 6 | MGL   | 1643   |

## Reaction No. 39870



| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:18.02(0.006SD) | 6 | MGL   | 1643   |

## Reaction No. 39871

| H+1 + imBenzHis-1 + Cu+2 + GlyPhe-1 = Cu+2_H+1_GlyPhe-1_imBenzHis-1 |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| No                                                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0.1   | KNO3   | lgK:21.8(0.05SD) | 6 | MGL   | 1643   |

| Reaction No. 39872                                        |    |       |        |                  |   |       |        |
|-----------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| imBenzHis-1 + Cu+2 + GlyPhe-1 = Cu+2_GlyPhe-1_imBenzHis-1 |    |       |        |                  |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.1   | KNO3   | lgK:17.1(0.03SD) | 6 | MGL   | 1643   |

| Reaction No. 41561                                    |    |       |                                    |               |   |       |        |
|-------------------------------------------------------|----|-------|------------------------------------|---------------|---|-------|--------|
| NPhenAnthranilic-1 + UO2+2 = UO2+2_NPhenAnthranilic-1 |    |       |                                    |               |   |       |        |
| No                                                    | t  | I-Str | Medium                             | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                     | 30 | 0.1   | Dioxan:Water<br>50:50Vol<br>NaClO4 | lgK:4.95(3SF) | 5 | MGL   | 2991   |

| Reaction No. 41943              |    |       |        |                |   |       |        |
|---------------------------------|----|-------|--------|----------------|---|-------|--------|
| H+1 + NMeDTTA-4 = H+1_NMeDTTA-4 |    |       |        |                |   |       |        |
| No                              | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                               | 20 | 0.1   | NaNO3  | lgK:10.89(4SF) | 5 | MHY   | 140    |

| Reaction No. 41944                     |    |       |        |               |   |       |        |
|----------------------------------------|----|-------|--------|---------------|---|-------|--------|
| H+1_NMeDTTA-4 + H+1 = H+1(2)_NMeDTTA-4 |    |       |        |               |   |       |        |
| No                                     | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                      | 20 | 0.1   | NaNO3  | lgK:7.39(3SF) | 5 | MHY   | 140    |

| Reaction No. 41945                        |    |       |        |               |   |       |        |
|-------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| H+1(2)_NMeDTTA-4 + H+1 = H+1(3)_NMeDTTA-4 |    |       |        |               |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                         | 20 | 0.1   | NaNO3  | lgK:3.65(3SF) | 5 | MHY   | 140    |

| Reaction No. 41946                        |    |       |        |               |   |       |        |
|-------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| H+1(3)_NMeDTTA-4 + H+1 = H+1(4)_NMeDTTA-4 |    |       |        |               |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                         | 20 | 0.1   | NaNO3  | lgK:2.84(3SF) | 5 | MHY   | 140    |

| Reaction No. 41947                |   |       |        |             |   |       |        |
|-----------------------------------|---|-------|--------|-------------|---|-------|--------|
| NMeDTTA-4 + Ba+2 = Ba+2_NMeDTTA-4 |   |       |        |             |   |       |        |
| No                                | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |



|   |    |     |               |                |   |     |     |
|---|----|-----|---------------|----------------|---|-----|-----|
| 1 | 20 | 0.1 | KCl           | lgK:7.21 (3SF) | 5 | MHY | 140 |
| 2 | 25 | 0   | Inf. Dilution | lgK:7.5 (2SF)  | 1 | ESC |     |

| Reaction No. 41948                        |    |       |               |                |   |       |        |
|-------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Ba+2 + H+1_NMeDTTA-4 = Ba+2_H+1_NMeDTTA-4 |    |       |               |                |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                         | 20 | 0.1   | KCl           | lgK:2.61 (3SF) | 5 | MHY   | 140    |
| 2                                         | 25 | 0     | Inf. Dilution | lgK:2.7 (2SF)  | 1 | ESC   |        |

| Reaction No. 41949                |    |       |        |                |   |       |        |
|-----------------------------------|----|-------|--------|----------------|---|-------|--------|
| NMeDTTA-4 + Ca+2 = Ca+2_NMeDTTA-4 |    |       |        |                |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KCl    | lgK:9.60 (3SF) | 5 | MHY   | 140    |

| Reaction No. 41950                        |    |       |        |                |   |       |        |
|-------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Ca+2 + H+1_NMeDTTA-4 = Ca+2_H+1_NMeDTTA-4 |    |       |        |                |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                         | 20 | 0.1   | KCl    | lgK:3.68 (3SF) | 5 | MHY   | 140    |

| Reaction No. 41951                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| NMeDTTA-4 + Cd+2 = Cd+2_NMeDTTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | NaNO3  | lgK:17.44 (4SF) | 5 | MHG   | 140    |

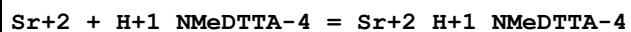
| Reaction No. 41952                |    |       |        |                |   |       |        |
|-----------------------------------|----|-------|--------|----------------|---|-------|--------|
| NMeDTTA-4 + Mg+2 = Mg+2_NMeDTTA-4 |    |       |        |                |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KCl    | lgK:7.31 (3SF) | 5 | MHY   | 140    |

| Reaction No. 41953                        |    |       |        |                |   |       |        |
|-------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Mg+2 + H+1_NMeDTTA-4 = Mg+2_H+1_NMeDTTA-4 |    |       |        |                |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                         | 20 | 0.1   | KCl    | lgK:2.92 (3SF) | 5 | MHY   | 140    |

| Reaction No. 41954                |    |       |        |                |   |       |        |
|-----------------------------------|----|-------|--------|----------------|---|-------|--------|
| NMeDTTA-4 + Sr+2 = Sr+2_NMeDTTA-4 |    |       |        |                |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KCl    | lgK:8.35 (3SF) | 5 | MHY   | 140    |

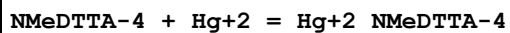
|   |    |   |               |               |   |     |  |
|---|----|---|---------------|---------------|---|-----|--|
| 2 | 25 | 0 | Inf. Dilution | lgK:8.2 (2SF) | 1 | ESC |  |
|---|----|---|---------------|---------------|---|-----|--|

## Reaction No. 41955



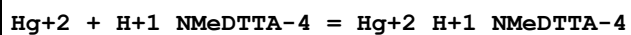
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | KCl           | lgK:3.20 (3SF) | 5 | MHY   | 140    |
| 2  | 25 | 0     | Inf. Dilution | lgK:3.1 (2SF)  | 1 | ESC   |        |

## Reaction No. 41956



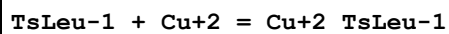
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | NaNO3  | lgK:7.21 (3SF) | 5 | MHG   | 140    |

## Reaction No. 41957



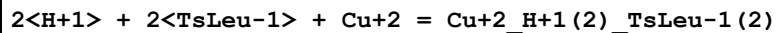
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | NaNO3  | lgK:2.61 (3SF) | 5 | MHG   | 140    |

## Reaction No. 42388



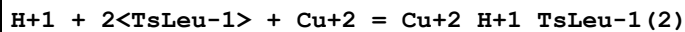
| No | t  | I-Str | Medium                               | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------------------------------------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | Methanol:Water<br>25:75Vol<br>NaClO4 | lgK:9.1 (0.04SD) | 5 | MGL   | 3896   |

## Reaction No. 42464



| No | t  | I-Str | Medium                               | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------------------------------------|--------------------|---|-------|--------|
| 1  | 25 | 0.1   | Methanol:Water<br>25:75Vol<br>NaClO4 | lgK:29.53 (0.02SD) | 5 | MGL   | 3896   |

## Reaction No. 42465



| No | t  | I-Str | Medium                               | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------------------------------------|--------------------|---|-------|--------|
| 1  | 25 | 0.1   | Methanol:Water<br>25:75Vol<br>NaClO4 | lgK:24.51 (0.01SD) | 5 | MGL   | 3896   |

## Reaction No. 42466

| 2<TsLeu-1> + Cu+2 = Cu+2_TsLeu-1(2) |    |       |                                      |                   |   |       |        |
|-------------------------------------|----|-------|--------------------------------------|-------------------|---|-------|--------|
| No                                  | t  | I-Str | Medium                               | Value (Dev)       | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | Methanol:Water<br>25:75Vol<br>NaClO4 | lgK:16.80(0.02SD) | 5 | MGL   | 3896   |

| Reaction No. 42474          |    |       |                                      |                   |   |       |        |
|-----------------------------|----|-------|--------------------------------------|-------------------|---|-------|--------|
| H+1 + TsLeu-1 = H+1_TsLeu-1 |    |       |                                      |                   |   |       |        |
| No                          | t  | I-Str | Medium                               | Value (Dev)       | W | Tech. | Ref. # |
| 1                           | 25 | 0.1   | NaClO4                               | lgK:12.00(0.02SD) | 5 | MGL   | 3896   |
| 2                           | 25 | 0.1   | Methanol:Water<br>25:75Vol<br>NaClO4 | lgK:12.33(0.02SD) | 5 | MGL   | 3896   |

| Reaction No. 42475                |    |       |                                      |                   |   |       |        |
|-----------------------------------|----|-------|--------------------------------------|-------------------|---|-------|--------|
| 2<H+1> + TsLeu-1 = H+1(2)_TsLeu-1 |    |       |                                      |                   |   |       |        |
| No                                | t  | I-Str | Medium                               | Value (Dev)       | W | Tech. | Ref. # |
| 1                                 | 25 | 0.1   | NaClO4                               | lgK:15.47(0.02SD) | 5 | MGL   | 3896   |
| 2                                 | 25 | 0.1   | Methanol:Water<br>25:75Vol<br>NaClO4 | lgK:16.34(0.02SD) | 5 | MGL   | 3896   |

| Reaction No. 45135                              |    |       |             |                 |   |       |        |
|-------------------------------------------------|----|-------|-------------|-----------------|---|-------|--------|
| Co+2_Cl-1(2) + Acridine = Co+2_Acridine_Cl-1(2) |    |       |             |                 |   |       |        |
| No                                              | t  | I-Str | Medium      | Value (Dev)     | W | Tech. | Ref. # |
| 1                                               | 23 |       | Cyclohexane | lgK:1.5(0.03SD) | 4 | MSP   | 906    |
| 2                                               | 23 |       | MeCN        | lgK:1.4(0.06SD) | 4 | MSP   | 906    |

| Reaction No. 45136                              |    |       |             |                  |   |       |        |
|-------------------------------------------------|----|-------|-------------|------------------|---|-------|--------|
| Co+2_Br-1(2) + Acridine = Co+2_Acridine_Br-1(2) |    |       |             |                  |   |       |        |
| No                                              | t  | I-Str | Medium      | Value (Dev)      | W | Tech. | Ref. # |
| 1                                               | 23 |       | Cyclohexane | lgK:1.68(0.02SD) | 4 | MSP   | 906    |

| Reaction No. 45137                          |    |       |                                   |                 |   |       |        |
|---------------------------------------------|----|-------|-----------------------------------|-----------------|---|-------|--------|
| 3<5MePhenanth> + Cd+2 = Cd+2_5MePhenanth(3) |    |       |                                   |                 |   |       |        |
| No                                          | t  | I-Str | Medium                            | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | Ethanol:Water<br>50:50Wgt<br>KNO3 | lgK:15.0(0.1SD) | 4 | MAG   | 906    |

| Reaction No. 45138 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Co+2_P3O10-5 + Pyrid2Azo4DiMeAniline = Co+2_Pyrid2Azo4DiMeAniline_P3O10-5 |    |       |        |                  |   |       |        |
|---------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| No                                                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0.3   | NaNO3  | lgK:3.0(0.07SD)  | 5 | MKN   | 999    |
| 2                                                                         | 25 | 0.3   | NaNO3  | lgK:3.10(0.01SD) | 5 | MSP   | 999    |

| Reaction No. 45139                                              |    |       |         |               |   |       |        |
|-----------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Pyrid2Azo4DiMeAniline + HgCH3+1 = HgCH3+1_Pyrid2Azo4DiMeAniline |    |       |         |               |   |       |        |
| No                                                              | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                               | 20 | 0.1   | Unknown | lgK:5.75(3SF) | 4 | MKN   | 906    |

| Reaction No. 45140                                                        |    |       |        |                |   |       |        |
|---------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Ni+2_P3O10-5 + Pyrid2Azo4DiMeAniline = Ni+2_Pyrid2Azo4DiMeAniline_P3O10-5 |    |       |        |                |   |       |        |
| No                                                                        | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0.3   | NaNO3  | lgK:3.5(0.2SD) | 5 | MKN   | 999    |
| 2                                                                         | 25 | 0.3   | NaNO3  | lgK:3.7(0.2SD) | 5 | MSP   | 999    |

| Reaction No. 45141                                                        |    |       |        |                 |   |       |        |
|---------------------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Zn+2_P3O10-5 + Pyrid2Azo4DiMeAniline = Zn+2_Pyrid2Azo4DiMeAniline_P3O10-5 |    |       |        |                 |   |       |        |
| No                                                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                         | 25 | 0.3   | NaNO3  | lgK:2.4(0.05SD) | 5 | MKN   | 999    |
| 2                                                                         | 25 | 0.3   | NaNO3  | lgK:2.5(0.03SD) | 5 | MSP   | 999    |

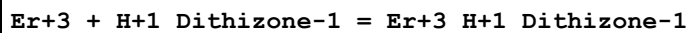
| Reaction No. 45142                         |    |       |                                     |                |   |       |        |
|--------------------------------------------|----|-------|-------------------------------------|----------------|---|-------|--------|
| H+1_Dithizone-1 + H+1 = H+1(2)_Dithizone-1 |    |       |                                     |                |   |       |        |
| No                                         | t  | I-Str | Medium                              | Value (Dev)    | W | Tech. | Ref. # |
| 1                                          | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:9.30(3SF)w | 4 | MPH   | 906    |

| Reaction No. 45143                            |    |       |                                     |                |   |       |        |
|-----------------------------------------------|----|-------|-------------------------------------|----------------|---|-------|--------|
| Ce+3 + H+1_Dithizone-1 = Ce+3_H+1_Dithizone-1 |    |       |                                     |                |   |       |        |
| No                                            | t  | I-Str | Medium                              | Value (Dev)    | W | Tech. | Ref. # |
| 1                                             | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:3.20(3SF)w | 4 | MPH   | 906    |

| Reaction No. 45144                            |   |       |        |             |   |       |        |
|-----------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Dy+3 + H+1_Dithizone-1 = Dy+3_H+1_Dithizone-1 |   |       |        |             |   |       |        |
| No                                            | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

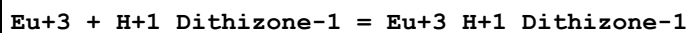
|   |    |     |                                     |                 |   |     |     |
|---|----|-----|-------------------------------------|-----------------|---|-----|-----|
| 1 | 20 | 0.1 | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:3.65 (3SF)w | 4 | MPH | 906 |
|---|----|-----|-------------------------------------|-----------------|---|-----|-----|

## Reaction No. 45145



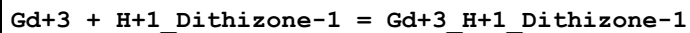
| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:3.80 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45146



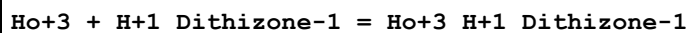
| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:3.65 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45147



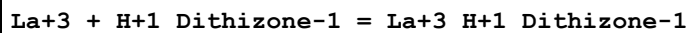
| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:3.45 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45148



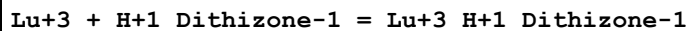
| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:3.70 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45149



| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:3.05 (3SF)w | 4 | MPH   | 906    |

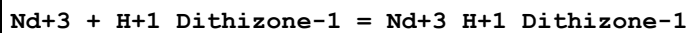
## Reaction No. 45150



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

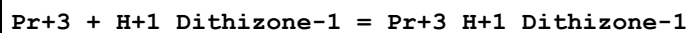
|   |    |     |                                     |                 |   |     |     |
|---|----|-----|-------------------------------------|-----------------|---|-----|-----|
| 1 | 20 | 0.1 | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:3.90 (3SF)w | 4 | MPH | 906 |
|---|----|-----|-------------------------------------|-----------------|---|-----|-----|

## Reaction No. 45151



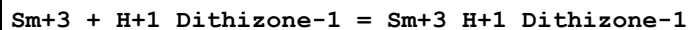
| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:3.40 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45152



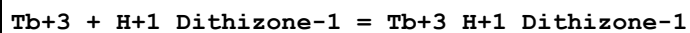
| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:3.30 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45153



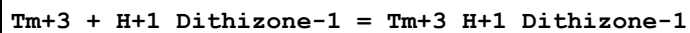
| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:3.50 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45154



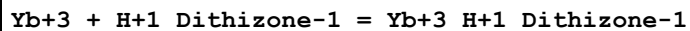
| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:3.55 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45155



| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:3.75 (3SF)w | 4 | MPH   | 906    |

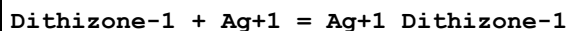
## Reaction No. 45156



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

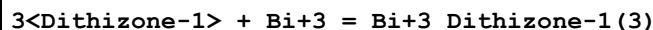
|   |    |     |                                     |                 |   |     |     |
|---|----|-----|-------------------------------------|-----------------|---|-----|-----|
| 1 | 20 | 0.1 | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:3.80 (3SF)w | 4 | MPH | 906 |
|---|----|-----|-------------------------------------|-----------------|---|-----|-----|

## Reaction No. 45157



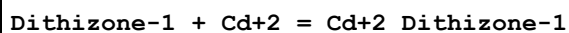
| No | t     | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|-------|-------|--------|-----------------|---|-------|--------|
| 1  | 24:26 | 0.1   | NaClO4 | lgK:7.0 (0.2SD) | 4 | MSP   | 906    |

## Reaction No. 45158



| No | t     | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-------|-------|---------------|-----------------|---|-------|--------|
| 1  | 24:26 | 0.1   | NaClO4        | lgK:32. (0.2SD) | 4 | MSP   | 906    |
| 2  | 25    | 0     | Inf. Dilution | lgK:32.5 (3SF)  | 1 | ESC   |        |

## Reaction No. 45159



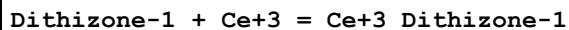
| No | t     | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|-------|-------|--------|-----------------|---|-------|--------|
| 1  | 24:26 | 0.1   | NaClO4 | lgK:7.8 (0.1SD) | 4 | MSP   | 906    |

## Reaction No. 45160



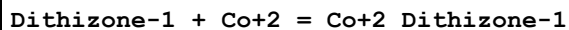
| No | t     | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|-------|-------|--------|------------------|---|-------|--------|
| 1  | 24:26 | 0.1   | NaClO4 | lgK:15.1 (0.2SD) | 4 | MSP   | 906    |

## Reaction No. 45161



| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:1.50 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45162



| No | t     | I-Str | Medium                              | Value (Dev)      | W | Tech. | Ref. # |
|----|-------|-------|-------------------------------------|------------------|---|-------|--------|
| 1  | 24:26 | 0.1   | NaClO4                              | lgK:7.5 (0.09SD) | 4 | MSP   | 906    |
| 2  | 25    | 0.1   | Dioxan:Water<br>50:50Wgt<br>Unknown | lgK:6.43 (3SF)   | 5 | MSP   | 906    |

| Reaction No. 45175                                                     |    |       |        |                    |   |       |        |
|------------------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| $\text{H}+1 + 2\text{Me2PhenNTA-3} = \text{H}+1\_2\text{Me2PhenNTA-3}$ |    |       |        |                    |   |       |        |
| No                                                                     | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                      | 20 | 0.1   | KCl    | lgK:11.07 (0.02SD) | 6 | MGL   | 2591   |

| Reaction No. 45176                                                                    |    |       |        |                   |   |       |        |
|---------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $\text{H}+1\_2\text{Me2PhenNTA-3} + \text{H}+1 = \text{H}+1(2)\_2\text{Me2PhenNTA-3}$ |    |       |        |                   |   |       |        |
| No                                                                                    | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                     | 20 | 0.1   | KCl    | lgK:2.66 (0.02SD) | 6 | MGL   | 2591   |

| Reaction No. 45177                                                                       |    |       |        |                  |   |       |        |
|------------------------------------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| $\text{H}+1(2)\_2\text{Me2PhenNTA-3} + \text{H}+1 = \text{H}+1(3)\_2\text{Me2PhenNTA-3}$ |    |       |        |                  |   |       |        |
| No                                                                                       | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                        | 20 | 0.1   | KCl    | lgK:1.5 (0.05SD) | 4 | MGL   | 2591   |

| Reaction No. 45178                                                       |    |       |        |                   |   |       |        |
|--------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $2\text{Me2PhenNTA-3} + \text{Mg}+2 = \text{Mg}+2\_2\text{Me2PhenNTA-3}$ |    |       |        |                   |   |       |        |
| No                                                                       | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                        | 20 | 0.1   | KCl    | lgK:5.17 (0.02SD) | 6 | MGL   | 2591   |

| Reaction No. 45179                                                       |    |       |        |                   |   |       |        |
|--------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $2\text{Me2PhenNTA-3} + \text{Ca}+2 = \text{Ca}+2\_2\text{Me2PhenNTA-3}$ |    |       |        |                   |   |       |        |
| No                                                                       | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                        | 20 | 0.1   | KCl    | lgK:7.46 (0.01SD) | 6 | MGL   | 2591   |

| Reaction No. 45180                                                       |    |       |               |                   |   |       |        |
|--------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $2\text{Me2PhenNTA-3} + \text{Sr}+2 = \text{Sr}+2\_2\text{Me2PhenNTA-3}$ |    |       |               |                   |   |       |        |
| No                                                                       | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                        | 20 | 0.1   | KCl           | lgK:5.38 (0.02SD) | 6 | MGL   | 2591   |
| 2                                                                        | 25 | 0     | Inf. Dilution | lgK:5.2 (2SF)     | 1 | ESC   |        |

| Reaction No. 45181                                                       |    |       |               |                   |   |       |        |
|--------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $2\text{Me2PhenNTA-3} + \text{Ba}+2 = \text{Ba}+2\_2\text{Me2PhenNTA-3}$ |    |       |               |                   |   |       |        |
| No                                                                       | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                        | 20 | 0.1   | KCl           | lgK:4.93 (0.02SD) | 6 | MGL   | 2591   |
| 2                                                                        | 25 | 0     | Inf. Dilution | lgK:5.1 (2SF)     | 1 | ESC   |        |

| Reaction No. 45182 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|



| 2Me2PhenNTA-3 + Zn+2 = Zn+2_2Me2PhenNTA-3 |    |       |        |                  |   |       |        |
|-------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| No                                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                         | 20 | 0.1   | KCl    | lgK:11.5(0.06SD) | 6 | MGL   | 2591   |

| Reaction No. 45183                    |       |       |        |                |   |       |        |
|---------------------------------------|-------|-------|--------|----------------|---|-------|--------|
| Dithizone-1 + Cu+2 = Cu+2_Dithizone-1 |       |       |        |                |   |       |        |
| No                                    | t     | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 24:26 | 0.1   | NaClO4 | lgK:9.4(0.1SD) | 5 | MSP   | 906    |

| Reaction No. 45184                    |       |       |        |                |   |       |        |
|---------------------------------------|-------|-------|--------|----------------|---|-------|--------|
| Dithizone-1 + Fe+2 = Fe+2_Dithizone-1 |       |       |        |                |   |       |        |
| No                                    | t     | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 24:26 | 0.1   | NaClO4 | lgK:4.8(0.2SD) | 5 | MSP   | 906    |

| Reaction No. 45185                          |       |       |        |                |   |       |        |
|---------------------------------------------|-------|-------|--------|----------------|---|-------|--------|
| 2<Dithizone-1> + Fe+2 = Fe+2_Dithizone-1(2) |       |       |        |                |   |       |        |
| No                                          | t     | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 24:26 | 0.1   | NaClO4 | lgK:9.0(0.1SD) | 5 | MSP   | 906    |

| Reaction No. 45186                    |       |       |        |                 |   |       |        |
|---------------------------------------|-------|-------|--------|-----------------|---|-------|--------|
| Dithizone-1 + Hg+2 = Hg+2_Dithizone-1 |       |       |        |                 |   |       |        |
| No                                    | t     | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                     | 24:26 | 0.1   | NaClO4 | lgK:20.6(0.2SD) | 4 | MSP   | 906    |

| Reaction No. 45187                    |       |       |        |                 |   |       |        |
|---------------------------------------|-------|-------|--------|-----------------|---|-------|--------|
| Dithizone-1 + Mn+2 = Mn+2_Dithizone-1 |       |       |        |                 |   |       |        |
| No                                    | t     | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                     | 24:26 | 0.1   | NaClO4 | lgK:4.9(0.09SD) | 5 | MSP   | 906    |

| Reaction No. 45188                          |       |       |        |                 |   |       |        |
|---------------------------------------------|-------|-------|--------|-----------------|---|-------|--------|
| 2<Dithizone-1> + Mn+2 = Mn+2_Dithizone-1(2) |       |       |        |                 |   |       |        |
| No                                          | t     | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 24:26 | 0.1   | NaClO4 | lgK:9.6(0.07SD) | 5 | MSP   | 906    |

| Reaction No. 45189                          |       |       |        |                  |   |       |        |
|---------------------------------------------|-------|-------|--------|------------------|---|-------|--------|
| 2<Dithizone-1> + Ni+2 = Ni+2_Dithizone-1(2) |       |       |        |                  |   |       |        |
| No                                          | t     | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                           | 24:26 | 0.1   | NaClO4 | lgK:14.2(0.05SD) | 5 | MSP   | 906    |

| Reaction No. 45190                    |       |       |        |                 |   |       |        |
|---------------------------------------|-------|-------|--------|-----------------|---|-------|--------|
| Dithizone-1 + Pb+2 = Pb+2_Dithizone-1 |       |       |        |                 |   |       |        |
| No                                    | t     | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                     | 23    | 0.1   | NaClO4 | lgK:12.46(4SF)  | 0 | MDS   | 906    |
| 2                                     | 24:26 | 0.1   | NaClO4 | lgK:7.3(0.09SD) | 5 | MSP   | 906    |

| Reaction No. 45191                    |       |       |        |                 |   |       |        |
|---------------------------------------|-------|-------|--------|-----------------|---|-------|--------|
| Dithizone-1 + Pd+2 = Pd+2_Dithizone-1 |       |       |        |                 |   |       |        |
| No                                    | t     | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                     | 24:26 | 0.1   | NaClO4 | lgK:11.4(0.1SD) | 5 | MSP   | 906    |

| Reaction No. 45192                          |       |       |        |                  |   |       |        |
|---------------------------------------------|-------|-------|--------|------------------|---|-------|--------|
| 2<Dithizone-1> + Pd+2 = Pd+2_Dithizone-1(2) |       |       |        |                  |   |       |        |
| No                                          | t     | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                           | 24:26 | 0.1   | NaClO4 | lgK:21.8(0.08SD) | 5 | MSP   | 906    |

| Reaction No. 45193                    |       |       |        |                |   |       |        |
|---------------------------------------|-------|-------|--------|----------------|---|-------|--------|
| Dithizone-1 + Sn+2 = Sn+2_Dithizone-1 |       |       |        |                |   |       |        |
| No                                    | t     | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 24:26 | 0.1   | NaClO4 | lgK:6.3(0.2SD) | 5 | MSP   | 906    |

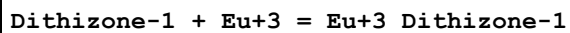
| Reaction No. 45194                          |       |       |        |                 |   |       |        |
|---------------------------------------------|-------|-------|--------|-----------------|---|-------|--------|
| 2<Dithizone-1> + Sn+2 = Sn+2_Dithizone-1(2) |       |       |        |                 |   |       |        |
| No                                          | t     | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 24:26 | 0.1   | NaClO4 | lgK:12.0(0.1SD) | 5 | MSP   | 906    |

| Reaction No. 45195                    |    |       |                                     |                |   |       |        |
|---------------------------------------|----|-------|-------------------------------------|----------------|---|-------|--------|
| Dithizone-1 + Dy+3 = Dy+3_Dithizone-1 |    |       |                                     |                |   |       |        |
| No                                    | t  | I-Str | Medium                              | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:2.05(3SF)w | 4 | MPH   | 906    |

| Reaction No. 45196                    |   |       |        |             |   |       |        |
|---------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Dithizone-1 + Er+3 = Er+3_Dithizone-1 |   |       |        |             |   |       |        |
| No                                    | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

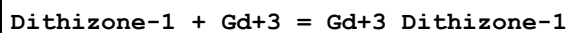
|   |    |     |                                     |                 |   |     |     |
|---|----|-----|-------------------------------------|-----------------|---|-----|-----|
| 1 | 20 | 0.1 | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:2.20 (3SF)w | 4 | MPH | 906 |
|---|----|-----|-------------------------------------|-----------------|---|-----|-----|

## Reaction No. 45197



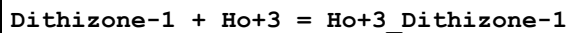
| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:2.00 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45198



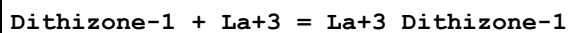
| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:1.80 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45199



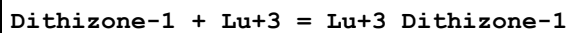
| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:2.15 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45200



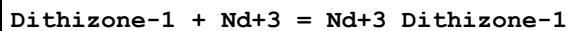
| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:1.40 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45201



| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:2.30 (3SF)w | 4 | MPH   | 906    |

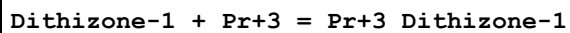
## Reaction No. 45202



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

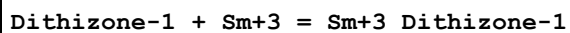
|   |    |     |                                     |                 |   |     |     |
|---|----|-----|-------------------------------------|-----------------|---|-----|-----|
| 1 | 20 | 0.1 | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:1.75 (3SF)w | 4 | MPH | 906 |
|---|----|-----|-------------------------------------|-----------------|---|-----|-----|

## Reaction No. 45203



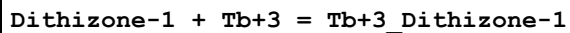
| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:1.65 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45204



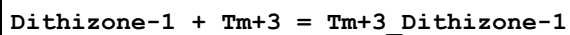
| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:1.90 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45205



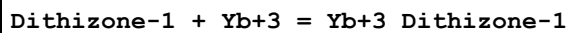
| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:1.95 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45206



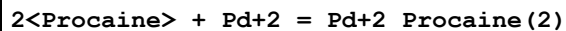
| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:2.25 (3SF)w | 4 | MPH   | 906    |

## Reaction No. 45207



| No | t  | I-Str | Medium                              | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------------------------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | Ethanol:Water<br>50:50Wgt<br>NaClO4 | lgK:2.30 (3SF)w | 4 | MPH   | 906    |

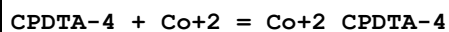
## Reaction No. 45208



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

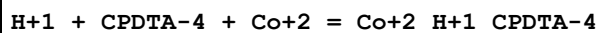
|   |    |  |  |                |   |     |     |
|---|----|--|--|----------------|---|-----|-----|
| 1 | 22 |  |  | lgK:7.88 (3SF) | 4 | MCO | 906 |
|---|----|--|--|----------------|---|-----|-----|

## Reaction No. 45209



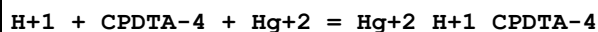
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 20 | 0.1   | NaNO3  | lgK:12.1 (0.06SD) | 4 | MPT   | 5655   |

## Reaction No. 45210



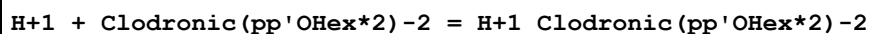
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 20 | 0.1   | NaNO3  | lgK:15.4 (0.04SD) | 4 | MPT   | 5655   |

## Reaction No. 45211



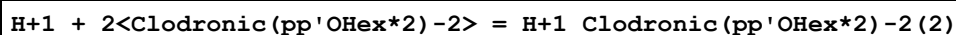
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 20 | 0.1   | NaNO3  | lgK:26.0 (0.08SD) | 4 | MPT   | 5655   |

## Reaction No. 46099



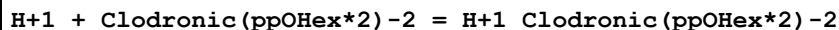
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 1     | Me4NC1 | lgK:2.9 (0.09MD) | 4 | MGL   | 4758   |

## Reaction No. 46100



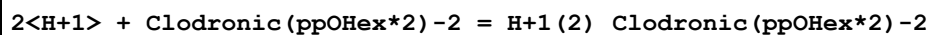
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 1     | Me4NC1 | lgK:7.41 (0.05MD) | 3 | MGL   | 4758   |

## Reaction No. 46101



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 1     | Me4NC1 | lgK:5.93 (0.02MD) | 5 | MGL   | 4758   |

## Reaction No. 46102



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 1     | Me4NC1 | lgK:7.18 (0.02MD) | 5 | MGL   | 4758   |

## Reaction No. 52403

| Ni+2_Dithizone-1(2) + Pyridine = Ni+2_Pyridine_Dithizone-1(2) |    |       |            |                  |   |       |        |
|---------------------------------------------------------------|----|-------|------------|------------------|---|-------|--------|
| No                                                            | t  | I-Str | Medium     | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                             | 19 |       | DiClEthane | lgK:1.344 (4SF)  | 4 | MSP   | 4661   |
| 2                                                             | 30 |       | DiClEthane | lgK:1.267 (4SF)  | 4 | MCL   | 4661   |
| 3                                                             | 19 |       | DiClEthane | dH:-26. (2SF) kJ | 4 | MCL   | 4661   |
| 4                                                             | 30 |       | DiClEthane | dH:-18. (2SF) kJ | 3 | MCL   | 4661   |
| 5                                                             | 30 |       | MeEtKetone | lgK:1.137 (4SF)  | 4 | MCL   | 4661   |
| 6                                                             | 30 |       | MeEtKetone | dH:-15. (2SF) kJ | 4 | MCL   | 4661   |

| Reaction No. 52404                                                        |    |       |            |                 |   |       |        |
|---------------------------------------------------------------------------|----|-------|------------|-----------------|---|-------|--------|
| Ni+2_Pyridine_Dithizone-1(2) + Pyridine = Ni+2_Pyridine(2)_Dithizone-1(2) |    |       |            |                 |   |       |        |
| No                                                                        | t  | I-Str | Medium     | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                         | 19 |       | DiClEthane | lgK:0.644 (3SF) | 4 | MSP   | 4661   |
| 2                                                                         | 19 |       | DiClEthane | dH:7. (1SF) kJ  | 4 | MCL   | 4661   |

| Reaction No. 52574                                                  |    |       |            |                  |   |       |        |
|---------------------------------------------------------------------|----|-------|------------|------------------|---|-------|--------|
| Ni+2_Dithizone-1(2) + 4MePyridine = Ni+2_4MePyridine_Dithizone-1(2) |    |       |            |                  |   |       |        |
| No                                                                  | t  | I-Str | Medium     | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                   | 19 |       | DiClEthane | lgK:1.544 (4SF)  | 4 | MCL   | 4661   |
| 2                                                                   | 30 |       | DiClEthane | lgK:1.544 (4SF)  | 4 | MCL   | 4661   |
| 3                                                                   | 19 |       | DiClEthane | dH:-26. (2SF) kJ | 4 | MCL   | 4661   |
| 4                                                                   | 30 |       | DiClEthane | dH:-22. (2SF) kJ | 4 | MCL   | 4661   |

| Reaction No. 61479                            |    |       |        |              |   |       |        |
|-----------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| H+1(3)_[18]22225N5 + H+1 = H+1(4)_[18]22225N5 |    |       |        |              |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                             | 35 | 0.2   | NaClO4 | lgK:2. (1SF) | 3 | MPT   | 3840   |

| Reaction No. 61480                            |    |       |        |              |   |       |        |
|-----------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| H+1(4)_[18]22225N5 + H+1 = H+1(5)_[18]22225N5 |    |       |        |              |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                             | 35 | 0.2   | NaClO4 | lgK:2. (1SF) | 3 | MPT   | 3840   |

| Reaction No. 62030                            |    |       |         |                |   |       |        |
|-----------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Ni+2 + H+1_imBenzHis-1 = Ni+2_H+1_imBenzHis-1 |    |       |         |                |   |       |        |
| No                                            | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | Unknown | lgK:2.32 (3SF) | 6 | CRV   | 552    |

| Reaction No. 62031                                                                                    |    |       |         |               |   |       |        |
|-------------------------------------------------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| $\text{Ni}+2_{\text{imBenzHis-1(2)}} + \text{H}+1 = \text{Ni}+2_{\text{H}+1_{\text{imBenzHis-1(2)}}}$ |    |       |         |               |   |       |        |
| No                                                                                                    | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                     | 25 | 0.1   | Unknown | lgK:4.07(3SF) | 6 | CRV   | 552    |

| Reaction No. 62032                                                                              |    |       |         |               |   |       |        |
|-------------------------------------------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| $\text{Cu}+2 + \text{H}+1_{\text{imBenzHis-1}} = \text{Cu}+2_{\text{H}+1_{\text{imBenzHis-1}}}$ |    |       |         |               |   |       |        |
| No                                                                                              | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                               | 25 | 0.1   | Unknown | lgK:4.53(3SF) | 6 | CRV   | 552    |

| Reaction No. 62033                                                                                       |    |       |         |               |   |       |        |
|----------------------------------------------------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| $\text{Cu}+2 + 2<\text{H}+1_{\text{imBenzHis-1}}> = \text{Cu}+2_{\text{H}+1(2)_{\text{imBenzHis-1(2)}}}$ |    |       |         |               |   |       |        |
| No                                                                                                       | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                        | 25 | 0.1   | Unknown | lgK:8.63(3SF) | 6 | CRV   | 552    |

| Reaction No. 62034                                                                                    |    |       |         |               |   |       |        |
|-------------------------------------------------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| $\text{Cu}+2_{\text{imBenzHis-1(2)}} + \text{H}+1 = \text{Cu}+2_{\text{H}+1_{\text{imBenzHis-1(2)}}}$ |    |       |         |               |   |       |        |
| No                                                                                                    | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                                     | 25 | 0.1   | Unknown | lgK:5.10(3SF) | 6 | CRV   | 552    |

| Reaction No. 62035                                                    |    |       |         |              |   |       |        |
|-----------------------------------------------------------------------|----|-------|---------|--------------|---|-------|--------|
| $\text{imBenzHis-1} + \text{Ag}+1 = \text{Ag}+1_{\text{imBenzHis-1}}$ |    |       |         |              |   |       |        |
| No                                                                    | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                     | 25 | 0.1   | Unknown | lgK:5.1(2SF) | 6 | CRV   | 552    |

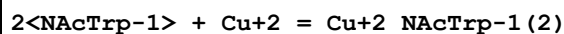
| Reaction No. 62036                                                                              |    |       |         |               |   |       |        |
|-------------------------------------------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| $\text{Ag}+1 + \text{H}+1_{\text{imBenzHis-1}} = \text{Ag}+1_{\text{H}+1_{\text{imBenzHis-1}}}$ |    |       |         |               |   |       |        |
| No                                                                                              | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                               | 25 | 0.1   | Unknown | lgK:3.32(3SF) | 6 | CRV   | 552    |

| Reaction No. 62037                                                                                       |    |       |         |              |   |       |        |
|----------------------------------------------------------------------------------------------------------|----|-------|---------|--------------|---|-------|--------|
| $\text{Ag}+1 + 2<\text{H}+1_{\text{imBenzHis-1}}> = \text{Ag}+1_{\text{H}+1(2)_{\text{imBenzHis-1(2)}}}$ |    |       |         |              |   |       |        |
| No                                                                                                       | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                                                        | 25 | 0.1   | Unknown | lgK:5.6(2SF) | 6 | CRV   | 552    |

| Reaction No. 62038                                                                |  |  |  |  |  |  |  |
|-----------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $2<\text{imBenzHis-1}> + 2<\text{Ag}+1> = \text{Ag}+1(2)_{\text{imBenzHis-1(2)}}$ |  |  |  |  |  |  |  |

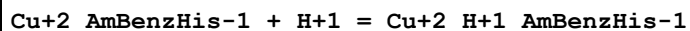
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:15.27 (4SF) | 6 | CRV   | 552    |

## Reaction No. 62152



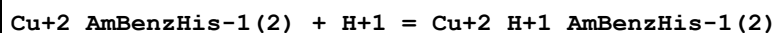
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 20 | 0.5   | Unknown | lgK:12.20 (4SF) | 6 | CRV   | 552    |

## Reaction No. 62156



| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:3.93 (3SF) | 6 | CRV   | 552    |

## Reaction No. 62157



| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:5.28 (3SF) | 4 | CRV   | 552    |
| 2  | 25 | 0.1   | Unknown | lgK:5.33 (3SF) | 6 | CRV   | 552    |

## Reaction No. 62856



| No | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|----------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:9.84 (3SF) | 4 | CRV   | 552    |

## Reaction No. 62857



| No | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|----------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:8.97 (3SF) | 0 | CRV   | 552    |

Note (1): Thermodynamically isolated

## Reaction No. 62858



| No | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|----------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | K+1 salt | lgK:9.01 (3SF) | 0 | CRV   | 552    |

Note (1): Thermodynamically isolated

## Reaction No. 62859





| No                                   | t  | I-Str | Medium   | Value (Dev)   | W | Tech. | Ref. # |
|--------------------------------------|----|-------|----------|---------------|---|-------|--------|
| 1                                    | 25 | 0.1   | K+1 salt | lgK:9.03(3SF) | 0 | CRV   | 552    |
| Note (1): Thermodynamically isolated |    |       |          |               |   |       |        |

| Reaction No. 66139                |    |       |        |                |   |       |        |
|-----------------------------------|----|-------|--------|----------------|---|-------|--------|
| 1PrEDTA-4 + Mg+2 = Mg+2_1PrEDTA-4 |    |       |        |                |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:10.15(4SF) | 6 | MPL   | 2011   |

| Reaction No. 66140                |    |       |               |               |   |       |        |
|-----------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 1PrEDTA-4 + Sr+2 = Sr+2_1PrEDTA-4 |    |       |               |               |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3          | lgK:9.68(3SF) | 6 | MPL   | 2011   |
| 2                                 | 25 | 0     | Inf. Dilution | lgK:9.8(2SF)  | 1 | ESC   |        |

| Reaction No. 66141                |    |       |               |               |   |       |        |
|-----------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 1PrEDTA-4 + Ba+2 = Ba+2_1PrEDTA-4 |    |       |               |               |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3          | lgK:8.50(3SF) | 6 | MPL   | 2011   |
| 2                                 | 25 | 0     | Inf. Dilution | lgK:8.8(2SF)  | 1 | ESC   |        |

| Reaction No. 66142                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 1PrEDTA-4 + La+3 = La+3_1PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:16.6(0.2SD) | 6 | MPL   | 1994   |

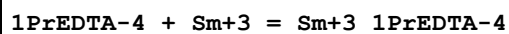
| Reaction No. 66143                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 1PrEDTA-4 + Ce+3 = Ce+3_1PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:17.1(0.1SD) | 6 | CRV   | 552    |

| Reaction No. 66144                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 1PrEDTA-4 + Pr+3 = Pr+3_1PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:17.5(0.1SD) | 6 | MPL   | 1994   |

| Reaction No. 66145                |  |  |  |  |  |  |  |
|-----------------------------------|--|--|--|--|--|--|--|
| 1PrEDTA-4 + Nd+3 = Nd+3_1PrEDTA-4 |  |  |  |  |  |  |  |

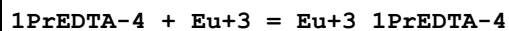
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:17.8(0.1SD) | 6 | MPL   | 1994   |

## Reaction No. 66146



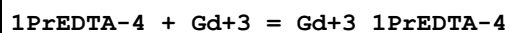
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:18.3(0.1SD) | 6 | MPL   | 1994   |

## Reaction No. 66147



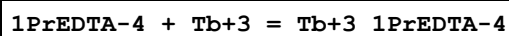
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:18.4(0.1SD) | 6 | MPL   | 1994   |

## Reaction No. 66148



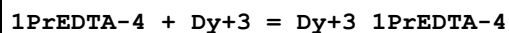
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:18.5(0.1SD) | 6 | MPL   | 1994   |

## Reaction No. 66149



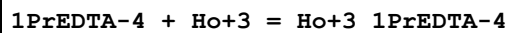
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:19.0(0.1SD) | 6 | MPL   | 1994   |

## Reaction No. 66150



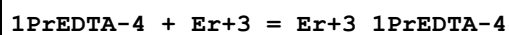
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:19.5(0.1SD) | 6 | MPL   | 1994   |

## Reaction No. 66151



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:19.8(0.1SD) | 6 | MPL   | 1994   |

## Reaction No. 66152



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:20.1(0.1SD) | 6 | MPL   | 1994   |

| Reaction No. 66153                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 1PrEDTA-4 + Tm+3 = Tm+3_1PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:20.5(0.2SD) | 6 | MPL   | 1994   |

| Reaction No. 66154                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 1PrEDTA-4 + Yb+3 = Yb+3_1PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:20.8(0.2SD) | 6 | MPL   | 1994   |

| Reaction No. 66155                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 1PrEDTA-4 + Lu+3 = Lu+3_1PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:21.0(0.2SD) | 6 | MPL   | 1994   |

| Reaction No. 66156                |    |       |        |                 |   |       |        |
|-----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 1PrEDTA-4 + Mn+2 = Mn+2_1PrEDTA-4 |    |       |        |                 |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:15.6(0.1SD) | 6 | MPL   | 1994   |

| Reaction No. 66157                |    |       |        |                  |   |       |        |
|-----------------------------------|----|-------|--------|------------------|---|-------|--------|
| 1PrEDTA-4 + Co+2 = Co+2_1PrEDTA-4 |    |       |        |                  |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | KNO3   | lgK:18.0(0.09SD) | 6 | MPL   | 1994   |

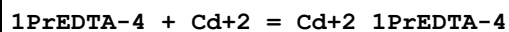
| Reaction No. 66158                |    |       |         |               |   |       |        |
|-----------------------------------|----|-------|---------|---------------|---|-------|--------|
| 1PrEDTA-4 + Ag+1 = Ag+1_1PrEDTA-4 |    |       |         |               |   |       |        |
| No                                | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | Unknown | lgK:8.05(3SF) | 6 | CRV   | 552    |

| Reaction No. 66159                        |    |       |         |               |   |       |        |
|-------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Ag+1_1PrEDTA-4 + H+1 = Ag+1_H+1_1PrEDTA-4 |    |       |         |               |   |       |        |
| No                                        | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                         | 20 | 0.1   | Unknown | lgK:5.80(3SF) | 6 | CRV   | 552    |

| Reaction No. 66160                |   |       |        |             |   |       |        |
|-----------------------------------|---|-------|--------|-------------|---|-------|--------|
| 1PrEDTA-4 + Zn+2 = Zn+2_1PrEDTA-4 |   |       |        |             |   |       |        |
| No                                | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

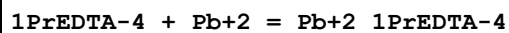
|   |    |     |      |                  |   |     |      |
|---|----|-----|------|------------------|---|-----|------|
| 1 | 20 | 0.1 | KNO3 | lgK:18.2(0.09SD) | 6 | MPL | 1994 |
|---|----|-----|------|------------------|---|-----|------|

## Reaction No. 66161



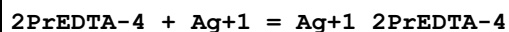
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:18.05(4SF) | 6 | MPL   | 2011   |

## Reaction No. 66162



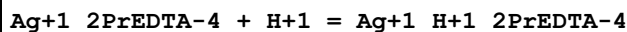
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 20 | 0.1   | KNO3   | lgK:19.3(0.09SD) | 6 | MPL   | 1994   |

## Reaction No. 66163



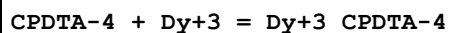
| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown | lgK:8.13(3SF) | 6 | CRV   | 552    |

## Reaction No. 66164



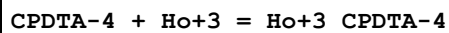
| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | Unknown | lgK:5.84(3SF) | 6 | CRV   | 552    |

## Reaction No. 66167



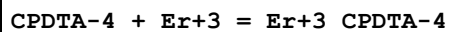
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:18.95(4SF) | 6 | CRV   | 552    |
| 2  | 20 | 0.1   | Unknown | dH:0.9(1SF)    | 6 | CRV   | 552    |

## Reaction No. 66168



| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:19.26(4SF) | 6 | CRV   | 552    |
| 2  | 20 | 0.1   | Unknown | dH:1.6(2SF)    | 6 | CRV   | 552    |

## Reaction No. 66169



| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:19.51(4SF) | 6 | CRV   | 552    |

|   |    |     |         |              |   |     |     |
|---|----|-----|---------|--------------|---|-----|-----|
| 2 | 20 | 0.1 | Unknown | dH:1.2 (2SF) | 6 | CRV | 552 |
|---|----|-----|---------|--------------|---|-----|-----|

| Reaction No. 66170            |    |       |         |                 |   |       |        |
|-------------------------------|----|-------|---------|-----------------|---|-------|--------|
| CPDTA-4 + Tm+3 = Tm+3_CPDTA-4 |    |       |         |                 |   |       |        |
| No                            | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                             | 25 | 0.1   | Unknown | lgK:19.71 (4SF) | 6 | CRV   | 552    |
| 2                             | 20 | 0.1   | Unknown | dH:-0.3 (1SF)   | 6 | CRV   | 552    |

| Reaction No. 66171            |    |       |         |                 |   |       |        |
|-------------------------------|----|-------|---------|-----------------|---|-------|--------|
| CPDTA-4 + Yb+3 = Yb+3_CPDTA-4 |    |       |         |                 |   |       |        |
| No                            | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                             | 25 | 0.1   | Unknown | lgK:19.93 (4SF) | 6 | CRV   | 552    |
| 2                             | 20 | 0.1   | Unknown | dH:-1.9 (2SF)   | 6 | CRV   | 552    |

| Reaction No. 66172            |    |       |         |                 |   |       |        |
|-------------------------------|----|-------|---------|-----------------|---|-------|--------|
| CPDTA-4 + Lu+3 = Lu+3_CPDTA-4 |    |       |         |                 |   |       |        |
| No                            | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                             | 25 | 0.1   | Unknown | lgK:20.20 (4SF) | 6 | CRV   | 552    |
| 2                             | 20 | 0.1   | Unknown | dH:0.2 (1SF)    | 6 | CRV   | 552    |

| Reaction No. 66173                    |    |       |         |                |   |       |        |
|---------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Co+2_CPDTA-4 + H+1 = Co+2_H+1_CPDTA-4 |    |       |         |                |   |       |        |
| No                                    | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 20 | 0.1   | Unknown | lgK:3.37 (3SF) | 4 | CRV   | 552    |

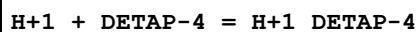
| Reaction No. 66174            |    |       |         |                 |   |       |        |
|-------------------------------|----|-------|---------|-----------------|---|-------|--------|
| CPDTA-4 + Cd+2 = Cd+2_CPDTA-4 |    |       |         |                 |   |       |        |
| No                            | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                             | 20 | 0.1   | Unknown | lgK:18.25 (4SF) | 4 | CRV   | 552    |

| Reaction No. 66175                    |    |       |         |                |   |       |        |
|---------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Hg+2_CPDTA-4 + H+1 = Hg+2_H+1_CPDTA-4 |    |       |         |                |   |       |        |
| No                                    | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 20 | 0.1   | Unknown | lgK:4. (0.9SD) | 4 | CRV   | 552    |

| Reaction No. 66176            |   |       |        |             |   |       |        |
|-------------------------------|---|-------|--------|-------------|---|-------|--------|
| CPDTA-4 + Pb+2 = Pb+2_CPDTA-4 |   |       |        |             |   |       |        |
| No                            | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

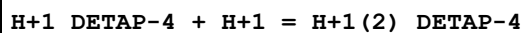
|   |    |     |         |                 |   |     |     |
|---|----|-----|---------|-----------------|---|-----|-----|
| 1 | 20 | 0.1 | Unknown | lgK:18.88 (4SF) | 4 | CRV | 552 |
|---|----|-----|---------|-----------------|---|-----|-----|

## Reaction No. 67525



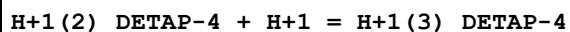
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:9.55 (3SF) | 5 | MGL   | 6529   |

## Reaction No. 67526



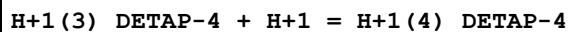
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:8.98 (3SF) | 5 | MGL   | 6529   |

## Reaction No. 67527



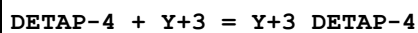
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:3.86 (3SF) | 5 | MGL   | 6529   |

## Reaction No. 67528



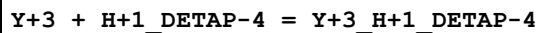
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:2.63 (3SF) | 5 | MGL   | 6529   |

## Reaction No. 67529



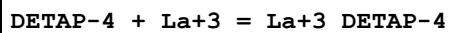
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:14.28 (4SF) | 5 | MGL   | 6529   |

## Reaction No. 67530



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:8.77 (3SF) | 5 | MGL   | 6529   |

## Reaction No. 67531



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:13.31 (4SF) | 5 | MGL   | 6529   |

## Reaction No. 67532

| La+3 + H+1_DETAP-4 = La+3_H+1_DETAP-4 |    |       |        |               |   |       |        |
|---------------------------------------|----|-------|--------|---------------|---|-------|--------|
| No                                    | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KNO3   | lgK:8.36(3SF) | 5 | MGL   | 6529   |

| Reaction No. 67533            |    |       |        |                |   |       |        |
|-------------------------------|----|-------|--------|----------------|---|-------|--------|
| DETAP-4 + Ce+3 = Ce+3_DETAP-4 |    |       |        |                |   |       |        |
| No                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                             | 25 | 0.1   | KNO3   | lgK:14.26(4SF) | 5 | MGL   | 6529   |

| Reaction No. 67534                    |    |       |        |               |   |       |        |
|---------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Ce+3 + H+1_DETAP-4 = Ce+3_H+1_DETAP-4 |    |       |        |               |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KNO3   | lgK:8.89(3SF) | 5 | MGL   | 6529   |

| Reaction No. 67535            |    |       |        |                |   |       |        |
|-------------------------------|----|-------|--------|----------------|---|-------|--------|
| DETAP-4 + Pr+3 = Pr+3_DETAP-4 |    |       |        |                |   |       |        |
| No                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                             | 25 | 0.1   | KNO3   | lgK:14.61(4SF) | 5 | MGL   | 6529   |

| Reaction No. 67536                    |    |       |        |               |   |       |        |
|---------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Pr+3 + H+1_DETAP-4 = Pr+3_H+1_DETAP-4 |    |       |        |               |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KNO3   | lgK:9.21(3SF) | 5 | MGL   | 6529   |

| Reaction No. 67537            |    |       |        |                |   |       |        |
|-------------------------------|----|-------|--------|----------------|---|-------|--------|
| DETAP-4 + Nd+3 = Nd+3_DETAP-4 |    |       |        |                |   |       |        |
| No                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                             | 25 | 0.1   | KNO3   | lgK:14.80(4SF) | 5 | MGL   | 6529   |

| Reaction No. 67538                    |    |       |        |               |   |       |        |
|---------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Nd+3 + H+1_DETAP-4 = Nd+3_H+1_DETAP-4 |    |       |        |               |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KNO3   | lgK:9.40(3SF) | 5 | MGL   | 6529   |

| Reaction No. 67539            |    |       |        |               |   |       |        |
|-------------------------------|----|-------|--------|---------------|---|-------|--------|
| DETAP-4 + Pm+3 = Pm+3_DETAP-4 |    |       |        |               |   |       |        |
| No                            | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                             | 25 | 0.1   | KNO3   | lgK:15.1(3SF) | 3 | MGL   | 6529   |

| Reaction No. 67540                                                                      |    |       |        |               |   |       |        |
|-----------------------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $\text{Pm}+3 + \text{H}+1\_ \text{DETAP}-4 = \text{Pm}+3\_ \text{H}+1\_ \text{DETAP}-4$ |    |       |        |               |   |       |        |
| No                                                                                      | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                                       | 25 | 0.1   | KNO3   | lgK:9.5 (2SF) | 3 | MGL   | 6529   |

| Reaction No. 67541                                            |    |       |        |                 |   |       |        |
|---------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{DETAP}-4 + \text{Sm}+3 = \text{Sm}+3\_ \text{DETAP}-4$ |    |       |        |                 |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.1   | KNO3   | lgK:15.29 (4SF) | 5 | MGL   | 6529   |

| Reaction No. 67542                                                                      |    |       |        |                |   |       |        |
|-----------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Sm}+3 + \text{H}+1\_ \text{DETAP}-4 = \text{Sm}+3\_ \text{H}+1\_ \text{DETAP}-4$ |    |       |        |                |   |       |        |
| No                                                                                      | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                       | 25 | 0.1   | KNO3   | lgK:9.59 (3SF) | 5 | MGL   | 6529   |

| Reaction No. 67543                                            |    |       |        |                 |   |       |        |
|---------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{DETAP}-4 + \text{Eu}+3 = \text{Eu}+3\_ \text{DETAP}-4$ |    |       |        |                 |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.1   | KNO3   | lgK:15.28 (4SF) | 5 | MGL   | 6529   |

| Reaction No. 67544                                                                      |    |       |        |                |   |       |        |
|-----------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Eu}+3 + \text{H}+1\_ \text{DETAP}-4 = \text{Eu}+3\_ \text{H}+1\_ \text{DETAP}-4$ |    |       |        |                |   |       |        |
| No                                                                                      | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                       | 25 | 0.1   | KNO3   | lgK:9.50 (3SF) | 5 | MGL   | 6529   |

| Reaction No. 67545                                            |    |       |        |                 |   |       |        |
|---------------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| $\text{DETAP}-4 + \text{Gd}+3 = \text{Gd}+3\_ \text{DETAP}-4$ |    |       |        |                 |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                             | 25 | 0.1   | KNO3   | lgK:14.21 (4SF) | 5 | MGL   | 6529   |

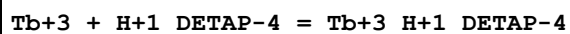
| Reaction No. 67546                                                                      |    |       |        |                |   |       |        |
|-----------------------------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Gd}+3 + \text{H}+1\_ \text{DETAP}-4 = \text{Gd}+3\_ \text{H}+1\_ \text{DETAP}-4$ |    |       |        |                |   |       |        |
| No                                                                                      | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                       | 25 | 0.1   | KNO3   | lgK:9.32 (3SF) | 5 | MGL   | 6529   |

| Reaction No. 67547                                            |  |  |  |  |  |  |  |
|---------------------------------------------------------------|--|--|--|--|--|--|--|
| $\text{DETAP}-4 + \text{Tb}+3 = \text{Tb}+3\_ \text{DETAP}-4$ |  |  |  |  |  |  |  |



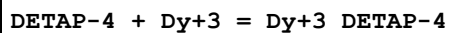
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:15.16 (4SF) | 5 | MGL   | 6529   |

## Reaction No. 67548



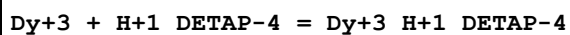
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:9.43 (3SF) | 5 | MGL   | 6529   |

## Reaction No. 67549



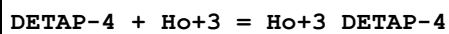
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:15.11 (4SF) | 5 | MGL   | 6529   |

## Reaction No. 67550



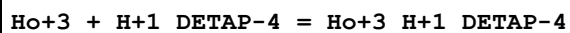
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:9.31 (3SF) | 5 | MGL   | 6529   |

## Reaction No. 67551



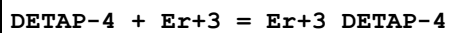
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:15.01 (4SF) | 5 | MGL   | 6529   |

## Reaction No. 67552



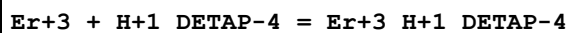
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:9.06 (3SF) | 5 | MGL   | 6529   |

## Reaction No. 67553



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:14.90 (4SF) | 5 | MGL   | 6529   |

## Reaction No. 67554



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:9.24 (3SF) | 5 | MGL   | 6529   |

| Reaction No. 67555            |    |       |        |                 |   |       |        |
|-------------------------------|----|-------|--------|-----------------|---|-------|--------|
| DETAP-4 + Tm+3 = Tm+3_DETAP-4 |    |       |        |                 |   |       |        |
| No                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                             | 25 | 0.1   | KNO3   | lgK:14.97 (4SF) | 5 | MGL   | 6529   |

| Reaction No. 67556                    |    |       |        |                |   |       |        |
|---------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Tm+3 + H+1_DETAP-4 = Tm+3_H+1_DETAP-4 |    |       |        |                |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KNO3   | lgK:9.29 (3SF) | 5 | MGL   | 6529   |

| Reaction No. 67557            |    |       |        |                 |   |       |        |
|-------------------------------|----|-------|--------|-----------------|---|-------|--------|
| DETAP-4 + Yb+3 = Yb+3_DETAP-4 |    |       |        |                 |   |       |        |
| No                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                             | 25 | 0.1   | KNO3   | lgK:15.17 (4SF) | 5 | MGL   | 6529   |

| Reaction No. 67558                    |    |       |        |                |   |       |        |
|---------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Yb+3 + H+1_DETAP-4 = Yb+3_H+1_DETAP-4 |    |       |        |                |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KNO3   | lgK:9.38 (3SF) | 5 | MGL   | 6529   |

| Reaction No. 67559            |    |       |        |                 |   |       |        |
|-------------------------------|----|-------|--------|-----------------|---|-------|--------|
| DETAP-4 + Lu+3 = Lu+3_DETAP-4 |    |       |        |                 |   |       |        |
| No                            | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                             | 25 | 0.1   | KNO3   | lgK:15.14 (4SF) | 5 | MGL   | 6529   |

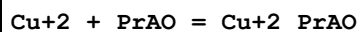
| Reaction No. 67560                    |    |       |        |                |   |       |        |
|---------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Lu+3 + H+1_DETAP-4 = Lu+3_H+1_DETAP-4 |    |       |        |                |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KNO3   | lgK:9.40 (3SF) | 5 | MGL   | 6529   |

| Reaction No. 73663    |    |       |        |                    |   |       |        |
|-----------------------|----|-------|--------|--------------------|---|-------|--------|
| H+1 + PrAO = H+1_PrAO |    |       |        |                    |   |       |        |
| No                    | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                     | 25 | 0.15  | NaCl   | lgK:8.97 (0.001SD) | 7 | MGL   | 18008  |

| Reaction No. 73664          |   |       |        |             |   |       |        |
|-----------------------------|---|-------|--------|-------------|---|-------|--------|
| 2<H+1> + PrAO = H+1(2)_PrAO |   |       |        |             |   |       |        |
| No                          | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

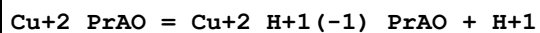
|   |    |      |      |                    |   |     |       |
|---|----|------|------|--------------------|---|-----|-------|
| 1 | 25 | 0.15 | NaCl | lgK:16.15(0.002SD) | 7 | MGL | 18008 |
|---|----|------|------|--------------------|---|-----|-------|

## Reaction No. 73665



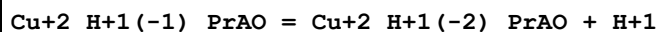
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaCl   | lgK:14.22(0.010SD) | 7 | MGL   | 18008  |

## Reaction No. 73667



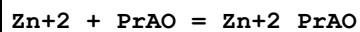
| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaCl   | lgK:11.35(0.008SD) | 7 | MGL   | 18008  |

## Reaction No. 73668



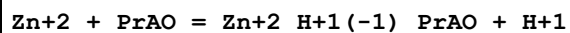
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaCl   | lgK:0.22(0.024SD) | 7 | MGL   | 18008  |

## Reaction No. 73669



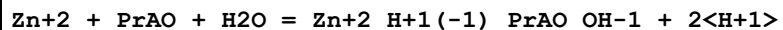
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaCl   | lgK:5.01(0.070SD) | 7 | MGL   | 18008  |

## Reaction No. 73670



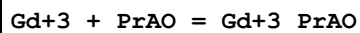
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaCl   | lgK:0.26(0.002SD) | 7 | MGL   | 18008  |

## Reaction No. 73671



| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaCl   | lgK:-8.79(0.006SD) | 7 | MGL   | 18008  |

## Reaction No. 73672



| No | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|--------|--------------------|---|-------|--------|
| 1  | 25 | 0.15  | NaCl   | lgK:15.53(0.030SD) | 7 | MGL   | 18008  |

## Reaction No. 73673

| Gd+3 + PrAO = Gd+3_H+1(-1)_PrAO + H+1 |    |       |        |                   |   |       |        |
|---------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| No                                    | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                     | 25 | 0.15  | NaCl   | lgK:7.64(0.033SD) | 7 | MGL   | 18008  |

| Reaction No. 73674                                  |    |       |        |                   |   |       |        |
|-----------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Gd+3 + PrAO + H2O = Gd+3_H+1(-1)_PrAO_OH-1 + 2<H+1> |    |       |        |                   |   |       |        |
| No                                                  | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                   | 25 | 0.15  | NaCl   | lgK:0.27(0.006SD) | 7 | MGL   | 18008  |

| Reaction No. 73686      |    |       |        |                     |   |       |        |
|-------------------------|----|-------|--------|---------------------|---|-------|--------|
| Ni+2 + PrAO = Ni+2_PrAO |    |       |        |                     |   |       |        |
| No                      | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                       | 25 | 0.15  | NaCl   | lgK:11.530(0.060SD) | 7 | MGL   |        |

| Reaction No. 73687                    |    |       |        |                    |   |       |        |
|---------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Ni+2 + PrAO = Ni+2_H+1(-1)_PrAO + H+1 |    |       |        |                    |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                     | 25 | 0.15  | NaCl   | lgK:7.621(0.032SD) | 7 | MGL   |        |

| Reaction No. 73688                       |    |       |        |                     |   |       |        |
|------------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| Ni+2 + PrAO = Ni+2_H+1(-2)_PrAO + 2<H+1> |    |       |        |                     |   |       |        |
| No                                       | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                        | 25 | 0.15  | NaCl   | lgK:-2.201(0.030SD) | 7 | MGL   |        |

## Data Assessment

| Weight | Assessment                                    |
|--------|-----------------------------------------------|
| 9      | highest accuracy                              |
| 8      | multi-determined; excellent dependability     |
| 7      | trustworthy; outstanding single determination |
| 6      | better than average reliability               |
| 5      | average single determination                  |
| 4      | some doubt                                    |
| 3      | poorly known                                  |
| 2      | guess to achieve consistency (or just as bad) |
| 1      | no information but consistent                 |

|   |                          |
|---|--------------------------|
| 0 | problematic/inconsistent |
|---|--------------------------|

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## Techniques

| Abbrev | Method                               |
|--------|--------------------------------------|
| CRV    | Critical review                      |
| EDH    | Debye-Hueckel corrections            |
| EES    | Personal estimate                    |
| EFE    | Free Energy relations                |
| EIU    | Critical IUPAC review (estim.)       |
| ELC    | Linear Combination of Reactions      |
| ENS    | Not specified                        |
| EOD    | Calculated from data of others       |
| ESC    | Standard conditions anchor           |
| MAG    | Silver electrode e.m.f.              |
| MCL    | Calorimetry                          |
| MCO    | Colorimetry (often for measuring pH) |
| MDS    | Distribution between two phases      |
| MFL    | Fluorescence                         |
| MGL    | Glass electrode                      |
| MHG    | Emf with amalgam electrode           |
| MHY    | Emf with hydrogen electrode          |
| MIS    | Ion selective electrode              |
| MIX    | Ion exchange                         |
| MKN    | Rate of reaction                     |
| MMR    | Nuclear magnetic resonance           |
| MMX    | Mixed techniques                     |
| MPH    | pH method, not specified             |
| MPL    | Polarography                         |
| MPS    | Potentiometry and Spectrophotometry  |
| MPT    | Potentiometry                        |
| MSL    | Solubility                           |
| MSP    | Spectroscopy                         |
| MTD    | Temperature difference               |

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